

BIO 004 · MODULE 4**Contents**

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Scan for the course home: the concept videos, Mastery OS, the calendar, and the exam schedule. Nothing behind this code is reprinted in this packet.

WHAT MODULE 4 COVERS

Module 4 is the organ systems module, and it is the largest block of the course at fifty five competencies. Respiratory opens it, because it finishes the story the cardiovascular module started: blood leaves the heart, and this is where it is reloaded.

From there the module works through the systems that move material into and out of the body. The lymphatic system drains what the capillaries leave behind. The digestive system takes food apart, and its accessory organs supply the tools. The urinary system filters the blood and makes urine. The reproductive systems make gametes and, in the female, support a pregnancy. Endocrine anatomy closes the module, because its glands are scattered through every system above it.

This is anatomy. Where function appears in these notes it is there because the structure does not make sense without it.

Course Notes.

1 min

PART A

9 chapters, in the order the module teaches them. Read a chapter before the class day it belongs to, then do the matching pre-work packet in Part B.

1 The Respiratory System

2 The Lymphatic System

3 The Alimentary Canal

4 The Accessory Digestive Organs

5 The Urinary System

6 The Male Reproductive System

7 The Female Reproductive System

8 Pregnancy and Birth

9 Endocrine Anatomy

CHAPTER 1

The Respiratory System

9 min

THE RESPIRATORY SYSTEM

A reference for the respiratory system video and lab. This chapter covers the upper and lower respiratory tracts, the larynx, the bronchial tree, the lungs and pleurae, the alveoli, and common respiratory disorders. The focus is on structure.

BY THE END

1. Distinguish the upper and lower respiratory tracts and the conducting and respiratory zones.
2. Identify the parts of the nasal cavity and the pharynx.
3. Identify the cartilages of the larynx and the parts of the bronchial tree.
4. Describe the lungs, the pleurae, and the structure of an alveolus.
5. Name common respiratory disorders.



WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The respiratory system
drsrennie-stack.github.io/new-build-bio4-solano/respiratory-concept-videos.html

The Respiratory System, an Overview

The respiratory system moves air in and out and brings it close to the blood. Its parts are grouped two ways, and the two groupings do not line up, which is exactly why both are tested.

THE TWO WAYS OF GROUPING THE PARTS

Structure	What it is
Respiratory system	The organs that move air and bring it close to the blood for gas exchange
Upper respiratory tract	The nose, nasal cavity, paranasal sinuses, and pharynx
Lower respiratory tract	The larynx, trachea, bronchi, and lungs
Conducting zone	The passages that only carry air, from the nose to the terminal bronchioles. It warms, moistens, and filters
Respiratory zone	The structures where gas exchange happens: the respiratory bronchioles, alveolar ducts, and alveoli
A note on scope	This chapter covers the structure of the system. The exchange of gases is a physiology topic and is not covered here

The Upper Respiratory Tract

The upper tract takes in air and conditions it, warming, moistening, and filtering it before it reaches the lungs.

THE NOSE AND NASAL CAVITY

Structure	What it is
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Structure	What it is
External nose	The visible part, framed by the nasal bones and the frontal process of the maxilla above and hyaline cartilage below
Root	The area between the eyebrows, where the nose meets the forehead
Bridge	The upper bony part of the nose
Dorsum nasi	The anterior margin, running from root to apex
Apex	The tip of the nose
Nares	The external openings, the nostrils
Nasal cavity	The space behind the nose, divided into right and left halves by the nasal septum
Nasal septum	Septal cartilage in front, the vomer and the perpendicular plate of the ethmoid behind
Nasal vestibule	The area just inside the nostrils, lined with skin, sebaceous glands, and coarse hairs that filter particles
Nasal conchae	The three scroll-shaped projections on the lateral wall that swirl the air and increase contact with the warm, moist lining
Nasal meatuses	The passages beneath each concha, named superior, middle, and inferior
Hard palate	The bony floor of the nasal cavity, separating it from the oral cavity
Soft palate	The muscular posterior floor, which lifts to close the nasopharynx during swallowing
Choanae	The posterior openings where the nasal cavity becomes the nasopharynx
Paranasal sinuses	Air-filled cavities in the frontal, ethmoid, sphenoid, and maxillary bones that drain into the nasal cavity

The mucosa of the nasal cavity

TWO MUCOSAE, TWO JOBS

Mucosa	Epithelium	Role
Olfactory mucosa	Olfactory epithelium, in a small patch in the roof	Houses the receptors for smell
Respiratory mucosa	Pseudostratified ciliated columnar with goblet cells	Secretes mucus, traps particles, and sweeps them toward the pharynx to be swallowed

HOLD ONTO THIS

Cold air makes the cilia sluggish, so the mucus stops moving backward and runs forward instead. That is why a nose runs in cold weather, and it is a structural answer to an everyday question.

The pharynx

THE THREE REGIONS, TOP TO BOTTOM

Region	Location	Carries	Epithelium and structures
Nasopharynx	Behind the nasal cavity, above the soft palate	Air only	Pseudostratified ciliated columnar. Holds the pharyngeal tonsil and the openings of the auditory tubes
Oropharynx	Behind the oral cavity, from the soft palate to the epiglottis	Air and food	Stratified squamous, for friction. Holds the palatine and lingual tonsils
Laryngopharynx	From the epiglottis to the esophagus	Air and food	Stratified squamous. The point where the air and food paths divide

The Larynx

The larynx, the voice box, connects the pharynx to the trachea. It keeps food out of the airway, holds the airway open, and houses the vocal cords. Nine cartilages build it, three single and three paired.

THE NINE CARTILAGES

Cartilage	Number	Description
Thyroid cartilage	Single	The large shield-shaped cartilage forming the laryngeal prominence, the Adam's apple
Cricoid cartilage	Single	The ring-shaped cartilage just below the thyroid cartilage, the only complete ring in the airway
Epiglottis	Single	The flexible spoon-shaped flap of elastic cartilage that folds over the laryngeal inlet during swallowing
Arytenoid cartilages	Paired	Anchor the vocal folds and move them. The most important of the paired cartilages
Corniculate cartilages	Paired	Sit on top of the arytenoids
Cuneiform cartilages	Paired	Lie in the fold between the arytenoid and the epiglottis

THE FOLDS AND THE OPENING

Structure	What it is
Vocal folds	The true vocal cords. Pearly white and avascular, they vibrate as air passes to produce sound
Vocal ligaments	The elastic bands within the vocal folds, running from the arytenoid cartilages to the thyroid cartilage
Vestibular folds	The false vocal cords, above the true folds. They close the airway during swallowing and play no part in sound
Glottis	The vocal folds together with the opening between them

HOLD ONTO THIS

Every laryngeal cartilage is hyaline except the epiglottis, which is elastic. It has to bend and spring back with every swallow, and hyaline cartilage would not.

The Trachea and Bronchial Tree

Below the larynx the airway becomes a branching tree, growing narrower and more numerous at every division. Follow the path of air in order.

1. **Trachea** the windpipe, held permanently open by C-shaped rings of hyaline cartilage, with the trachealis muscle closing the gap posteriorly against the esophagus
2. **Carina** the internal ridge where the trachea forks into the two main bronchi
3. **Main bronchi** the primary bronchi, one entering each lung. The right is wider and more vertical
4. **Lobar bronchi** the secondary bronchi, one to each lobe of the lung
5. **Segmental bronchi** the tertiary bronchi, one to each bronchopulmonary segment
6. **Bronchioles** the smallest conducting airways, with smooth muscle walls and no cartilage
7. **Terminal bronchioles** the last airways of the conducting zone, just before gas exchange begins

HOLD ONTO THIS

Cartilage decreases and smooth muscle increases as the tree gets smaller. The switch point is the bronchiole, which is why bronchioles can constrict and bronchi cannot, and why asthma is a disease of the small airways.

The Lungs and Pleurae

The two lungs fill most of the thoracic cavity. They are not identical: the left lung gives up room for the heart.

COMPARE THE TWO LUNGS

Lung	Lobes	Fissures	Note
Right lung	Three: superior, middle, and inferior	A horizontal fissure and an oblique fissure	The larger of the two
Left lung	Two: superior and inferior	An oblique fissure only	Smaller, with a cardiac notch that makes room for the heart

THE SURFACES AND THE COVERING

Structure	What it is
Apex	The narrow superior tip of a lung, rising just above the clavicle
Base	The broad inferior surface of a lung, resting on the diaphragm
Hilum	The indentation on the medial surface where the bronchus and vessels enter and leave
Root of the lung	The bundle at the hilum: main bronchus, pulmonary artery, pulmonary veins, bronchial vessels, nerves, and lymphatics
Parietal pleura	The serous membrane lining the thoracic wall
Visceral pleura	The serous membrane covering the outer surface of the lung
Pleural cavity	The thin, fluid-filled space between the two pleurae that lets the lung glide as it expands
Cardiac notch	The indentation in the left lung that accommodates the heart
Lingula	The tongue-shaped projection of the left superior lobe, below the cardiac notch

The Respiratory Zone and the Alveoli

Past the terminal bronchioles the airway becomes the respiratory zone, where the walls thin out into air sacs and gas exchange becomes possible.

THE STRUCTURES OF THE RESPIRATORY ZONE

Structure	What it is
Respiratory bronchioles	The first airways of the respiratory zone, with a few alveoli budding from their walls
Alveolar ducts	Passages whose walls are lined entirely with alveoli
Alveolar sac	A cluster of alveoli that share a common space
Alveoli	The tiny air sacs that are the site of gas exchange. The lungs hold hundreds of millions of them
Type I alveolar cells	The thin, flat simple squamous cells that form most of the alveolar wall, where gases cross
Type II alveolar cells	The cuboidal cells that secrete surfactant, which lowers surface tension and keeps the alveoli from collapsing
Alveolar macrophages	The dust cells that patrol the alveolar surface and engulf debris
Respiratory membrane	The thin barrier between the alveolar air and the blood: alveolar epithelium, fused basement membranes, and capillary endothelium
Pulmonary capillaries	The dense net of capillaries wrapping each alveolus

Respiratory Disorders

Respiratory disorders mostly block airflow or damage the surface where gases cross.

COMPARE THE COMMON ONES BY THE STRUCTURE EACH ONE AFFECTS

Disorder	Structure affected	What it is
Asthma	The bronchioles	Inflammation and narrowing of the bronchioles, causing wheezing and breathlessness
COPD	The airways and alveoli	Chronic obstructive pulmonary disease, a long-term blockage of airflow that includes emphysema and chronic bronchitis
Emphysema	The alveoli	Destruction of the alveolar walls, which reduces the surface available for gas exchange
Pneumonia	The alveoli	An infection that inflames the alveoli and fills them with fluid
Pneumothorax	The pleural cavity	Air in the pleural cavity, which breaks the seal and lets the lung collapse
Lung cancer	The lung tissue	A malignant tumor of the lung, strongly linked to smoking

WHY THE DESIGN HAS WEAK POINTS

The **epiglottis** normally seals the airway during a swallow. When something slips past it, a coughing fit is the airway's attempt to clear it. Because the right main bronchus is wider and more vertical than the left, an inhaled object most often lands in the right lung.

The **alveoli** give the lungs an enormous surface, but it is fragile. Emphysema destroys alveolar walls and merges many small sacs into a few large ones, trading a huge exchange surface for a small one. The change is structural and permanent.

The **pleural cavity** is sealed and holds only a thin film of fluid. If the chest wall or lung is punctured, air enters that space, the seal is lost, and the lung collapses, a pneumothorax. The structure that lets the lung glide is also what makes it vulnerable.

The **cricoid cartilage** is the only complete ring in the airway, which is why the space just above it, the cricothyroid membrane, is the landmark for an emergency airway. Anatomy chooses that spot, not convenience.

CHAPTER 2

The Lymphatic System

6 min

THE LYMPHATIC SYSTEM

A reference for the lymphatic system video and lab. This chapter covers lymph and the pathway it follows, the lymphatic vessels, lymph nodes, the lymphatic organs, and common disorders of the system.

BY THE END

1. Define lymph and trace the pathway it follows back to the blood.
2. Describe the structure of lymphatic vessels and a lymph node.
3. Sort the lymphatic organs into primary and secondary.
4. Name common disorders of the lymphatic system.



WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The lymphatic systemdrsrennie-stack.github.io/new-build-bio4-solano/lymphatic-concept-videos.html

The Lymphatic System, an Overview

The lymphatic system is a one-way drainage network. It collects fluid that leaks out of the blood capillaries, filters it, and returns it to the bloodstream, while also housing the body's defenses.

WHAT THE SYSTEM IS AND WHAT IT DOES

Structure	What it is
Lymphatic system	A network of vessels, tissues, and organs that drains tissue fluid and defends the body
Interstitial fluid	The fluid in the spaces between cells, filtered out of the blood capillaries
Lymph	The interstitial fluid once it has entered the lymphatic vessels; a clear, watery fluid
Three functions	Draining excess interstitial fluid back to the blood, absorbing dietary fats from the small intestine, and housing the immune defenses
A one-way system	Lymphatic vessels carry lymph in one direction only, toward the heart; the system has no pump of its own

The Pathway of Lymph

Lymph travels a fixed route from the tissue spaces back to the bloodstream, growing into larger and larger vessels along the way. Follow it in order.

1. **Lymphatic capillaries** blind-ended tubes that take up excess interstitial fluid from the tissue spaces
2. **Lymphatic collecting vessels** larger vessels, with valves, that carry the lymph onward
3. **Lymph nodes** the lymph is filtered as it passes through nodes spaced along the vessels
4. **Lymphatic trunks** collecting vessels merge into large trunks, each draining a major region of the body
5. **Lymphatic ducts** the trunks empty into two ducts: the right lymphatic duct and the larger thoracic duct
6. **Subclavian veins** the ducts return the lymph to the bloodstream where the subclavian and internal jugular veins meet

Lymphatic Vessels

Lymphatic vessels resemble veins, but they begin as blind-ended capillaries and they move fluid without a pump.

THE VESSELS, FROM THE SMALLEST TO THE DUCTS

Structure	What it is
Lymphatic capillaries	The smallest lymphatic vessels, blind-ended and more permeable than blood capillaries
Overlapping endothelial cells	The cells of a lymphatic capillary overlap as one-way flap valves, letting fluid in but not back out
Lacteals	Specialized lymphatic capillaries in the lining of the small intestine that absorb dietary fat
Lymphatic valves	One-way valves in the collecting vessels that keep lymph moving in a single direction
How lymph moves	With no pump, lymph is pushed along by the squeeze of nearby skeletal muscles and by breathing movements
Right lymphatic duct	The smaller duct, draining lymph from the upper right quadrant of the body
Thoracic duct	The larger duct, draining the rest of the body; it begins at a sac called the cisterna chyli

Lymph Nodes

Lymph nodes are the filters of the system. Hundreds of them lie along the lymphatic vessels, cleaning the lymph before it returns to the blood.

THE PARTS OF A NODE AND THE VESSELS THAT SERVE IT

Structure	What it is
Lymph node	A small, bean-shaped organ that filters lymph and houses immune cells
Capsule	The dense connective tissue covering of a node
Cortex	The outer region of a node, packed with lymphocytes in rounded clusters
Medulla	The inner region of a node
Afferent vessels	The lymphatic vessels that carry lymph into a node
Efferent vessels	The lymphatic vessels that carry filtered lymph out of a node, leaving at the hilum
Node clusters	Nodes are especially concentrated in the cervical, axillary, and inguinal regions

HOLD ONTO THIS

Afferent vessels arrive, efferent vessels exit. The same pair of words names flow in and flow out everywhere in the body, so learning it once here pays off again in the kidney and the nervous system.

The Lymphatic Organs

The lymphatic organs are sorted into primary organs, where lymphocytes form and mature, and secondary organs, where lymphocytes meet and respond to invaders. Compare them.

THE PRIMARY AND SECONDARY LYMPHATIC ORGANS

Organ	Group	Location	Role
Red bone marrow	Primary	The spongy bone of the axial skeleton and limb girdles	Where all lymphocytes form, and where B cells mature
Thymus	Primary	The mediastinum, behind the sternum	Where T cells mature
Lymph nodes	Secondary	Clustered along the lymphatic vessels	Filter the lymph
Spleen	Secondary	The upper left abdomen, behind the stomach	Filters the blood
Tonsils	Secondary	A ring around the entrance to the throat	Guard against inhaled and swallowed pathogens
MALT	Secondary	The mucous membranes of the digestive, respiratory, and other tracts	Guards the body's mucosal surfaces

HOLD ONTO THIS

Only two organs are primary, the red bone marrow and the thymus, because those are the only places lymphocytes are made and matured. Everything else on the list is secondary, a place where a finished lymphocyte meets what it was built to fight.

The Spleen, Thymus, and Tonsils

Three of the lymphatic organs are worth a closer look at their structure.

The spleen**THE SPLEEN AND ITS TWO TISSUES**

Structure	What it is
Spleen	The largest lymphatic organ, in the upper left abdomen behind the stomach
White pulp	The lymphocyte-rich tissue of the spleen that carries out its immune functions
Red pulp	The tissue of the spleen that filters the blood and removes worn-out erythrocytes

The thymus and tonsils**THE THYMUS AND THE RING OF TONSILS**

Structure	What it is
Thymus	The primary organ where T cells mature; large in childhood and shrinking after puberty
Tonsils	A ring of lymphatic tissue guarding the throat, including the palatine, lingual, and pharyngeal tonsils
Pharyngeal tonsil	The single tonsil on the posterior wall of the nasopharynx, also called the adenoid

Disorders of the Lymphatic System

When drainage fails or the lymphatic tissue is diseased, the problems show up as swelling or as changes in the nodes and organs. Compare the common ones.

COMPARE THE COMMON ONES BY THE STRUCTURE EACH ONE AFFECTS

Disorder	Structure affected	What it is
Edema	The interstitial space	Swelling from excess interstitial fluid when lymphatic drainage cannot keep up
Lymphedema	The lymphatic vessels	Long-term tissue swelling from blocked or missing lymphatic vessels, sometimes after lymph node removal
Lymphadenopathy	The lymph nodes	Enlarged lymph nodes, often a sign the nodes are fighting an infection
Lymphoma	Lymphatic tissue	A cancer arising in lymphatic tissue, including Hodgkin and non-Hodgkin lymphoma
Splenomegaly	The spleen	An enlarged spleen, which can follow an infection or a blood disorder
Tonsillitis	The tonsils	Inflammation of the tonsils, common in childhood

THE SYSTEM A CLINICIAN READS THROUGH THE SKIN

Swollen glands are **lymphadenopathy**: lymph nodes enlarge as they trap and fight pathogens, which is why a clinician feels for tender nodes in the neck, armpit, and groin during an exam.

Because lymph passes through nodes in order, the lymphatic system also maps how some cancers spread. The first node a region drains into is the **sentinel node**, and checking it tells a surgeon whether a cancer has begun to travel. Removing many nodes can block drainage and leave lasting lymphedema in the limb.

The **spleen** sits just under the lower left ribs. Its soft tissue is easily torn, so a blow to that area can rupture the spleen and cause heavy internal bleeding, since the organ filters a large volume of blood.

CHAPTER 3

The Alimentary Canal

8 min

THE DIGESTIVE SYSTEM, THE ALIMENTARY CANAL

Covers the wall of the digestive tube, the peritoneum, and the regions of the canal from the mouth to the anus, with the named features of the stomach and the intestines. The focus is on the structures and the job each one does.

- BY THE END**
1. Distinguish the alimentary canal from the accessory organs and name the four layers of its wall.
 2. Trace the pathway of food from the mouth to the anus.
 3. Identify the named features of the stomach, small intestine, and large intestine.



WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The digestive system
drsrennie-stack.github.io/new-build-bio4-solano/gi-concept-videos.html

The Alimentary Canal, an Overview

The digestive system has two groups of organs. The alimentary canal is the continuous tube that food travels through; the accessory organs sit alongside it and empty their secretions in. Covers the canal.

THE TWO GROUPS OF DIGESTIVE ORGANS

Structure	What it is
Digestive system	The organs that take in food and eliminate waste; the alimentary canal plus the accessory organs
Alimentary canal	The gastrointestinal tract; a continuous muscular tube from the mouth to the anus
Organs of the canal	The mouth, most of the pharynx, the esophagus, the stomach, the small intestine, and the large intestine
Accessory organs	The teeth, tongue, salivary glands, liver, gallbladder, and pancreas; covered on their own sheet
Lumen	The open interior space of the tube, continuous with the outside world at the mouth and anus
Length	About 16 to 23 feet in a living person, longer once muscle tone is lost

The Wall of the Alimentary Canal

From the lower esophagus to the anal canal, the wall of the tube is built from the same four layers, deep to superficial. Compare them.

THE FOUR LAYERS OF THE WALL

Layer	What it contains
Mucosa	The inner lining; an epithelium in contact with food, the lamina propria of areolar connective tissue beneath it, and a thin muscularis mucosae
Submucosa	Areolar connective tissue with blood and lymphatic vessels, and the submucosal nerve plexus
Muscularis	Usually an inner circular and an outer longitudinal layer of smooth muscle, with the myenteric nerve plexus between them
Serosa	The outermost layer; a serous membrane, also called the visceral peritoneum, replaced by a fibrous adventitia on the esophagus

Two nerve networks in the wall make up the enteric nervous system, the nerve supply of the gut.

THE ENTERIC NERVOUS SYSTEM

Structure	What it is
Submucosal plexus	The nerve network in the submucosa that controls the secretions of the gut
Myenteric plexus	The nerve network between the muscle layers that controls the movement of the gut

HOLD ONTO THIS

Four layers everywhere, with two exceptions worth holding. The esophagus wears a fibrous adventitia where the rest of the tube has a serosa, and the stomach carries a third layer of muscle where the rest carries two.

The Peritoneum

The peritoneum is the large serous membrane of the abdomen. Its folds anchor the digestive organs and carry their vessels and nerves.

THE LAYERS OF THE PERITONEUM

Structure	What it is
Parietal peritoneum	The layer that lines the wall of the abdominal cavity
Visceral peritoneum	The layer that covers the organs; it is the serosa of those organs
Peritoneal cavity	The slim, fluid-filled space between the parietal and visceral layers
Retroperitoneal organs	Organs behind the peritoneum, not within its cavity; the kidneys, pancreas, duodenum, and parts of the colon

Five large folds of peritoneum weave among the organs. Compare what each one connects.

THE FIVE FOLDS OF PERITONEUM

Fold	What it connects
Greater omentum	The largest fold; a fatty apron hanging from the stomach over the intestines
Lesser omentum	Connects the stomach and duodenum to the liver, and carries vessels to the liver
Falciform ligament	Attaches the liver to the anterior abdominal wall and the diaphragm
Mesentery	A fan-shaped fold binding the jejunum and ileum to the posterior abdominal wall
Mesocolon	Binds the transverse and sigmoid colon to the posterior abdominal wall

The Mouth, Pharynx, and Esophagus

Food enters at the mouth, is swallowed through the pharynx, and travels the esophagus to the stomach.

FROM THE ORAL CAVITY TO THE STOMACH

Structure	What it is
Oral cavity	The mouth; bounded by the cheeks, the lips, the palate, and the tongue
Hard palate	The bony anterior roof of the mouth, formed by the maxillae and palatine bones
Soft palate	The muscular posterior roof of the mouth
Uvula	The finger-like projection of the soft palate that closes off the nasopharynx during swallowing
Fauces	The opening from the mouth into the oropharynx
Palatine and lingual tonsils	Lymphatic tissue between the palatal arches and at the base of the tongue
Pharynx	The funnel-shaped passage with three regions: the nasopharynx, oropharynx, and laryngopharynx
Esophagus	The collapsible muscular tube, about 10 inches long, that carries food from the laryngopharynx to the stomach
Esophageal hiatus	The opening in the diaphragm that the esophagus passes through
Upper and lower esophageal sphincters	The rings of muscle that guard the top and bottom of the esophagus; the lower is also called the cardiac sphincter

The Stomach

The stomach is a J-shaped widening of the tube below the diaphragm. It is a mixing chamber and a holding reservoir.

THE REGIONS AND NAMED FEATURES OF THE STOMACH

Structure	What it is
Cardia	The region that surrounds the opening from the esophagus
Fundus	The rounded portion that bulges above and to the left of the cardia
Body	The large central portion of the stomach
Pyloric part	The region that connects the body to the duodenum; it has the pyloric antrum, pyloric canal, and pylorus
Pyloric sphincter	The ring of smooth muscle between the pylorus and the duodenum
Lesser curvature	The concave superior border of the stomach
Greater curvature	The convex inferior border of the stomach
Rugae	The folds of the inner stomach lining that flatten as the stomach fills
Gastric pits and glands	The pits in the stomach lining that lead to the gastric glands, the secreting glands of the stomach
Muscularis of the stomach	Three layers of smooth muscle, an inner oblique, a middle circular, and an outer longitudinal; one more than the rest of the tube

The Small Intestine

The small intestine is the long, coiled region where most absorption happens. Its lining is folded again and again to expand the surface area.

THE REGIONS OF THE SMALL INTESTINE AND THE FOLDS OF ITS LINING

Structure	What it is
Duodenum	The first and shortest region, about 12 inches, curving from the pyloric sphincter; mostly retroperitoneal
Jejunum	The middle region, about 3 feet long
Ileum	The longest region, about 6 feet, joining the large intestine at the ileocecal sphincter
Ileocecal sphincter	The valve of smooth muscle where the ileum meets the large intestine
Circular folds	The plicae circulares; permanent ridges of the mucosa and submucosa that increase surface area
Villi	Finger-like projections of the mucosa that give the lining a velvety look and greatly increase surface area
Lacteal	The lymphatic capillary inside each villus that absorbs dietary fats
Microvilli	The microscopic projections of the absorptive cells that form the brush border

HOLD ONTO THIS

The small intestine amplifies its surface three times over, each fold smaller than the last: circular folds of the wall, villi of the mucosa, and microvilli on the single absorptive cell.

The Large Intestine

The large intestine is the wider, final region of the canal. It absorbs water and forms and stores the feces.

THE REGIONS AND NAMED FEATURES OF THE LARGE INTESTINE

Structure	What it is
Cecum	The pouch at the start of the large intestine, just below the ileocecal valve
Vermiform appendix	The narrow, worm-like tube of lymphatic tissue attached to the cecum
Colon	The long portion with four parts: the ascending, transverse, descending, and sigmoid colon
Rectum	The straight portion, about 6 inches, anterior to the sacrum and coccyx
Anal canal	The last 2 to 3 cm of the tract, ending at the anus
Internal and external anal sphincters	The involuntary smooth muscle and voluntary skeletal muscle that guard the anus
Teniae coli	Three thickened bands of longitudinal muscle running the length of the colon
Haustra	The pouches that give the colon its puckered appearance, formed by the teniae coli
Omental appendices	Small fat-filled pouches of peritoneum attached to the teniae coli

The Pathway of Food

Food moves through the alimentary canal in one direction, through each region in turn.

1. **Mouth** food is taken in, chewed, and mixed with saliva
2. **Pharynx** the swallowed food passes through the oropharynx and laryngopharynx
3. **Esophagus** muscular contractions carry the food down to the stomach
4. **Stomach** food is churned and held as it becomes a liquid
5. **Small intestine** the duodenum, jejunum, and ileum, where most absorption occurs
6. **Large intestine** the cecum, colon, rectum, and anal canal, where water is absorbed and feces form
7. **Anus** the feces are eliminated

Common Disorders of the Alimentary Canal

Compare the common disorders of the canal by the structure each one affects.

THE COMMON DISORDERS OF THE CANAL

Disorder	What it is
Gastroesophageal reflux disease (GERD)	Stomach contents flow back up through a weak lower esophageal sphincter and irritate the esophagus
Hiatal hernia	Part of the stomach pushes up through the esophageal hiatus of the diaphragm
Peptic ulcer	An eroded sore in the wall of the stomach or duodenum
Appendicitis	Inflammation of the vermiform appendix; rupture can spread infection to the peritoneum
Diverticulosis	Small pouches that bulge outward through weak spots in the colon wall
Inflammatory bowel disease	Chronic inflammation of the intestinal wall, as in Crohn disease and ulcerative colitis
Hemorrhoids	Swollen, dilated veins in the wall of the anal canal and rectum
Colorectal cancer	A malignant tumor of the colon or rectum

THE LOCATION OF A PROBLEM NAMES THE SYMPTOM

Because each region of the canal has its own job, where a problem sits predicts what the patient feels. A weak **lower esophageal sphincter** brings heartburn behind the breastbone; an inflamed **appendix** brings pain to the lower right abdomen.

The four-layer wall explains how a problem deepens. A **peptic ulcer** begins in the mucosa, and the danger is that it erodes through the submucosa and muscularis. If it reaches the serosa and perforates, the contents of the lumen spill into the peritoneal cavity.

CHAPTER 4

The Accessory Digestive Organs

7 min

ACCESSORY ORGANS OF DIGESTION

Covers the teeth and tongue, the salivary glands, the liver, the gallbladder and the path of bile, and the pancreas. The focus is on the structures and the job each one does.

BY THE END

1. Name the six accessory digestive organs and the role of each.
2. Identify the structures of the liver, gallbladder, and pancreas.
3. Trace the path of bile from the liver to the small intestine.



WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Accessory digestive organs

drsrennie-stack.github.io/new-build-bio4-solano/accessory-digestive-concept-videos.html

The Accessory Organs, an Overview

The accessory digestive organs are not part of the food tube itself. The teeth and tongue work on food in the mouth; the rest add secretions to the canal through ducts.

THE SIX ACCESSORY ORGANS AND HOW THEY WORK

Structure	What it is
Accessory digestive organs	The teeth, tongue, salivary glands, liver, gallbladder, and pancreas
Teeth and tongue	The accessory organs that make direct contact with food in the mouth
Glandular accessory organs	The salivary glands, liver, and pancreas, which produce secretions delivered through ducts
Storage organ	The gallbladder, which stores and concentrates bile until it is needed

The Teeth and Tongue

The teeth and the tongue prepare food in the mouth, breaking it down and moving it for chewing and swallowing.

THE STRUCTURES OF THE TEETH AND THE TONGUE

Structure	What it is
Teeth	The accessory organs that cut and grind food during chewing
Tongue	A digestive organ of skeletal muscle covered with mucous membrane; it forms the floor of the mouth
Extrinsic tongue muscles	Muscles that originate outside the tongue and move it side to side and in and out
Intrinsic tongue muscles	Muscles within the tongue that alter its shape and size for speech and swallowing
Lingual frenulum	The midline fold of mucous membrane anchoring the tongue to the floor of the mouth
Papillae	The projections on the tongue surface that carry taste buds and aid the grip on food
Lingual glands	Small glands in the tongue that secrete mucus and watery fluid

The Salivary Glands

Saliva keeps the mouth moist, cleanses it, and begins the breakdown of food. Three large pairs of glands produce most of it. Compare them.

THE THREE LARGE PAIRS OF SALIVARY GLANDS

Gland	Location
Parotid glands	Anterior and inferior to the ears, between the skin and the masseter muscle; the duct opens by the second upper molar
Submandibular glands	In the floor of the mouth, along the inner side of the mandible; the ducts open near the lingual frenulum
Sublingual glands	Beneath the tongue, above the submandibular glands; the ducts open onto the floor of the mouth

THE MINOR GLANDS AND THEIR PRODUCT

Structure	What it is
Minor salivary glands	Small labial, buccal, palatal, and lingual glands scattered through the lining of the mouth
Saliva	The watery secretion that moistens, cleanses, and lubricates food and begins its breakdown

The Liver

The liver is the largest gland in the body. It sits below the diaphragm, mostly in the upper right abdomen, and it makes bile.

THE LOBES, LIGAMENTS, AND MICROSCOPIC STRUCTURES OF THE LIVER

Structure	What it is
Liver	The large gland below the diaphragm; it produces bile
Right and left lobes	The two principal lobes of the liver, separated by the falciform ligament
Quadrante and caudate lobes	Two smaller lobes on the posterior and inferior surface
Falciform ligament	The fold of peritoneum that suspends the liver from the anterior abdominal wall and diaphragm
Ligamentum teres	The round ligament; a fibrous cord in the free edge of the falciform ligament, a remnant of the fetal umbilical vein
Hepatocytes	The functional cells of the liver, arranged in plates called hepatic laminae
Hepatic sinusoids	The wide, leaky capillaries that run between the plates of hepatocytes
Bile canaliculi	The tiny channels between hepatocytes that collect the bile they make
Portal triad	The trio found at the corner of each liver lobule: a bile duct, a branch of the hepatic artery, and a branch of the hepatic portal vein

HOLD ONTO THIS

The portal triad is three things at the corner of every lobule, and the traffic runs both ways: the branch of the hepatic artery and the branch of the hepatic portal vein bring blood in, while the bile duct carries bile out.

The Gallbladder and the Path of Bile

The gallbladder is a small sac on the underside of the liver that stores bile. Bile travels a branching set of ducts to reach the small intestine.

THE GALLBLADDER AND THE BILE DUCTS

Structure	What it is
Gallbladder	The pear-shaped sac on the posterior surface of the liver that stores and concentrates bile
Fundus, body, and neck	The three regions of the gallbladder, from the broad inferior end to the tapered superior end
Cystic duct	The duct that carries bile to and from the gallbladder
Common bile duct	The duct formed where the common hepatic duct and the cystic duct join
Hepatopancreatic ampulla	The ampulla of Vater, the short widened duct where the bile duct and pancreatic duct meet before the duodenum

Bile drains away from the liver cells along a path of widening ducts to reach the duodenum.

1. **Bile canaliculi** tiny channels between hepatocytes collect the bile
2. **Hepatic ducts** the bile is gathered into the right and left hepatic ducts, which merge into the common hepatic duct
3. **Cystic duct and gallbladder** bile may detour into the gallbladder for storage through the cystic duct
4. **Common bile duct** the common hepatic and cystic ducts join to form the common bile duct
5. **Hepatopancreatic ampulla** the common bile duct joins the pancreatic duct at the ampulla
6. **Duodenum** bile enters the small intestine at the major duodenal papilla, past the sphincter of Oddi

The Pancreas

The pancreas lies behind the stomach. It is both an exocrine gland, secreting into the duodenum, and an endocrine gland.

THE REGIONS, DUCTS, AND CLUSTERS OF THE PANCREAS

Structure	What it is
Pancreas	The gland posterior to the stomach, retroperitoneal in position
Head, body, and tail	The three regions of the pancreas, from the wide end in the curve of the duodenum to the tapered end
Pancreatic duct	The main duct, also called the duct of Wirsung; it joins the common bile duct
Accessory duct	The smaller duct, also called the duct of Santorini, that empties separately into the duodenum
Major duodenal papilla	The raised opening in the duodenal lining where the hepatopancreatic ampulla delivers bile and pancreatic juice
Sphincter of Oddi	The muscular ring that controls flow through the hepatopancreatic ampulla into the duodenum
Acini	The exocrine clusters of the pancreas that produce pancreatic juice; about 99 percent of the gland
Pancreatic islets	The endocrine clusters of the pancreas that produce hormones; about 1 percent of the gland

HOLD ONTO THIS

Each pancreatic duct carries two names. The main one is the duct of Wirsung and the smaller one is the duct of Santorini, and only the main one joins the common bile duct.

Common Disorders of the Accessory Organs

Compare the common disorders of the accessory organs by the structure each one affects.

THE COMMON DISORDERS OF THE ACCESSORY ORGANS

Disorder	What it is
Gallstones	Hardened deposits that form in the gallbladder and can block the bile ducts
Cholecystitis	Inflammation of the gallbladder, often caused by a blocking gallstone
Hepatitis	Inflammation of the liver, commonly from a viral infection
Cirrhosis	The replacement of healthy liver tissue with scar tissue, with loss of liver function
Jaundice	A yellowing of the skin and eyes from a buildup of bilirubin, often from liver or bile duct disease
Pancreatitis	Inflammation of the pancreas
Mumps	A viral infection that inflames and enlarges the parotid glands

SHARED DUCTS, SHARED PROBLEMS

The bile duct and the pancreatic duct meet at the **hepatopancreatic ampulla** before they reach the duodenum. Because they share that final passage, a gallstone lodged at the ampulla can block bile and pancreatic juice at once, which is one way gallstones trigger pancreatitis.

The liver's blood supply is unusual. It receives nutrient-rich blood from the **hepatic portal vein** as well as oxygenated blood from the **hepatic artery**. That heavy blood flow filters the body's nutrients, but it also makes the liver a common site for cancer to spread to.

CHAPTER 5

The Urinary System

8 min

THE URINARY SYSTEM

Covers the gross anatomy of the kidney, the nephron, the juxtaglomerular apparatus and filtration membrane, the blood supply, and the ureters, bladder, and urethra. The focus is on the structures and the job each one does.

BY THE END

1. Describe the gross anatomy of the kidney and its tissue layers.
2. Identify the parts of a nephron and the layers of the filtration membrane.
3. Trace the blood supply of the kidney and the path of urine.



WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The urinary system

drsrennie-stack.github.io/new-build-bio4-solano/urinary-concept-videos.html

The Urinary System, an Overview

The urinary system filters the blood and makes urine. The kidneys do the work; the rest of the system carries and stores the urine.

THE PARTS OF THE SYSTEM AND WHAT EACH ONE DOES

Structure	What it is
Urinary system	The kidneys, ureters, urinary bladder, and urethra
Kidneys	The paired organs that filter the blood and form urine
Ureters	The paired tubes that carry urine from the kidneys to the bladder
Urinary bladder	The muscular sac that stores urine
Urethra	The tube that carries urine from the bladder to the outside
Roles of the kidney	Filtering wastes from the blood, balancing water, ions, and pH, and helping regulate blood pressure

The Kidneys, Gross Anatomy

The kidneys are bean-shaped organs against the back wall of the abdomen. Three tissue layers cushion and anchor each one, and the interior sorts into a cortex and a medulla.

THE POSITION, THE COVERINGS, AND THE REGIONS INSIDE

Structure	What it is
Location	Retroperitoneal, between T12 and L3; the right kidney sits slightly lower because of the liver
Renal hilum	The notch on the medial border where the ureter, vessels, and nerves enter and leave
Renal capsule	The smooth, fibrous inner layer that maintains the kidney's shape
Adipose capsule	The layer of fat around the capsule that cushions the kidney and holds it in place
Renal fascia	The outer layer of connective tissue that anchors the kidney to the abdominal wall
Renal cortex	The outer region of kidney tissue, just beneath the capsule
Renal columns	Extensions of the cortex that dip down between the renal pyramids
Renal medulla	The inner region of kidney tissue
Renal pyramids	The cone-shaped masses of the medulla, with their bases toward the cortex
Renal papilla	The tip of a renal pyramid, which drains urine into a minor calyx
Parenchyma	The functional tissue of the kidney, the cortex and pyramids together, which holds the nephrons

The Nephron

The nephron is the functional unit of the kidney, the structure that actually makes urine. It has two main parts: the renal corpuscle and the renal tubule.

THE PARTS OF A NEPHRON

Structure	What it is
Nephron	The microscopic functional unit of the kidney; each kidney holds about one million
Renal corpuscle	The filtering head of the nephron; the glomerulus inside the glomerular capsule
Glomerulus	The tuft of leaky, fenestrated capillaries where the blood is filtered
Glomerular capsule	Bowman's capsule; the cup that surrounds the glomerulus and collects the filtrate
Podocytes	The cells of the inner, visceral layer of the capsule, whose foot-like processes form filtration slits
Capsular space	The space inside the capsule where the filtrate collects before entering the tubule
Proximal convoluted tubule	The first, coiled segment of the renal tubule, lined with a brush border of microvilli
Nephron loop	The loop of Henle; the U-shaped segment with a descending limb and an ascending limb
Distal convoluted tubule	The second coiled segment of the renal tubule
Collecting duct	The duct that gathers fluid from many nephrons and carries it toward the renal papilla

HOLD ONTO THIS

The segments of the renal tubule are named in the order the filtrate meets them: capsular space, proximal convoluted tubule, nephron loop, distal convoluted tubule, collecting duct. Learn the segments in that order and the path of the filtrate through the nephron comes free.

Nephrons come in two types, sorted by how deep their loop reaches.

COMPARE THEM

Type	Features
Cortical nephrons	About eighty five percent of nephrons; short loops that barely enter the medulla, wrapped by peritubular capillaries
Juxtamedullary nephrons	About fifteen percent of nephrons; long loops that reach deep into the medulla, served by the vasa recta; they concentrate the urine

The Juxtaglomerular Apparatus and Filtration Membrane

Where the distal tubule touches its own glomerulus, a small structure monitors and adjusts filtration. The filtration membrane itself is a three-layer sieve.

THE JUXTAGLOMERULAR APPARATUS

Structure	What it is
Juxtaglomerular apparatus	The structure where the distal convoluted tubule contacts the arterioles of its own renal corpuscle
Macula densa	The cells of the distal tubule that sense the sodium and chloride concentration of the fluid
Juxtaglomerular cells	Modified smooth muscle cells in the afferent arteriole that release renin when blood pressure is low

Blood is filtered across a three-layer membrane.

COMPARE WHAT EACH LAYER HOLDS BACK

Layer	What it holds back
Fenestrated endothelium	The pored wall of the glomerular capillary; it lets plasma pass but blocks blood cells
Basement membrane	The layer between the capillary and the podocytes; it blocks large plasma proteins
Slit membranes	The filtration slits between the podocyte processes; they block medium-sized proteins

The Blood Supply of the Kidney

Blood reaches the nephrons through a branching chain of arteries and leaves through a matching set of veins. Trace the arterial path.

1. **Renal artery** branches from the abdominal aorta and enters at the hilum
2. **Segmental and interlobar arteries** carry blood between the renal pyramids
3. **Arcuate arteries** arch along the bases of the pyramids, between the cortex and medulla
4. **Interlobular arteries** branch up into the cortex
5. **Afferent arteriole** delivers blood into the glomerulus
6. **Glomerulus** the capillary tuft where filtration occurs
7. **Efferent arteriole** carries blood out of the glomerulus
8. **Peritubular capillaries and vasa recta** wrap the renal tubules, then drain into veins that lead to the renal vein

HOLD ONTO THIS

Blood meets two capillary beds in the kidney, one after the other: the glomerulus, where filtration happens, and then the peritubular capillaries and vasa recta around the tubules. The efferent arteriole is the vessel between them.

The Ureters, Bladder, and Urethra

Once urine leaves the nephrons it is collected, carried, stored, and finally released by the rest of the system.

FROM THE RENAL PAPILLA TO THE OUTSIDE

Structure	What it is
Minor and major calyces	The cup-shaped chambers that collect urine from the renal papillae and channel it into the renal pelvis
Renal pelvis	The funnel-shaped cavity that collects urine and passes it into the ureter
Ureters	The muscular tubes that carry urine from the kidneys to the bladder by peristalsis
Trigone	The smooth triangular area at the base of the bladder, marked by the two ureter openings and the urethral opening
Detrusor muscle	The smooth muscle layer of the bladder wall that contracts to empty it
Internal urethral sphincter	The involuntary smooth muscle ring at the bladder outlet
External urethral sphincter	The voluntary skeletal muscle ring that gives conscious control over urination
Male urethra	Longer; it passes through the prostate as the prostatic urethra, through the perineum as the membranous urethra, and through the penis as the spongy urethra
Female urethra	Shorter; it opens between the clitoris and the vaginal opening

The Path of Urine

Urine leaves the nephron and flows through a fixed series of structures to the outside.

1. **Collecting duct** gathers the fluid from many nephrons
2. **Minor and major calyces** collect the urine draining from the renal papillae
3. **Renal pelvis** funnels the urine toward the ureter
4. **Ureter** carries the urine down to the bladder
5. **Urinary bladder** stores the urine
6. **Urethra** carries the urine out of the body

Common Disorders of the Urinary System

Compare the common disorders of the urinary system by the structure each one affects.

COMMON DISORDERS OF THE URINARY SYSTEM

Disorder	What it is
Urinary tract infection	An infection of the urethra, bladder, or kidneys, usually by bacteria
Kidney stones	Hardened mineral deposits that form in the kidney and can lodge in the ureter
Glomerulonephritis	Inflammation of the glomeruli that impairs filtration
Renal failure	Loss of the kidney's ability to filter the blood, acute or chronic
Nephroptosis	A floating kidney; the kidney drops from its normal position when its fat support is lost
Urinary incontinence	Loss of voluntary control over urination, as stress, urge, overflow, or functional incontinence
Polycystic kidney disease	An inherited disorder in which many fluid-filled cysts enlarge the kidneys

POSITION AND PATHWAY PREDICT THE PROBLEM

The kidney depends on its **adipose capsule** to stay in place. When that fat is lost, the kidney can drop, kinking the ureter and blocking the flow of urine. The anatomy of the support explains the symptom.

Urine travels one narrow path. A **kidney stone** that forms in the renal pelvis causes no pain until it enters the **ureter**, where it scrapes and blocks a tube only millimeters wide. The location along that path predicts where the pain is felt.

The two urethras differ, and so does the risk. The **female urethra** is short, which gives bacteria a shorter climb to the bladder and is one reason urinary tract infections are more common in women.

CHAPTER 6

The Male Reproductive System

6 min

THE REPRODUCTIVE SYSTEM, MALE

Covers the scrotum and testes, the duct system and the path of sperm, the structure of a sperm cell, the accessory glands, and the penis. The focus is on the structures and the job each one does.

- BY THE END**
1. Identify the structures of the scrotum and the testes.
 2. Trace the path of sperm through the duct system.
 3. Name the accessory glands and the parts of a sperm cell and the penis.



WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The male reproductive system
drsrennie-stack.github.io/new-build-bio4-solano/reproductive-male-concept-videos.html

The Male Reproductive System, an Overview

The male reproductive system produces sperm and delivers it. Its parts sort into the gonads, a duct system, the accessory glands, and the supporting structures.

THE FOUR GROUPS OF PARTS

Structure	What it is
Testes	The male gonads; they produce sperm and secrete hormones
Duct system	The epididymis, ductus deferens, ejaculatory ducts, and urethra, which store, mature, and transport sperm
Accessory glands	The seminal vesicles, prostate, and bulbourethral glands, which add fluid to form semen
Supporting structures	The scrotum and the penis, which help deliver sperm into the female tract

The Scrotum and Testes

The testes hang outside the body in the scrotum, which keeps them a few degrees cooler than the core. Inside each testis, coiled tubules produce the sperm.

THE SCROTUM OUTSIDE AND THE TESTIS WITHIN

Structure	What it is
Scrotum	The pouch of skin and subcutaneous tissue that supports the testes; a raphe marks its surface and a scrotal septum divides it within
Dartos muscle	The smooth muscle of the scrotum that wrinkles the skin to reduce heat loss
Cremaster muscle	The skeletal muscle that raises and lowers the testes to regulate their temperature
Tunica vaginalis	The serous membrane, derived from the peritoneum, that partly covers each testis
Tunica albuginea	The white fibrous capsule that divides each testis into two hundred to three hundred lobules
Seminiferous tubules	The tightly coiled tubes within the lobules where sperm are produced
Sustentacular cells	The Sertoli cells lining the seminiferous tubules; they nourish developing sperm and form the blood-testis barrier
Interstitial cells	The Leydig cells between the tubules; they produce testosterone

HOLD ONTO THIS

Two muscles set the temperature of the testes, and they are different tissues. The dartos is smooth muscle that wrinkles the scrotal skin; the cremaster is skeletal muscle that raises and lowers the testes themselves.

The Duct System and the Path of Sperm

After they form, sperm travel a fixed series of ducts that store them, mature them, and carry them out.

THE DUCTS, IN ORDER, AND THE CORD THAT CARRIES THEM

Structure	What it is
Epididymis	The comma-shaped, coiled duct on the testis where sperm are stored and gain motility
Ductus deferens	The vas deferens; the duct that carries sperm from the epididymis up toward the urethra
Ejaculatory duct	The short duct formed where the ductus deferens meets the seminal vesicle duct; it empties into the prostatic urethra
Urethra	The shared final duct; it passes through the prostate, the perineum, and the penis
Spermatic cord	The bundle that contains the ductus deferens, blood vessels, nerves, and the cremaster muscle

Sperm follow one route from the testis to the outside.

1. **Seminiferous tubules** sperm are produced inside the testis
2. **Epididymis** sperm are stored and gain the ability to move
3. **Ductus deferens** sperm are carried up out of the scrotum
4. **Ejaculatory duct** the ductus deferens joins the seminal vesicle duct
5. **Urethra** sperm pass through the prostatic, membranous, and spongy urethra to the outside

HOLD ONTO THIS

Only the seminiferous tubules make sperm. Every other structure on this route stores, matures, or carries them, so sort the male structures into one that produces and the rest that transport.

The Structure of a Sperm Cell

A mature sperm cell is built for one job, reaching and entering an oocyte. It has a head and a tail.

THE HEAD AND THE PARTS OF THE TAIL

Structure	What it is
Head	The part of the sperm that holds the nucleus with its twenty three chromosomes
Acrosome	The enzyme-filled cap over the head that helps the sperm penetrate the oocyte
Neck	The part of the tail just behind the head, containing the centrioles
Middle piece	The part of the tail packed with mitochondria, which power movement
Principal piece	The longest part of the tail
End piece	The tapering terminal part of the tail

The Accessory Glands and Semen

Three sets of accessory glands add fluid to the sperm. The mixture of sperm and these fluids is semen.

COMPARE THE GLANDS

Gland	What it adds
Seminal vesicles	An alkaline fluid rich in fructose that nourishes the sperm; the largest share of semen
Prostate	A milky, slightly acidic fluid that helps the sperm stay active
Bulbourethral glands	The Cowper's glands; an alkaline mucus that neutralizes urine in the urethra and lubricates

AND THE MIXTURE ITSELF

Structure	What it is
Semen	The mixture of sperm and the fluids of the three accessory glands

The Penis

The penis delivers semen and carries the urethra. It is built around three columns of erectile tissue.

THE REGIONS OF THE PENIS AND ITS ERECTILE COLUMNS

Structure	What it is
Root	The attached portion of the penis, containing the bulb and the crura
Body	The free, movable portion of the penis
Glans penis	The enlarged, acorn-shaped tip, an extension of the corpus spongiosum
Prepuce	The foreskin; the loose fold of skin over the glans
Corpora cavernosa	The two dorsal columns of erectile tissue
Corpus spongiosum	The single ventral column of erectile tissue that surrounds the spongy urethra
Erectile tissue	The spongy tissue of the columns that fills with blood during an erection

Common Disorders of the Male Reproductive System

Compare the common disorders of the male reproductive system by the structure each one affects.

COMMON DISORDERS OF THE MALE REPRODUCTIVE SYSTEM

Disorder	What it is
Cryptorchidism	One or both testes fail to descend into the scrotum
Hydrocele	A collection of serous fluid in the tunica vaginalis around a testis
Varicocele	An abnormal swelling of the veins that drain the testis
Benign prostatic hyperplasia	A non-cancerous enlargement of the prostate that can squeeze the urethra
Prostate cancer	A malignant tumor of the prostate gland
Testicular cancer	A malignant tumor of the testis
Inguinal hernia	A loop of intestine pushes through a weak point in the abdominal wall at the inguinal canal

WHY THE TESTES HANG OUTSIDE

Sperm production needs a temperature a few degrees below the body core, which is why the testes sit in the **scrotum** and the **cremaster** and **dartos** muscles adjust their position. In **cryptorchidism**, an undescended testis stays at body temperature, and if it is not corrected the warmth can destroy the sperm-forming cells.

The duct system shares its final stretch with the urinary tract. Because the **urethra** passes straight through the **prostate**, an enlarged prostate squeezes the urethra and makes urination difficult, which is how a reproductive organ produces a urinary symptom.

CHAPTER 7**The Female Reproductive System**

7 min

THE REPRODUCTIVE SYSTEM, FEMALE

Covers the ovaries, the uterine tubes and the path of the ovum, the uterus, the vagina and external genitalia, and the mammary glands. The focus is on the structures and the job each one does.

BY THE END

1. Identify the structures of the ovary and the uterine tubes.
2. Name the regions, ligaments, and wall layers of the uterus.
3. Identify the external genitalia and the structures of the mammary gland.

**WATCH AFTER YOU READ THIS CHAPTER****Concept videos: The female reproductive system**

drsrennie-stack.github.io/new-build-bio4-solano/reproductive-female-concept-videos.html

The Female Reproductive System, an Overview

The female reproductive system produces oocytes, receives sperm, and supports a developing fetus. Its organs sort into the gonads, the genital tract, the external genitalia, and the mammary glands.

THE ORGANS OF THE SYSTEM

Structure	What it is
Ovaries	The female gonads; they produce oocytes and secrete hormones
Uterine tubes	The fallopian tubes; they carry the oocyte toward the uterus
Uterus	The muscular organ where a fertilized ovum implants and a fetus develops
Vagina	The canal that receives the penis and serves as the birth canal
Vulva	The external genitalia, collectively called the vulva or pudendum
Mammary glands	The milk-producing glands of the breasts

The Ovaries

The ovaries are paired, almond-shaped organs held beside the uterus by ligaments. Inside, the developing oocytes sit in structures called follicles.

THE OVARY, ITS ATTACHMENTS, AND ITS INTERIOR

Structure	What it is
Ovaries	The paired, almond-shaped female gonads, one on each side of the uterus
Broad ligament	The fold of peritoneum that drapes over the uterus and tubes; it attaches to the ovary by the mesovarium
Ovarian ligament	The cord that anchors the ovary to the uterus
Suspensory ligament	The fold that attaches the ovary to the pelvic wall
Germinal epithelium	The surface layer of low cuboidal epithelium covering the ovary
Tunica albuginea	The whitish capsule of dense connective tissue just beneath the germinal epithelium
Ovarian cortex	The region beneath the tunica albuginea that contains the ovarian follicles
Ovarian medulla	The deeper region of looser connective tissue holding blood vessels and nerves
Ovarian follicles	The structures in the cortex, each holding an oocyte at a stage of development
Mature follicle	The large, fluid-filled follicle ready to release its oocyte at ovulation
Corpus luteum	The yellow body; the structure left after ovulation, which later degenerates into a scar called the corpus albicans

HOLD ONTO THIS

The ovary is read from the outside in: germinal epithelium, tunica albuginea, cortex with the follicles, then medulla with the vessels. The follicles you are asked to find are always in the cortex.

The Uterine Tubes and the Path of the Ovum

The uterine tubes catch the released oocyte and carry it toward the uterus. Fertilization, when it happens, usually happens here.

THE REGIONS OF THE UTERINE TUBE

Structure	What it is
Infundibulum	The funnel-shaped end of the tube, open to the pelvic cavity near the ovary
Fimbriae	The fringe of finger-like projections at the infundibulum that sweep the oocyte into the tube
Ampulla	The widest and longest portion of the tube, where fertilization usually occurs
Isthmus	The narrow, thick-walled portion that joins the tube to the uterus
Lining of the tube	Ciliated and secretory cells whose cilia and fluid move the oocyte toward the uterus

The ovum follows one route from the ovary to the uterus.

1. **Ovary** the mature follicle releases the secondary oocyte at ovulation
2. **Fimbriae and infundibulum** the fimbriae sweep the oocyte into the funnel of the tube
3. **Ampulla** the oocyte travels through the widest part of the tube, where sperm may fertilize it
4. **Isthmus** the oocyte or early embryo passes through the narrow part of the tube
5. **Uterus** the oocyte or embryo reaches the uterine cavity

The Uterus

The uterus is a thick-walled, pear-shaped organ between the bladder and the rectum. Ligaments hold it in place and three layers build its wall.

THE REGIONS OF THE UTERUS AND THE LIGAMENTS THAT HOLD IT

Structure	What it is
Fundus	The dome-shaped superior portion of the uterus, above the entry of the tubes
Body	The tapered central portion of the uterus
Cervix	The narrow inferior portion that opens into the vagina
Uterine cavity	The interior space of the body of the uterus
Cervical canal	The interior of the cervix, opening to the uterine cavity at the internal os and to the vagina at the external os
Broad ligaments	The paired folds of peritoneum attaching the uterus to the pelvic walls
Uterosacral ligaments	The paired folds connecting the uterus to the sacrum
Cardinal ligaments	The paired ligaments extending from the pelvic wall to the cervix and vagina
Round ligaments	The paired fibrous cords running from the uterus to the labia majora

The three layers of the uterine wall

The wall of the uterus has three layers. Compare them.

THE LAYERS OF THE UTERINE WALL

Layer	What it is
Perimetrium	The outer serous layer, part of the visceral peritoneum
Myometrium	The thick middle layer of smooth muscle that contracts during childbirth
Endometrium	The inner vascular lining; its stratum functionalis sheds in menstruation and its stratum basalis rebuilds it

The Vagina and External Genitalia

The vagina connects the cervix to the outside. The external genitalia, the vulva, surround its opening.

THE VAGINA AND THE STRUCTURES OF THE VULVA

Structure	What it is
Vagina	The fibromuscular canal from the cervix to the exterior; the birth canal
Fornix	The recess of the vagina that surrounds the cervix
Hymen	The thin fold of mucous membrane that partly closes the vaginal opening
Mons pubis	The cushion of fatty tissue over the pubic symphysis
Labia majora	The two larger, hair-covered folds of skin; homologous to the scrotum
Labia minora	The two smaller, hairless folds within the labia majora
Clitoris	The small mass of erectile tissue at the anterior junction of the labia minora; homologous to the penis
Vestibule	The region between the labia minora, holding the vaginal and urethral openings
Greater vestibular glands	The Bartholin's glands beside the vaginal opening, which secrete lubricating mucus
Perineum	The diamond-shaped region holding the external genitalia and anus, split into a urogenital triangle and an anal triangle

The Mammary Glands

The mammary glands lie in the breasts, over the chest muscles. They are modified sweat glands that produce milk.

THE STRUCTURES OF THE MAMMARY GLAND

Structure	What it is
Mammary glands	The modified sweat glands of the breast that produce milk
Lobes	The 15 to 20 or more compartments of the mammary gland, each divided into lobules
Lobules	The subdivisions of a lobe, containing the milk-secreting alveoli
Alveoli	The small sacs of glandular tissue that secrete milk
Lactiferous ducts	The ducts that carry milk from the lobes toward the nipple
Nipple	The pigmented projection where the lactiferous ducts open
Areola	The circular pigmented area of skin around the nipple
Suspensory ligaments	The Cooper's ligaments that support the breast

HOLD ONTO THIS

The mammary gland nests in one direction: alveoli inside lobules, lobules inside lobes, lobes draining by lactiferous ducts to the nipple. Name them in that order and you have named the path milk takes.

Common Disorders of the Female Reproductive System

Compare the common disorders of the female reproductive system by the structure each one affects.

THE COMMON DISORDERS

Disorder	What it is
Endometriosis	Endometrial tissue grows outside the uterus, on the ovaries or pelvic lining
Ovarian cysts	Fluid-filled sacs that form on or in an ovary
Uterine fibroids	Non-cancerous tumors of the smooth muscle of the myometrium
Pelvic inflammatory disease	An infection that spreads through the uterus, tubes, and ovaries
Cervical cancer	A malignant tumor of the cervix
Ectopic pregnancy	An embryo implants outside the uterus, most often in a uterine tube
Breast cancer	A malignant tumor of the mammary gland tissue

OPEN PATHWAYS AND SHED LININGS

The female tract has an open route to the abdomen. The **infundibulum** opens into the pelvic cavity rather than sealing onto the ovary, which is how an oocyte is caught, but also how infection can climb from the vagina through the uterus and tubes to cause **pelvic inflammatory disease**.

The same open route explains **ectopic pregnancy**. If a fertilized ovum is delayed in the uterine tube, it can implant there instead of in the uterus, in a narrow tube never built to hold it.

The **endometrium** is built in two layers for a reason. The stratum functionalis is shed and rebuilt each cycle from the permanent stratum basalis, so the lining can renew itself month after month.

CHAPTER 8

Pregnancy and Birth

6 min

PREGNANCY AND BIRTH

Covers fertilization and implantation, the stages of prenatal development, the placenta and fetal membranes, the gravid uterus, and the stages of labor.

BY THE END

1. Describe fertilization and implantation.
2. Order the stages of prenatal development from zygote to fetus.
3. Name the placenta and the fetal membranes and what each does.
4. Describe the gravid uterus and the three stages of labor.



WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Pregnancy and birth
drsrennie-stack.github.io/new-build-bio4-solano/pregnancy-concept-videos.html

Pregnancy and Birth, an Overview

Pregnancy begins at fertilization and ends at birth. Over those months a single cell becomes a fetus, supported by a temporary set of structures built just for the pregnancy.

THE VOCABULARY OF A PREGNANCY

Structure	What it is
Pregnancy	The period from fertilization to birth, during which a fetus develops in the uterus
Gestation	The length of a pregnancy, about 38 weeks from fertilization, or 40 weeks counted from the last menstrual period
Embryo	The developing offspring during the first 8 weeks after fertilization
Fetus	The developing offspring from the 9th week until birth
Trimesters	The three roughly three-month periods that pregnancy is divided into

Fertilization and Implantation

Pregnancy begins when a sperm and an egg unite. The new cell divides as it travels, and about a week later it embeds itself in the wall of the uterus.

FROM ONE CELL TO AN IMPLANTED BLASTOCYST

Structure	What it is
Fertilization	The union of a sperm and an egg, normally in the ampulla of the uterine tube
Zygote	The single cell formed at fertilization
Cleavage	The series of rapid cell divisions the zygote undergoes as it travels toward the uterus
Morula	The solid ball of cells produced by cleavage
Blastocyst	The hollow ball of cells that reaches the uterus, with an inner cell mass and an outer layer
Inner cell mass	The cluster of cells of the blastocyst that becomes the embryo
Trophoblast	The outer layer of the blastocyst that becomes the fetal part of the placenta
Implantation	The embedding of the blastocyst into the endometrium, about a week after fertilization

HOLD ONTO THIS

Fertilization happens in the ampulla, implantation happens in the endometrium, and the week between them is spent travelling and dividing. Two places, one journey.

Stages of Prenatal Development

From one cell to a newborn, development passes through a fixed set of stages. Follow them in order.

1. **Zygote** the single cell formed when a sperm and an egg unite
2. **Cleavage** rapid cell divisions produce a solid ball of cells, the morula
3. **Blastocyst** a hollow ball of cells that reaches the uterus and implants in the endometrium
4. **Embryo** from implantation through week 8; the organs and body systems form
5. **Fetus** from week 9 until birth; the body grows and the systems mature
6. **Birth** the fetus is delivered, ending the gestation

The Placenta and Fetal Membranes

A pregnancy builds temporary structures that support the fetus and are discarded at birth. Compare the placenta and the fetal membranes.

THE TEMPORARY STRUCTURES OF PREGNANCY

Structure	What it is	Role
Placenta	The disc-shaped organ that forms during pregnancy	Exchanges oxygen, nutrients, and wastes between the mother's blood and the fetus's blood
Umbilical cord	The cord linking the fetus to the placenta	Carries the two umbilical arteries and the umbilical vein
Amnion	The innermost fetal membrane	Forms the fluid-filled sac that surrounds the fetus
Amniotic fluid	The fluid within the amnion	Cushions the fetus, lets it move, and allows even growth
Chorion	The outermost fetal membrane	Contributes the fetal portion of the placenta
Yolk sac	An early fetal membrane	Forms the earliest blood cells and the germ cells

The Gravid Uterus and Fetal Position

The uterus changes more than any other organ during pregnancy, and the fetus settles into position for birth.

THE PREGNANT UTERUS AND HOW THE FETUS LIES

Structure	What it is
Gravid uterus	The uterus during pregnancy, which enlarges enormously to hold the growing fetus
The expanding uterus	By full term it reaches up to the rib cage, pushing the abdominal organs aside
Cervical changes	The cervix stays firm and closed through pregnancy, then softens and thins as birth approaches
Vertex position	The head-down position most fetuses settle into by late pregnancy, ready for birth
Breech position	A position in which the buttocks or feet, rather than the head, are nearest the cervix
Lightening	The descent of the fetus lower into the pelvis as birth nears

Labor and Birth

Labor is the process of birth. It is divided into three stages, which follow in a fixed order.

1. **Dilation stage** regular contractions thin and open the cervix until it is fully dilated; the longest stage
2. **Expulsion stage** strong contractions push the fetus through the cervix and vagina; it ends with the birth of the baby
3. **Placental stage** contractions continue and expel the placenta, now called the afterbirth

THE VOCABULARY OF BIRTH

Structure	What it is
Labor	The process by which the fetus and placenta are expelled from the uterus
Afterbirth	The placenta and attached membranes, expelled in the placental stage
Cesarean section	Surgical delivery of the fetus through an incision in the abdominal and uterine walls

Disorders of Pregnancy

Compare the common conditions of pregnancy by the structure or stage each one affects.

THE COMMON CONDITIONS OF PREGNANCY

Condition	Affected structure or stage	What it is
Ectopic pregnancy	The implantation site	The embryo implants outside the uterus, most often in a uterine tube; a structural emergency
Miscarriage	The pregnancy	The loss of a pregnancy before the fetus can survive on its own
Placenta previa	The placenta	The placenta lies over the cervical opening and can block the birth canal
Preeclampsia	The mother's blood pressure	A serious condition of later pregnancy marked by high blood pressure
Gestational diabetes	Blood sugar control	High blood sugar that first develops during pregnancy
Breech presentation	Fetal position	The fetus is positioned buttocks or feet first, which can complicate a vaginal delivery

TEMPORARY STRUCTURES, LASTING CLUES

The **placenta** is the exchange organ of pregnancy, and where it sits matters. In **placenta previa** it lies over the cervix, blocking the path the baby would take, which is why position is checked on ultrasound and may call for a cesarean section.

The **amniotic sac** normally stays sealed until labor. Its rupture, felt as the water breaking, releases the amniotic fluid and signals that birth is near. A sample of that fluid can also be drawn, in amniocentesis, to study the fetus.

The fetal membranes and placenta are built for one pregnancy and discarded at birth as the **afterbirth**. Examining the delivered placenta afterward gives clues to how the pregnancy went, a temporary organ read like a record.

CHAPTER 9

Endocrine Anatomy

7 min

ENDOCRINE ANATOMY

Covers the endocrine glands, where each one sits in the body, and the gross anatomy of the pituitary, thyroid, parathyroid, adrenal, and pancreatic glands. Hormone action belongs to physiology and is not covered here.

BY THE END

1. Name the major endocrine glands and locate each one in the body.
2. Describe the gross anatomy of the pituitary gland and its connection to the hypothalamus.
3. Identify the parts of the thyroid, parathyroid, adrenal, and pancreatic glands.
4. Name organs that are not endocrine glands but still contain endocrine tissue.



WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Endocrine anatomy

drsrennie-stack.github.io/new-build-bio4-solano/endocrine-concept-videos.html

The Endocrine System, an Overview

The endocrine system is a scattered set of glands and tissues that signal the body by releasing hormones into the blood. Covers where those glands sit and how each one is built.

THE BASIC VOCABULARY

Structure	What it is
Endocrine system	The glands and tissues that release hormones into the blood to signal distant cells
Hormone	A chemical messenger released by an endocrine cell that travels in the blood to act on target cells
Endocrine gland	A ductless gland that secretes its product directly into the surrounding fluid and blood
Exocrine gland	By contrast, a gland that secretes through a duct onto a surface; not part of the endocrine system
Target cell	A cell that carries the receptor for a given hormone and therefore responds to it

The Major Endocrine Glands

The pure endocrine glands do nothing but make hormones. They are spread from the brain to the pelvis. Compare them by where each one sits.

WHERE EACH GLAND SITS

Gland	Location	Note
Hypothalamus	The floor and walls of the third ventricle, in the diencephalon of the brain	Links the nervous and endocrine systems and directs the pituitary
Pituitary gland	In the sella turcica of the sphenoid bone, hanging below the hypothalamus	The small gland the hypothalamus controls
Pineal gland	The posterior roof of the third ventricle, in the brain	A small, cone-shaped gland
Thyroid gland	The anterior neck, just below the larynx, wrapped around the trachea	The largest purely endocrine gland
Parathyroid glands	The posterior surface of the thyroid gland, usually four small glands	Tiny glands embedded behind the thyroid

Gland	Location	Note
Thymus	The mediastinum, behind the sternum	Large in childhood, shrinks after puberty
Adrenal glands	One capping the superior pole of each kidney	Also called the suprarenal glands
Pancreatic islets	Scattered through the pancreas, behind the stomach	The endocrine portion of the pancreas
Ovaries	The female pelvis, one on each side of the uterus	The female gonads
Testes	The scrotum	The male gonads

The Pituitary Gland

The pituitary gland, or hypophysis, is about the size of a pea. It sits in the sella turcica and hangs from the hypothalamus by a stalk. Its two lobes come from two different tissues.

THE PITUITARY AND ITS CONNECTIONS

Structure	What it is
Hypophysis	Another name for the pituitary gland
Sella turcica	The saddle-shaped pocket of the sphenoid bone that cradles and protects the pituitary
Infundibulum	The slender stalk that connects the pituitary to the hypothalamus above it
Hypophyseal portal system	The set of blood vessels that carries hypothalamic signals directly to the anterior lobe

The two lobes

Compare the two lobes by the tissue they are made of and how each connects to the hypothalamus.

THE LOBES OF THE PITUITARY GLAND

Lobe	Other name	Tissue	Link to the hypothalamus
Anterior lobe	Adenohypophysis	Glandular epithelial tissue	The hypophyseal portal system, a set of blood vessels
Posterior lobe	Neurohypophysis	Nervous tissue, an extension of the hypothalamus itself	A tract of nerve fibers running down through the infundibulum

HOLD ONTO THIS

The posterior lobe is nervous tissue, so it connects by nerve fibers. The anterior lobe is glandular epithelium, so it connects by blood vessels. The tissue tells you the link.

The Thyroid and Parathyroid Glands

The thyroid gland sits across the front of the trachea, and the four small parathyroid glands are embedded in its back surface.

The thyroid gland

THE THYROID, FROM THE GROSS GLAND TO THE FOLLICLE

Structure	What it is
Thyroid gland	The largest purely endocrine gland, two lateral lobes lying against the trachea below the larynx
Isthmus	The bridge of thyroid tissue, crossing the front of the trachea, that connects the two lobes
Thyroid follicles	The microscopic hollow spheres that make up the gland
Colloid	The protein-rich fluid stored in the center of each follicle
Follicular cells	The cells forming the wall of a follicle
Parafollicular cells	Also called C cells, scattered between the follicles

The parathyroid glands

THE PARATHYROID GLANDS AND THEIR CELLS

Structure	What it is
Parathyroid glands	Usually four small, round glands on the posterior surface of the thyroid lobes
Chief cells	The main hormone-producing cell of the parathyroid glands

The Adrenal Glands

An adrenal, or suprarenal, gland caps the top of each kidney. Each gland is really two glands in one: an outer cortex and an inner medulla, formed from different tissues.

THE TWO PARTS OF AN ADRENAL GLAND

Structure	What it is
Adrenal cortex	The outer region and the bulk of the gland, glandular tissue arranged in three layered zones
Adrenal medulla	The inner core, modified nervous tissue that belongs to the sympathetic nervous system

The cortex is built in three zones. Follow them in order, from the surface inward.

1. **Zona glomerulosa** the thin outer zone, just beneath the connective tissue capsule
2. **Zona fasciculata** the thick middle zone, the widest of the three, its cells in straight columns
3. **Zona reticularis** the inner zone, a branching network of cells against the medulla

The Pancreas and the Gonads

The pancreas and the gonads are mixed organs. Each does an endocrine job alongside another role, so only part of the organ is endocrine tissue.

THE MIXED ORGANS

Structure	What it is
Pancreas	A soft gland behind the stomach, both an exocrine and an endocrine organ
Pancreatic islets	Also called the islets of Langerhans, the small clusters of endocrine cells scattered through the pancreas
Acini	By contrast, the exocrine clusters of the pancreas that make digestive enzymes; not endocrine tissue
Ovaries	The paired female gonads in the pelvis, endocrine glands as well as reproductive organs
Testes	The paired male gonads in the scrotum, endocrine glands as well as reproductive organs

Organs with Endocrine Tissue

Beyond the pure endocrine glands, many organs whose main job is something else still carry pockets of endocrine cells. Compare them.

ORGANS THAT CONTAIN ENDOCRINE TISSUE

Organ	Endocrine tissue it contains
Hypothalamus	Neurosecretory cells; a neuroendocrine organ that controls the pituitary gland
Heart	Hormone-producing cells in the wall of the atria
Kidneys	Cells that release a hormone driving red blood cell production, plus an enzyme acting on blood pressure
Stomach and small intestine	Scattered endocrine cells in the lining of the digestive tract
Liver	Cells that contribute to a growth-related hormone and to blood pressure control
Skin	Cells that begin the production of vitamin D, which acts as a hormone
Adipose tissue	Fat cells that release a hormone signaling the body's energy stores
Placenta	A temporary organ of pregnancy that releases hormones supporting the pregnancy

WHERE A GLAND SITS SHAPES ITS DISEASE

The **pituitary gland** is boxed inside the bony **sella turcica**, so a pituitary tumor has nowhere to expand but upward, where it presses on the optic chiasm. A loss of peripheral vision can be the first sign that a gland the size of a pea has enlarged.

The **thyroid gland** lies just under the skin of the anterior neck, so an enlarged thyroid, a goiter, is visible and can be felt. The same superficial position lets a clinician palpate the gland and makes thyroid surgery accessible, though the tiny **parathyroid glands** on its back surface must be carefully spared.

Each **adrenal gland** rests on the superior pole of a kidney, which is why imaging ordered for the kidney also captures the adrenal, and why an adrenal mass is sometimes found by accident.

Pre-work.

1 min

PART B

5 packets, in the order they are due. Work them in sequence: read the Part A chapter, organize it into mini visuals, draw it from memory, apply it with your notes closed, then retrieve later in the week by redrawing from memory. Context before content. This is not a night-before task.

1 The Respiratory and Lymphatic Systems

2 The Digestive System

3 Urinary and Endocrine

4 Reproductive Systems and Pregnancy

5 Module 4 Integration

THESE PAGES PRINT LANDSCAPE. TURN THE PACKET SIDEWAYS.

Pre-work • The Respiratory and Lymphatic Systems

The Respiratory System and The Lymphatic System. Read both Part A chapters first. The respiratory chapter carries over from the Module 3 packet, so read it as a refresher rather than a first pass, and give the new lymphatic material the longer half of your time. Both systems are pathway systems, so the order of the parts is the content.

Module 4 • 1 of 5

Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1 Start with the reading.

Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.

2 Organize what you read.

In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.

3 Draw it.

In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.

4 Apply it.

Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.

5 Come back to it later in the week

, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

3 min

PANEL 1 OF 3

Turn the respiratory and lymphatic chapters into mini visuals. Eight chunks. Two chains carry the two pathways, and the pathways are where most of the marks are.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 GRID

Two ways of grouping the parts

Structure down the side, nose to alveolus in order. Upper or lower in one column, conducting or respiratory in the other. Circle every row where the two columns disagree, because those rows are the questions.

2 LADDER

The three regions of the pharynx

Three rungs, nasopharynx at the top. Beside each write what it carries, its epithelium, and its tonsil. Mark the rung where the epithelium changes and the rung where the air and food paths finally divide.

3 HUB

The larynx

The larynx at the hub. Spoke to the three single cartilages and the three paired ones, then to the vocal folds, the vestibular folds, and the glottis. Mark the one cartilage that is elastic and the one that is a complete ring.

4 CHAIN

The airway, trachea to alveolus

Trachea through terminal bronchiole, then on to the respiratory bronchiole, alveolar duct, and alveolus. Under the chain draw two bars, cartilage falling and smooth muscle rising, and mark the box where the conducting zone ends.

5 SPLIT

Right lung against left lung

Lobes, fissures, size, and the feature unique to each side. Then hang the parietal pleura, the visceral pleura, and the pleural cavity off both branches, since those three are the same on both sides.

6 CHAIN

The pathway of lymph, tissue space to bloodstream

Six boxes with arrows, lymphatic capillary through to the subclavian vein. Mark the box where the lymph is filtered, and write beside the last box that this is the single point where a one-way system finally meets the blood.

7 HUB

A lymph node

One bean-shaped node at the hub. Spoke to the capsule, cortex, medulla, and hilum. Draw several afferent vessels entering through the capsule and one efferent vessel leaving at the hilum, then write the three regions where nodes cluster.

8 GRID

Primary against secondary lymphatic organs

Organ down the side, six rows. Group, location, and role across the top. Only two rows can say primary, and the reason they do is the answer to the whole grid.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

3 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

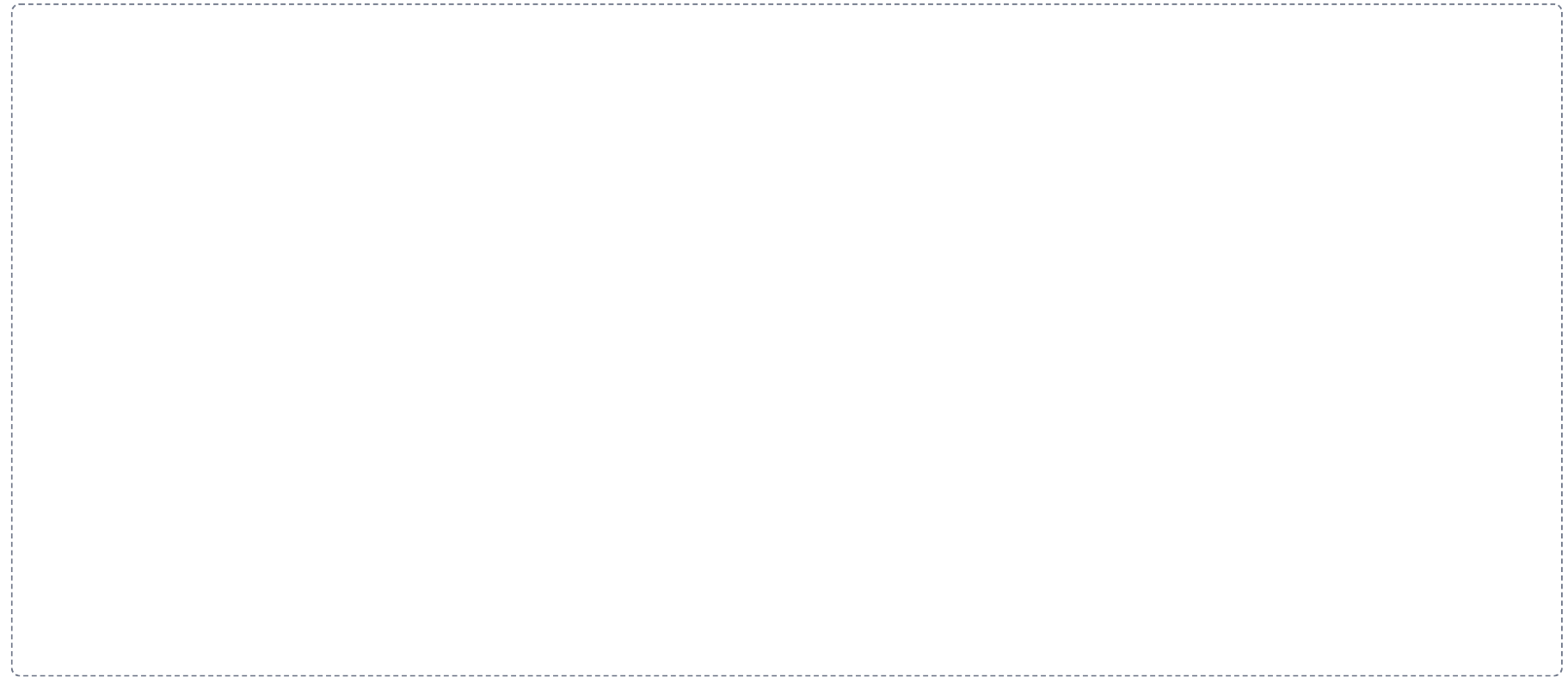
DRAWING 1

The lungs, the pleurae, and the bronchial tree

BRAIN DUMP FIRST

Two minutes. Both lungs with every lobe and fissure, the tree inside them, and the membranes around them.

Draw the trachea with its C-shaped cartilage rings and the carina below. Branch into both main bronchi and make the right one visibly wider and more vertical, then continue into lobar and segmental bronchi. Outline both lungs around the tree: three lobes and two fissures on the right, two lobes, one fissure, a cardiac notch, and the lingula on the left. Mark the apex, base, and hilum on each, then draw the parietal pleura against the chest wall, the visceral pleura on the lung surface, and the pleural cavity between them.



In your notebook, head the page **M4-1 The lungs, the pleurae, and the bronchial tree** so you can find it later.
Correct it in a second color.

YOU KNOW IT WHEN

You can say which lung an inhaled bead lands in, and point at the two features on your own drawing that decided it.

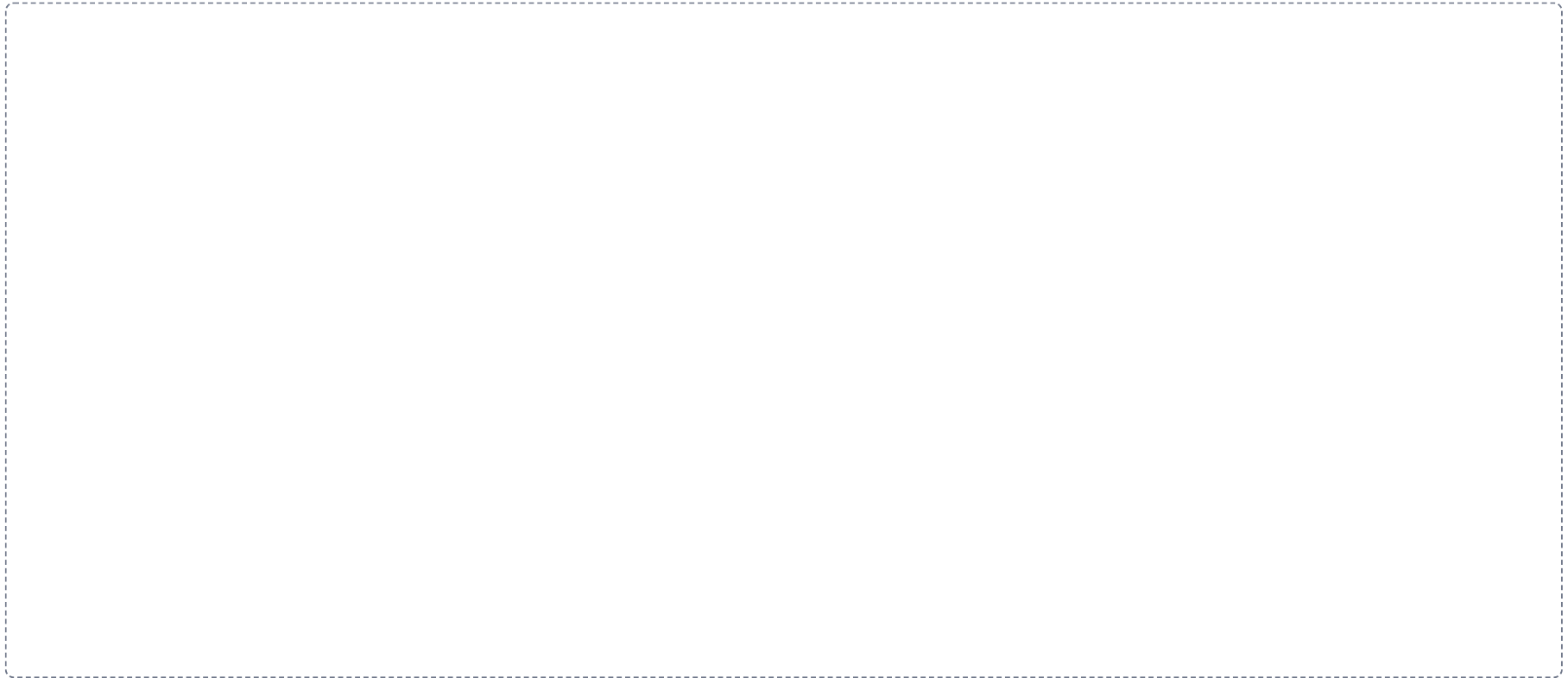
DRAWING 2

An alveolus and the respiratory membrane

BRAIN DUMP FIRST

One minute. Every cell of the alveolar wall and the three layers a gas crosses.

Draw one alveolar sac with several alveoli opening into it, at large scale. Line the wall with flat type I cells, add a few cuboidal type II cells, and put an alveolar macrophage on the surface. Wrap a pulmonary capillary against the outside, then enlarge the point where the two touch and label the alveolar epithelium, the fused basement membranes, and the capillary endothelium.



In your notebook, head the page **M4-2 An alveolus and the respiratory membrane** so you can find it later.
Correct it in a second color.

YOU KNOW IT WHEN

You can point at what emphysema destroys on your own drawing and say why the loss is permanent.

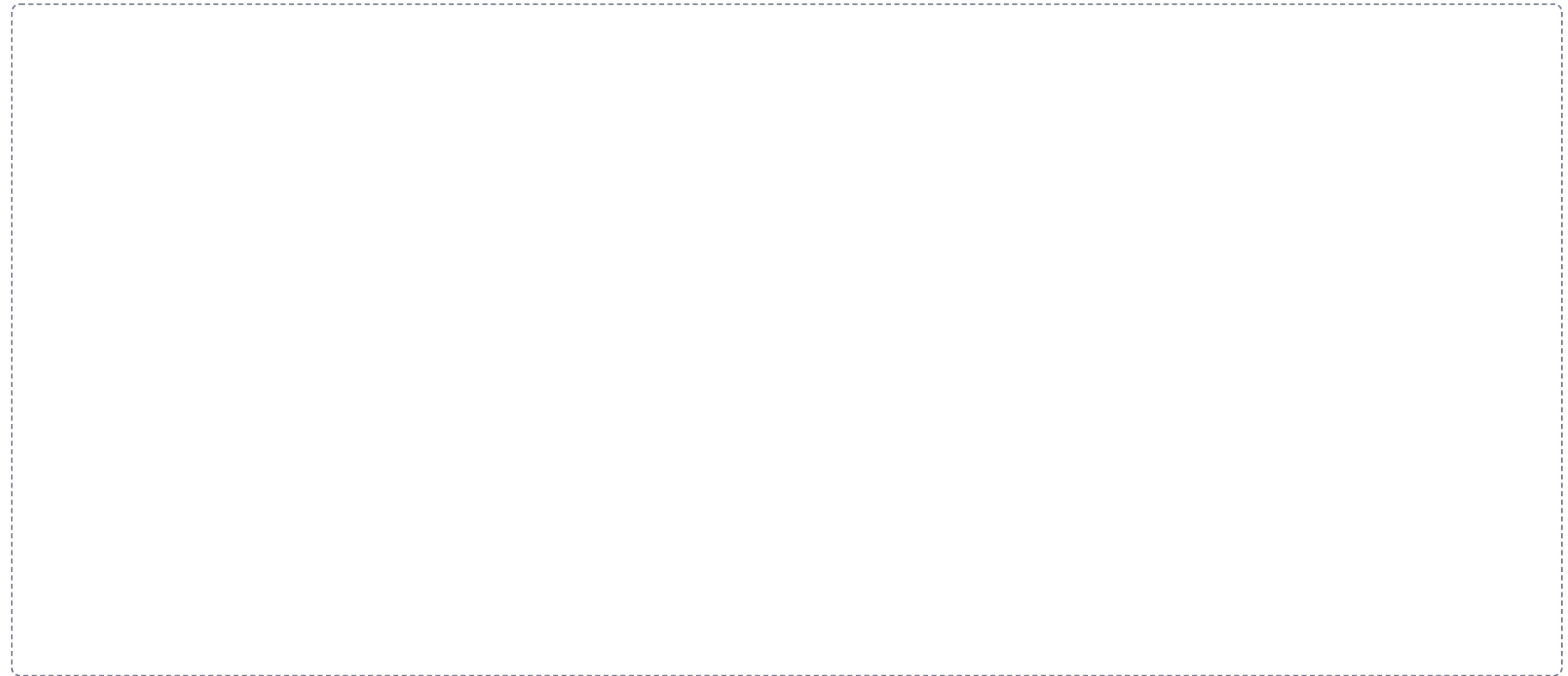
DRAWING 3

The lymphatic system, and a node in section

BRAIN DUMP FIRST

Two minutes. Where the lymph goes, which duct drains what, and what happens inside a node.

Draw a body outline. Shade the upper right quadrant one way and the rest of the body another, then draw the right lymphatic duct draining the shaded quadrant and the thoracic duct draining everything else, both emptying where the subclavian and internal jugular veins meet. Add the cisterna chyli at the start of the thoracic duct and mark the cervical, axillary, and inguinal clusters. Beside the outline draw one node cut open, with its capsule, cortex, medulla, several afferent vessels, and one efferent vessel at the hilum.



In your notebook, head the page **M4-3 The lymphatic system, and a node in section** so you can find it later.
Correct it in a second color.

YOU KNOW IT WHEN

You can name which duct returns lymph from any region someone points to, and say why the two ducts are so unequal in size.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

After surgery for breast cancer a patient has had the axillary lymph nodes on one side removed. Months later that arm is persistently swollen. Name the condition and explain it from the structure of the system.

Answer. Lymphedema.

Reasoning. Start with what the lymphatic system does, not with the surgery. Lymphatic capillaries take up the fluid that leaks out of the blood capillaries and return it through collecting vessels, nodes, trunks, and ducts to the subclavian veins, so the whole thing is a one-way drain with no alternate route. Remove the axillary nodes and you remove the vessels that ran through them, and the lymph of that entire limb loses the road it was using. Fluid keeps arriving in the tissue spaces at the old rate but leaves more slowly, so it accumulates, and the swelling is the surplus. Ordinary edema would settle once its cause passed, which is what rules it out here, because the missing structure is not rebuilt. Notice that the arm swells and the trunk does not, and that tells you exactly where along the route the drainage was cut.

Set A · Name it

1. The scroll-shaped projections that swirl air in the nasal cavity are the _____.
2. The only complete cartilage ring in the airway is the _____.
3. The specialized lymphatic capillaries that absorb dietary fat in the small intestine are the _____.
4. The sac at the beginning of the thoracic duct is the _____.
5. Name the two primary lymphatic organs and say what matures in each.

Set B · Predict it

1. The chest wall is punctured and air enters the pleural cavity. Predict what happens to the lung and name the condition.

2. A patient loses the use of the skeletal muscles of one leg. Predict the effect on lymph return from that limb and say why.

3. A premature infant lacks surfactant. Predict what happens to the alveoli between breaths and name the cell that should have made it.

4. A blow to the lower left ribs is followed by heavy internal bleeding. Name the organ and explain why it bleeds so much.

Set C · Tell it apart

1. The larynx is in the lower respiratory tract but in the conducting zone. Explain how both are true.

2. Distinguish edema from lymphedema by the structure at fault in each.

3. Compare a lymphatic capillary with a blood capillary by how each one begins and how permeable each one is.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

Two ways of grouping the parts

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

2

The three regions of the pharynx

LADDER

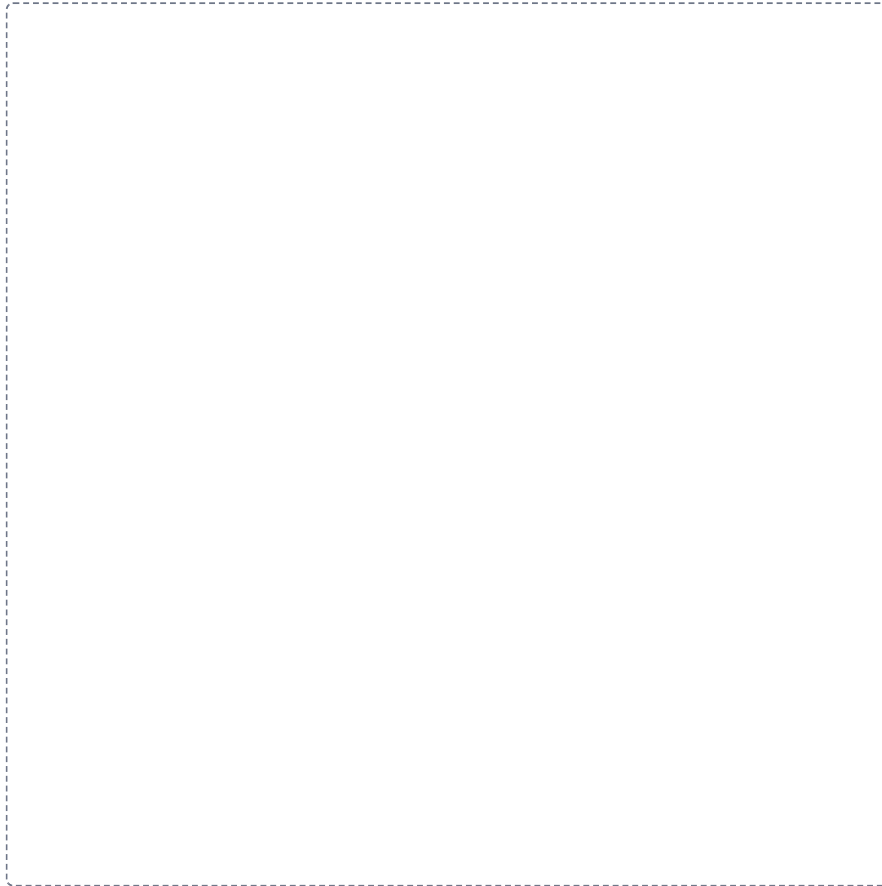
Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

3

The larynx

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.



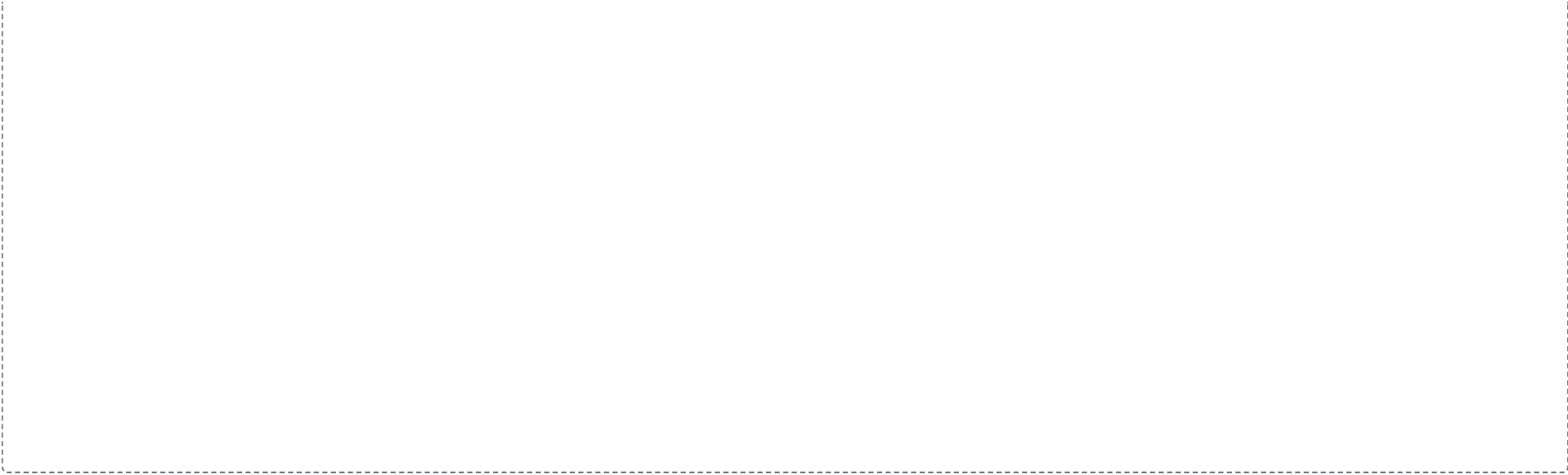
4

The airway, trachea to alveolus

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

A large dashed rectangular box intended for the student to draw a flowchart showing the steps from the trachea to the alveolus. The box is empty and occupies most of the page below the instructions.

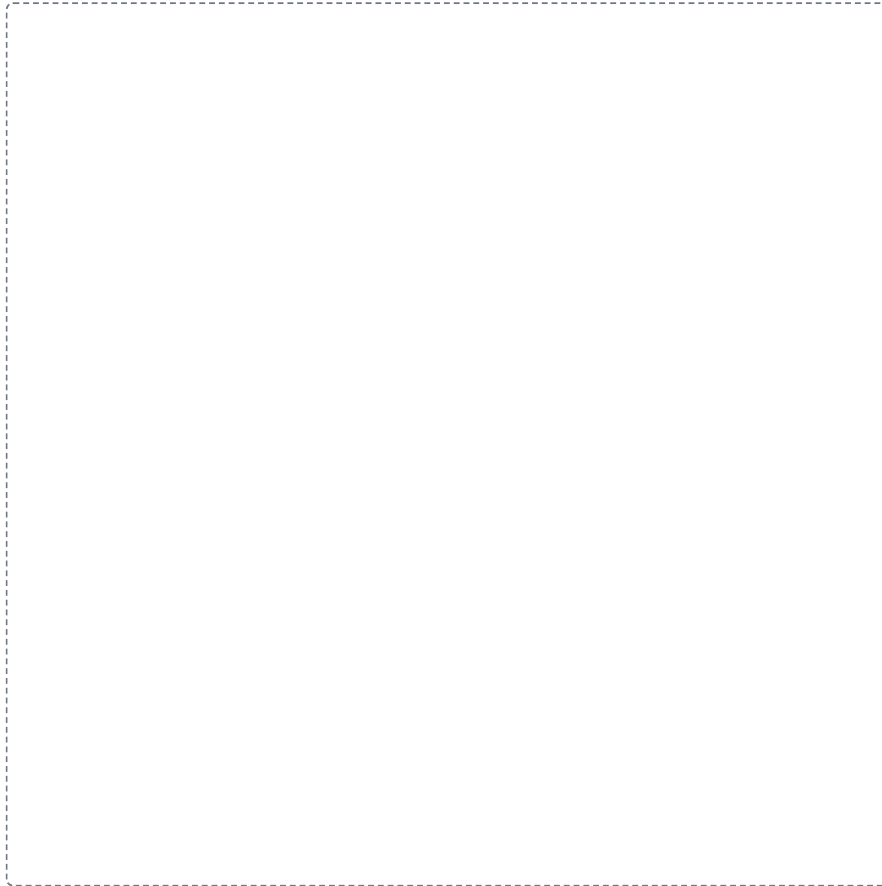


5

Right lung against left lung

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.



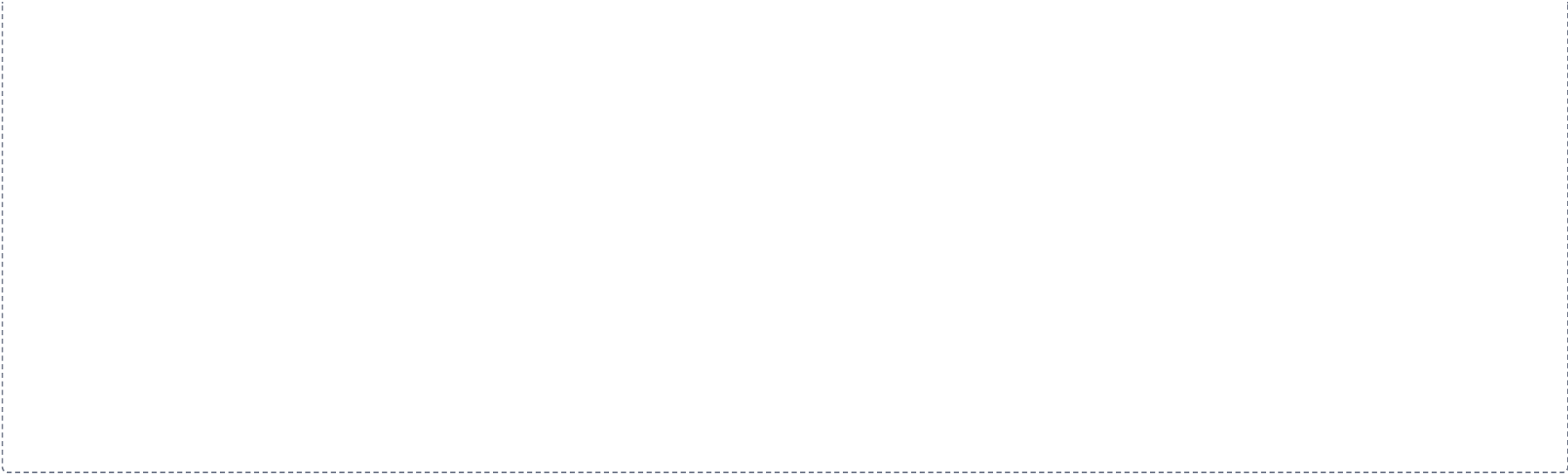
6

The pathway of lymph, tissue space to bloodstream

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.



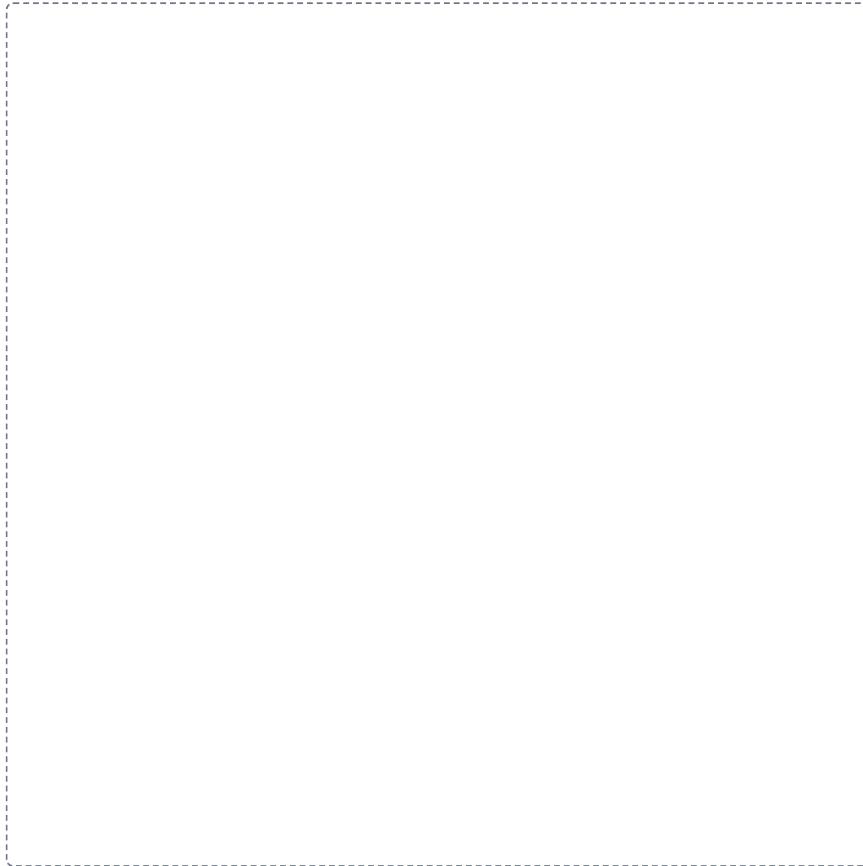


7

A lymph node

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

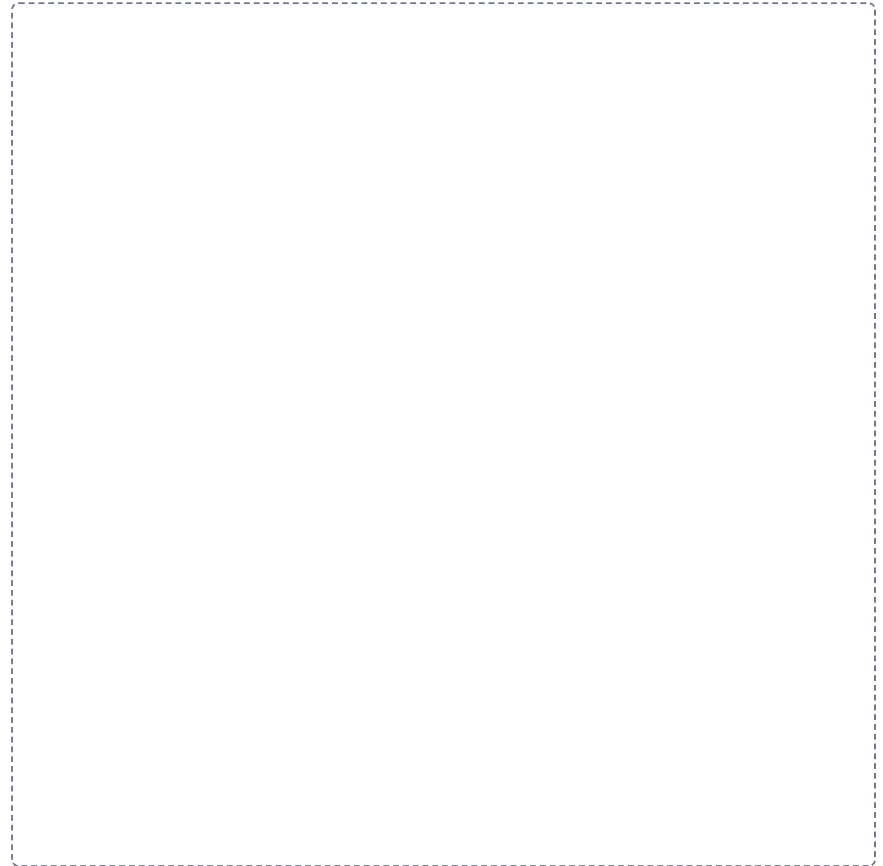


8

Primary against secondary lymphatic organs

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.



Pre-work • The Digestive System

The Alimentary Canal and The Accessory Digestive Organs. Read both Part A chapters first. The canal is one continuous tube with one wall plan and a short list of local exceptions, so learn the plan before you learn the regions, or you will memorize the same four layers six times.

Module 4 • 2 of 5

Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1 Start with the reading.

Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.

2 Organize what you read.

In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.

3 Draw it.

In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.

4 Apply it.

Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.

5 Come back to it later in the week

, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

3 min

PANEL 1 OF 3

Turn the two digestive chapters into mini visuals. Eight chunks. Two chains carry the two routes, the food and the bile, and those two are the ones you will be asked to trace.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 SPLIT

The canal against the accessory organs

One split. On one branch the organs food actually passes through, on the other the six organs that stay outside the tube and deliver into it. Write beside each accessory organ how it reaches the lumen, by direct contact or by a duct.

2 LADDER

The four layers of the wall

Four rungs, mucosa at the top down to serosa. Beside the mucosa rung write its own three sublayers, and beside the muscularis rung write its two directions of smooth muscle. Then mark the two exceptions: where the serosa becomes an adventitia, and where the muscularis gains a third layer.

3 CHAIN

The pathway of food, mouth to anus

Seven boxes with arrows, mouth to anus. Under each box write in three words what happens to the food there, and mark every sphincter it has to pass on the way out.

4 HUB

The stomach

One J-shaped stomach at the hub. Spoke to the cardia, fundus, body, and pyloric part, the two curvatures, the sphincter at each end, the rugae, and the gastric pits with their glands. Draw the three muscle layers as three arrows pointing three different ways.

5 LADDER

Three levels of surface amplification

Three rungs, the largest fold at the top and the smallest at the bottom: circular folds, villi, microvilli. Beside each write which layers of the wall it is built from, and hang the lacteal and the capillary bed off the villus rung.

6 HUB

The large intestine

One large intestine at the hub, drawn as a frame around the small intestine. Spoke to the cecum and appendix, the four parts of the colon in order, the rectum, and the anal canal with its two sphincters. Then mark the three features that name the colon on sight: teniae coli, haustra, and omental appendices.

7 HUB

The liver, from lobe to lobule

The liver at the hub. Spoke out to the four lobes, the falciform ligament, and the ligamentum teres in its free edge, then spoke inward to one lobule: hepatocytes in plates, sinusoids running between them, bile canaliculi, and a portal triad at the corner. Write beside the triad which two members bring blood in and which one carries bile out.

8 CHAIN

The path of bile, canaliculus to duodenum

Six boxes with arrows, bile canaliculi through to the duodenum. Draw the gallbladder as a side branch off the cystic duct rather than as a box in the line, because bile only detours there. Mark the ampulla and the sphincter that guards it.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

3 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

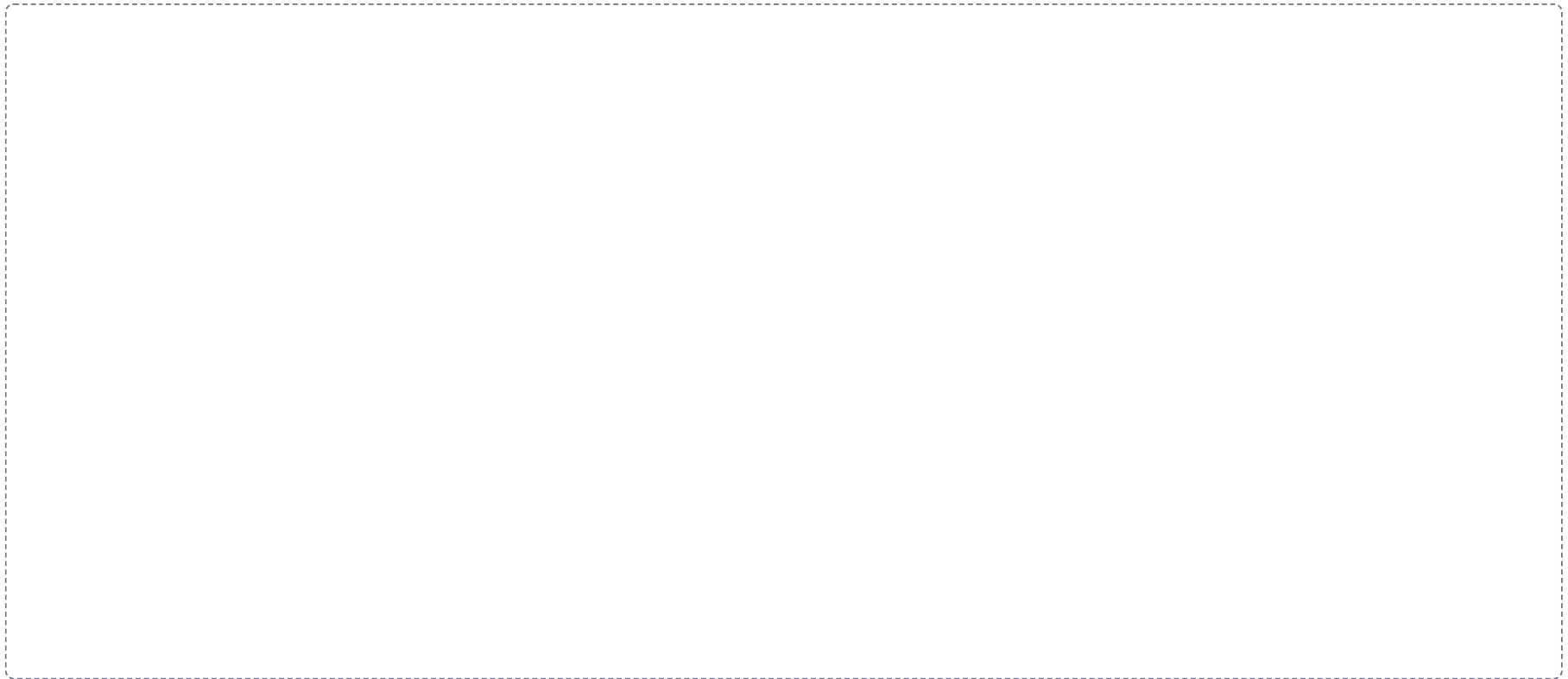
DRAWING 1

The abdominal organs and the folds of peritoneum

BRAIN DUMP FIRST

Two minutes. The tube in place, the membranes that hold it, and which organs sit behind them.

Draw the abdomen in sagittal section with the diaphragm across the top. Lay in the stomach, the small intestine, and the colon, then draw the parietal peritoneum along the wall and the visceral peritoneum over the organs, with the peritoneal cavity between them. Add the greater omentum hanging as an apron from the greater curvature, the lesser omentum between the stomach and the liver, the falciform ligament running to the anterior wall, the mesentery holding the jejunum and ileum, and the mesocolon. Then shade the retroperitoneal organs behind the parietal layer and label each one.



In your notebook, head the page **M4-7 The abdominal organs and the folds of peritoneum** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can say for any abdominal organ whether it lies inside the peritoneal cavity or behind it, and name the fold that anchors it.

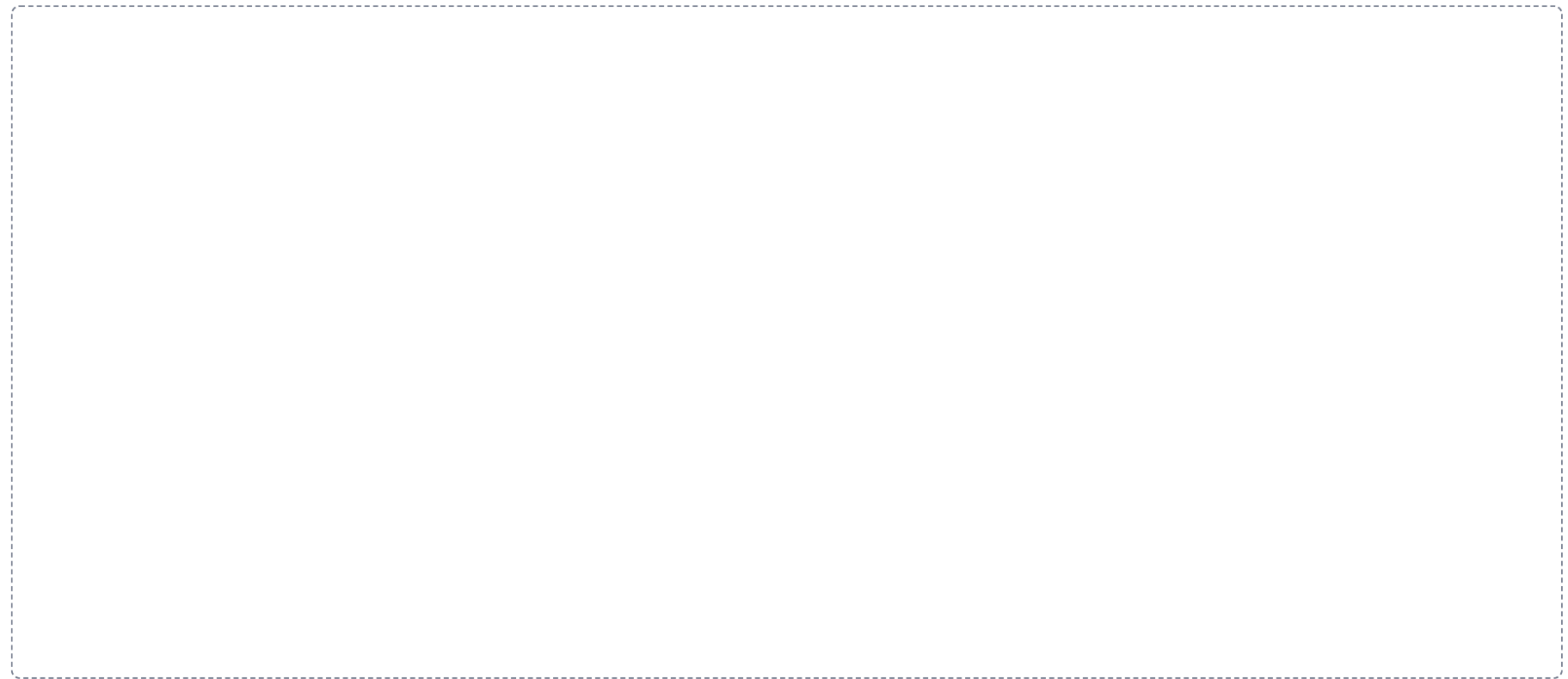
DRAWING 2

The stomach, cut open

BRAIN DUMP FIRST

One minute. Every named region and border, and the wall in cross section.

Draw the stomach J-shaped with the anterior wall cut away. Mark the cardia at the esophagus, the fundus bulging above and to the left, the body, and the pyloric part with its antrum, canal, and pylorus. Label the lesser curvature along the concave border and the greater curvature along the convex one, and put the lower esophageal sphincter and the pyloric sphincter at the two ends. Draw the rugae running through the empty stomach, then take a small square of the wall and enlarge it to show the four layers with three muscle layers instead of two, and a gastric pit leading down into a gastric gland.



In your notebook, head the page **M4-8 The stomach, cut open** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain from your own drawing why the stomach carries a third muscle layer that the rest of the tube does not.

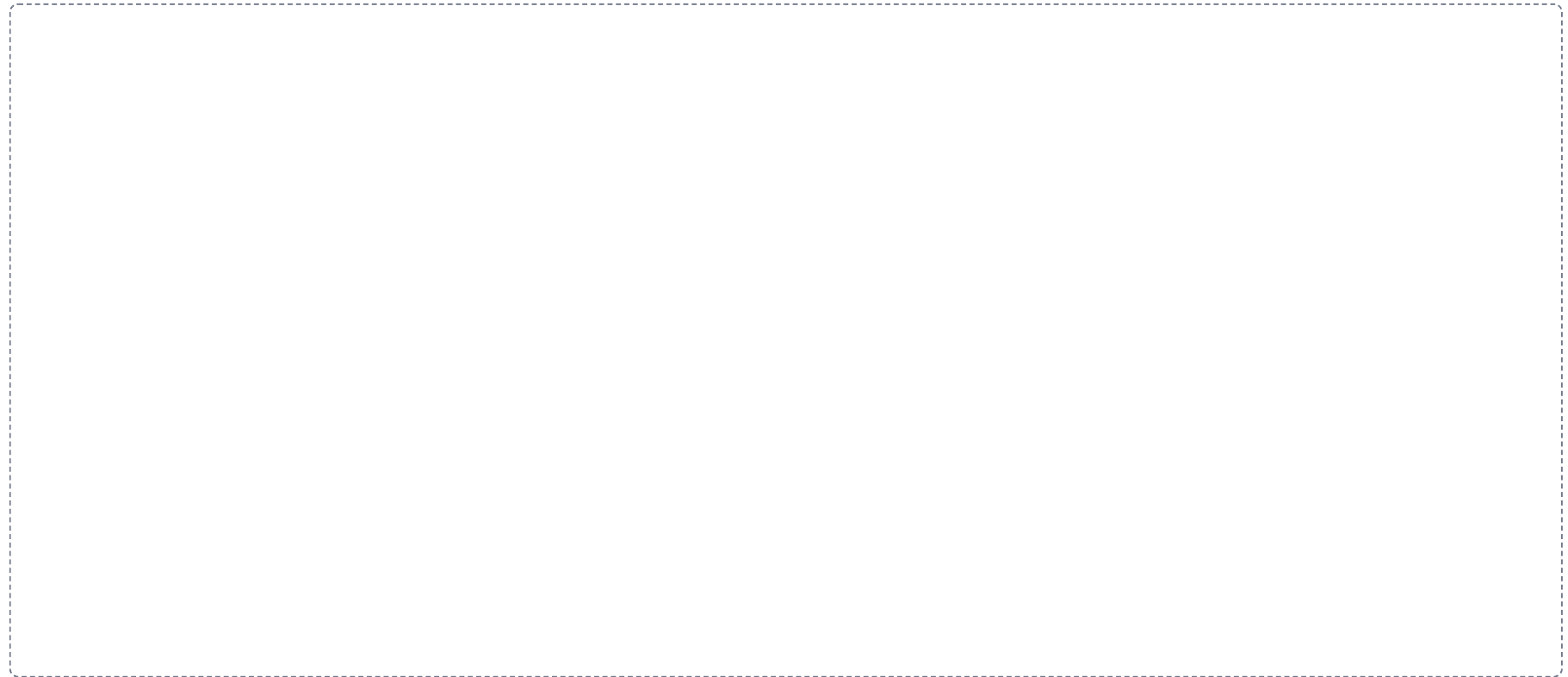
DRAWING 3

The liver, gallbladder, pancreas, and the duct tree

BRAIN DUMP FIRST

Two minutes. Three organs and every duct that joins them to the duodenum.

Draw the liver with its right and left lobes and the falciform ligament between them, the gallbladder tucked under the right lobe, the duodenum curving below, and the pancreas behind, with its head in that curve and then its body and tail. Now draw the ducts: the right and left hepatic ducts joining as the common hepatic duct, the cystic duct running out to the gallbladder, those two forming the common bile duct, the pancreatic duct joining it at the hepatopancreatic ampulla, and the accessory duct entering the duodenum on its own. Mark the major duodenal papilla and the sphincter of Oddi.



In your notebook, head the page **M4-9 The liver, gallbladder, pancreas, and the duct tree** so you can find it later.
Correct it in a second color.

YOU KNOW IT WHEN

You can put one gallstone anywhere on your duct tree and say exactly what backs up behind it.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A patient has yellow skin and eyes, pale stools, and an inflamed pancreas. Imaging finds a single stone. Name the one place it is lodged and explain why one stone produces all three findings.

Answer. In the hepatopancreatic ampulla.

Reasoning. Work backward from what is blocked rather than forward from the stone. Yellow skin and pale stools say that bile is not reaching the duodenum, so the obstruction sits somewhere on the bile path. An inflamed pancreas says that pancreatic juice is not leaving either, so the obstruction is also on the pancreatic path. A stone in the cystic duct would trap bile inside the gallbladder and leave the pancreas alone, and a stone in the common hepatic duct would sit above the point where the pancreatic duct joins, so neither one fits both findings. Only a single place on the map belongs to both routes: the ampulla where the common bile duct and the pancreatic duct meet, just before the major duodenal papilla. One stone, one shared passage, two organs backed up behind it.

Set A · Name it

1. The nerve plexus lying between the two muscle layers, which controls the movement of the gut, is the _____.
2. The fatty apron of peritoneum hanging from the greater curvature of the stomach is the _____.
3. The lymphatic capillary inside each intestinal villus is the _____.
4. The three thickened bands of longitudinal muscle running the length of the colon are the _____.
5. Name the three members of a portal triad and say which way each one carries its fluid.

Set B · Predict it

1. The lower esophageal sphincter weakens. Predict the symptom and name the condition.

2. The gallbladder is removed. Predict what happens to bile, working from the path it takes without that detour.

3. A peptic ulcer erodes through the last layer of the stomach wall. Predict what follows and name every layer it crossed.

4. Disease flattens the circular folds and villi of the small intestine. Predict the effect and explain it from surface area.

Set C · Tell it apart

1. The esophagus is the one region of the canal without a serosa. Say what it has instead and why the difference matters if it is perforated.

2. Distinguish the exocrine tissue of the pancreas from its endocrine tissue by structure, product, and route out.

3. Explain why an inflamed appendix hurts in the lower right abdomen while a weak esophageal sphincter is felt behind the breastbone.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

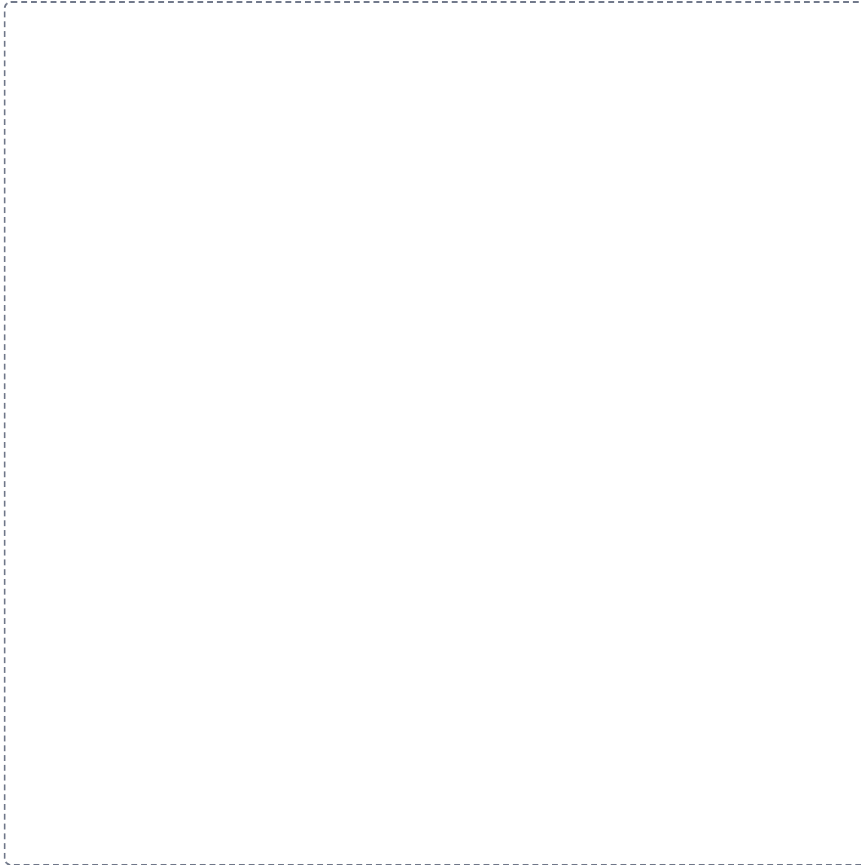
Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

The canal against the accessory organs

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

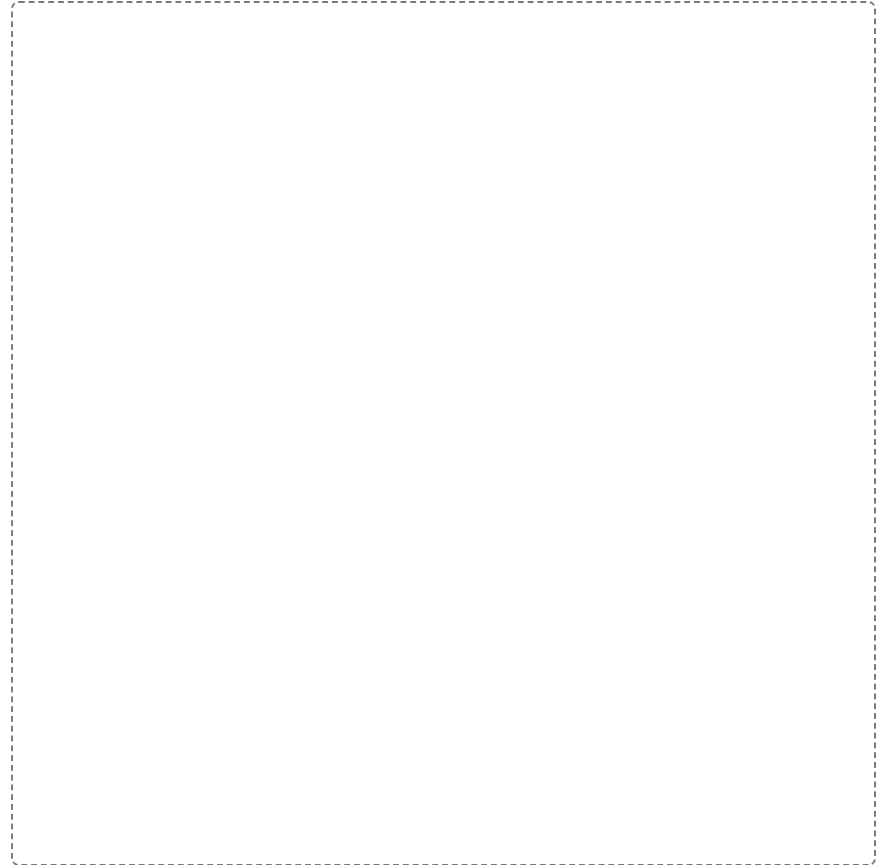


2

The four layers of the wall

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.



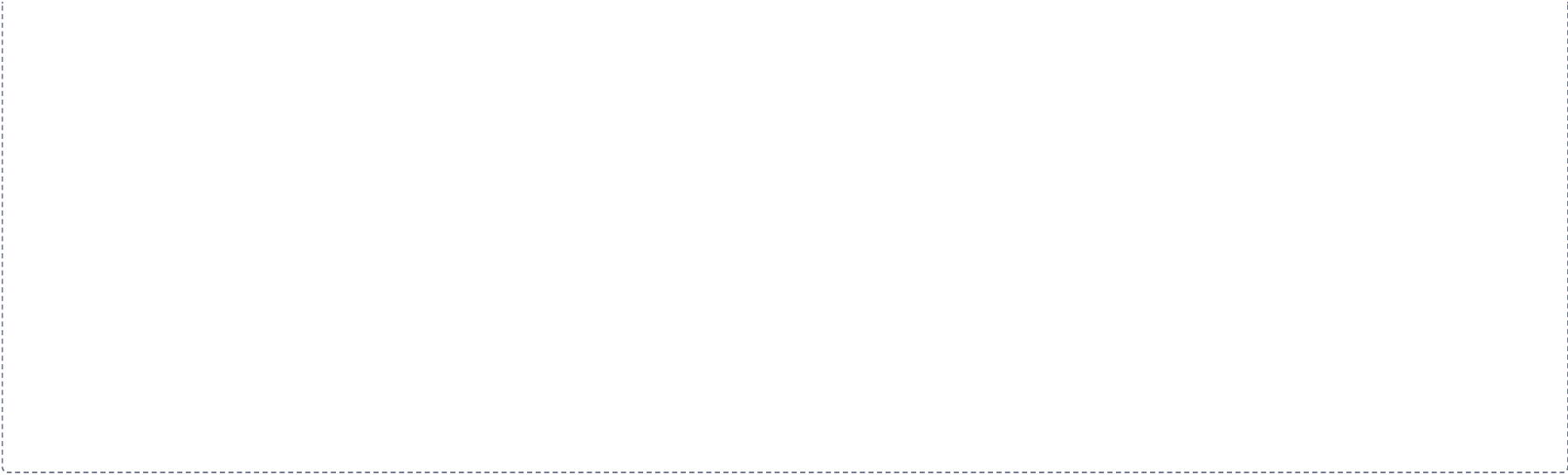
3

The pathway of food, mouth to anus

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.



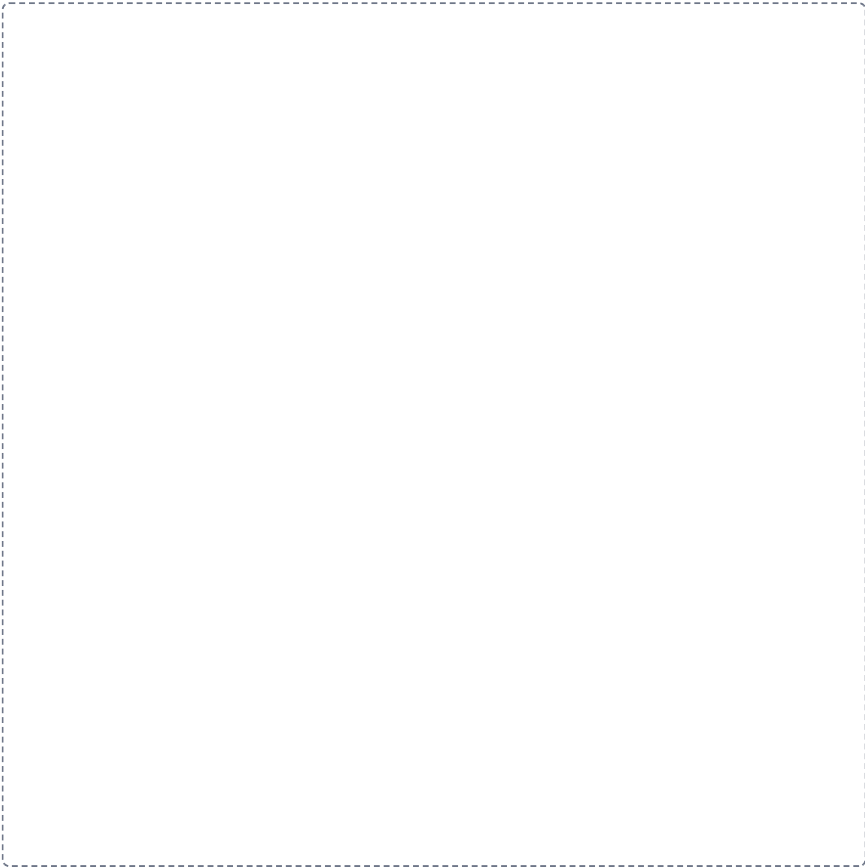


4

The stomach

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

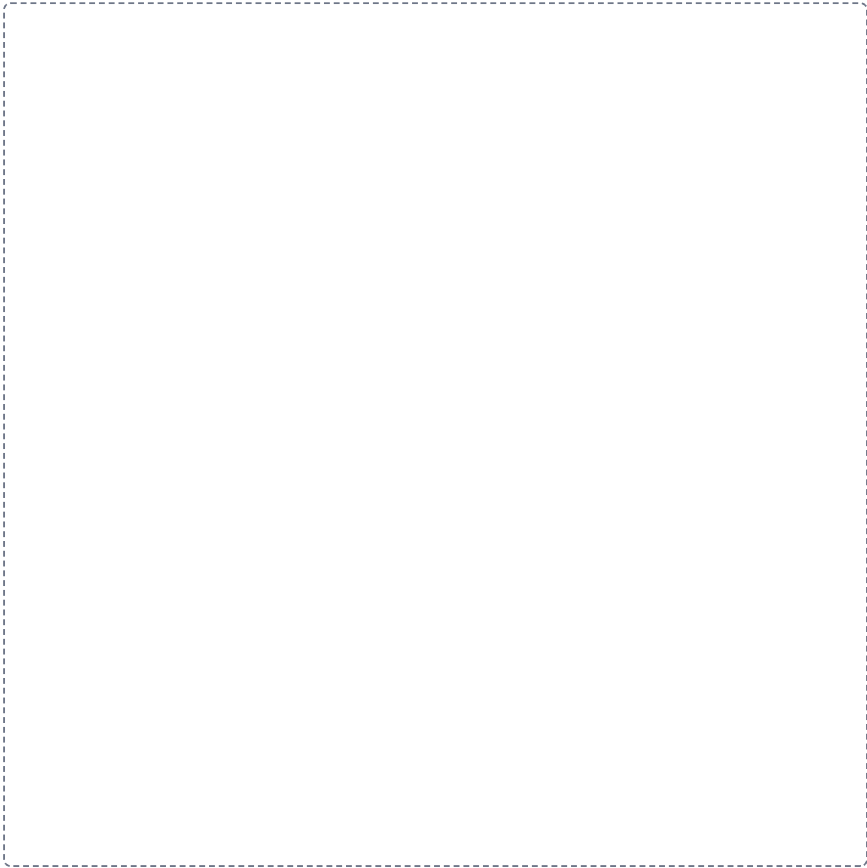


5

Three levels of surface amplification

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

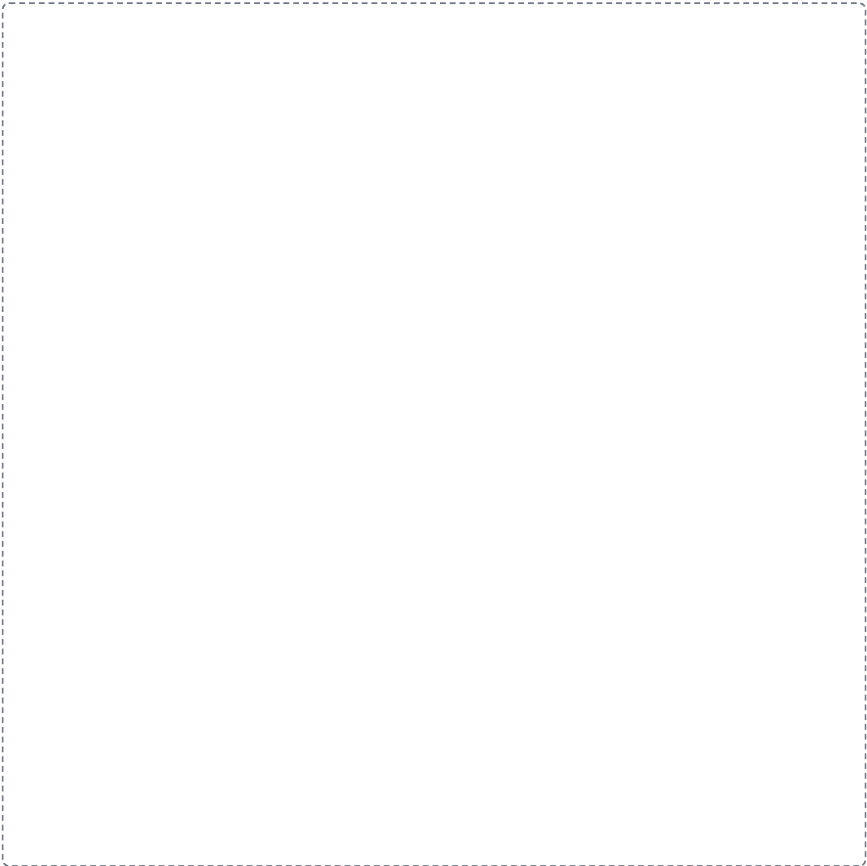


6

The large intestine

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

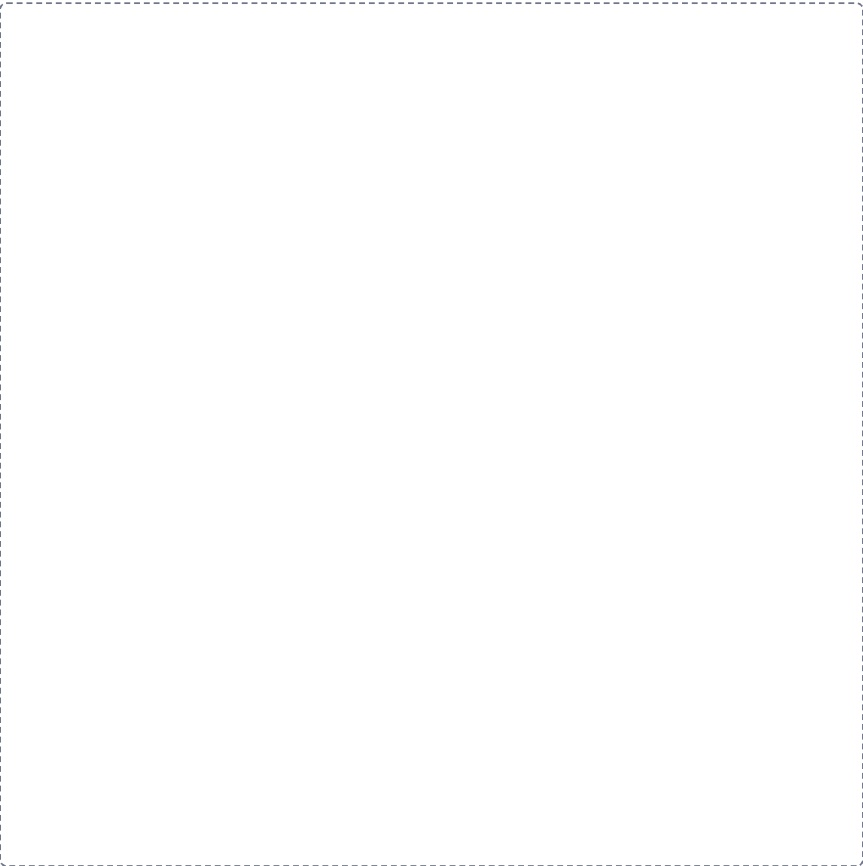


7

The liver, from lobe to lobule

HUB

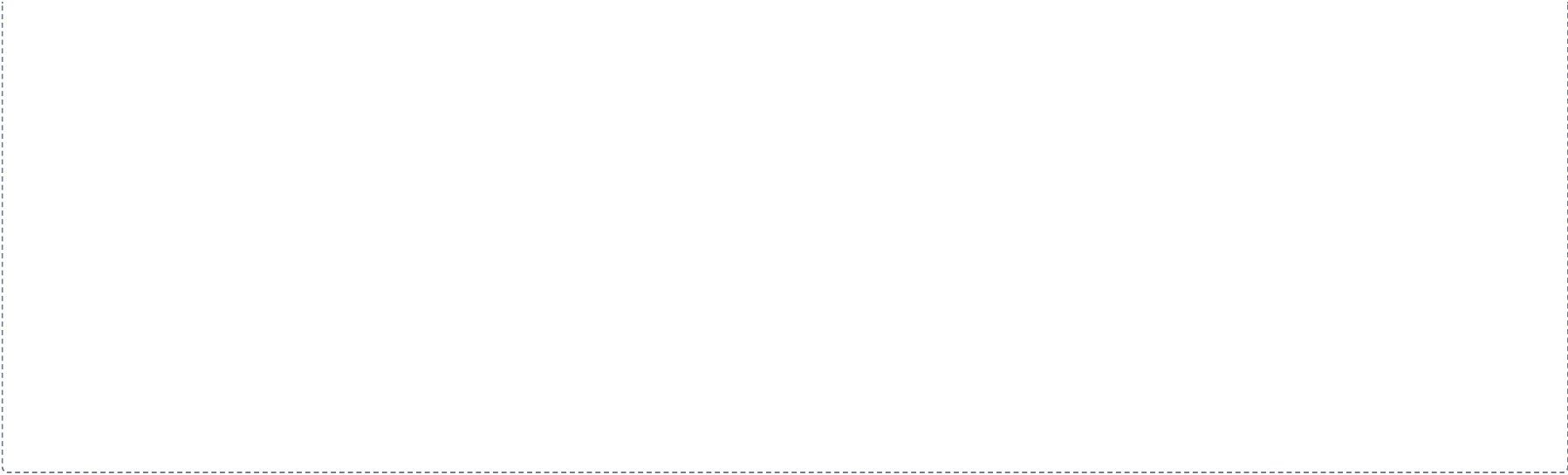
Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.



The path of bile, canaliculus to duodenum

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.



Pre-work • Urinary and Endocrine

The Urinary System and Endocrine Anatomy. Read both Part A chapters first. The two chapters share a sheet because an adrenal gland caps each kidney and the kidney is itself an organ with endocrine tissue, so learning them together saves you a second pass over the same region.

Module 4 • 3 of 5

Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1 Start with the reading.

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2 Organize what you read.

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3 Draw it.

In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.

4 Apply it.

Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.

5 Come back to it later in the week

, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

3 min

PANEL 1 OF 3

Turn the urinary and endocrine chapters into mini visuals. Eight chunks. The kidney takes six of them, because it is the one organ in this module you have to hold at three scales at once.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 LADDER

The kidney from the outside in

Five rungs, renal fascia at the top down to the medulla. Beside each covering write the one job it does. On the last two rungs draw the pyramids with their bases toward the cortex and the renal columns dipping down between them.

2 HUB

The kidney in frontal section

One bean at the hub, hilum on the medial side. Spoke to the cortex, the columns, the medulla, the pyramids, the papillae, the minor and major calyces, the renal pelvis, and the ureter. Draw each papilla pointing into its own minor calyx, because that is the joint most people draw backward.

3 CHAIN

The blood supply, aorta to renal vein

Eight boxes with arrows, renal artery through to the renal vein. Circle the two capillary beds and box the single vessel that sits between them, then write beside each bed what happens there.

4 HUB

The renal corpuscle and the juxtaglomerular apparatus

One corpuscle at large scale in the middle. Spoke to the glomerulus, the glomerular capsule with its two layers, the podocytes and their filtration slits, and the capsular space. Then draw the afferent and efferent arterioles and the distal tubule curving back to touch them, and label the macula densa and the juxtaglomerular cells at that contact.

5 LADDER

The three layers of the filtration membrane

Three rungs, in the order plasma crosses them. Beside each rung write what it lets through and what it holds back. The rungs get finer as you go down, and that gradient is the whole answer.

6 CHAIN

From capsular space to the outside

One long chain with arrows: capsular space, the three tubule segments, the collecting duct, then papilla, calyces, pelvis, ureter, bladder, urethra. Draw a line across the chain at the point where the fluid stops being called filtrate and starts being called urine.

7 GRID

Where every endocrine gland sits

Gland down the side, ten rows, running from the top of the body to the bottom. Location and one note across the top. Add a third column: is this a pure endocrine gland, or an organ with a second job.

8 SPLIT

The two lobes of the pituitary

One split. Other name, tissue, and link to the hypothalamus on each branch. The tissue column predicts the link column, so draw an arrow between those two cells on both branches.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

3 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

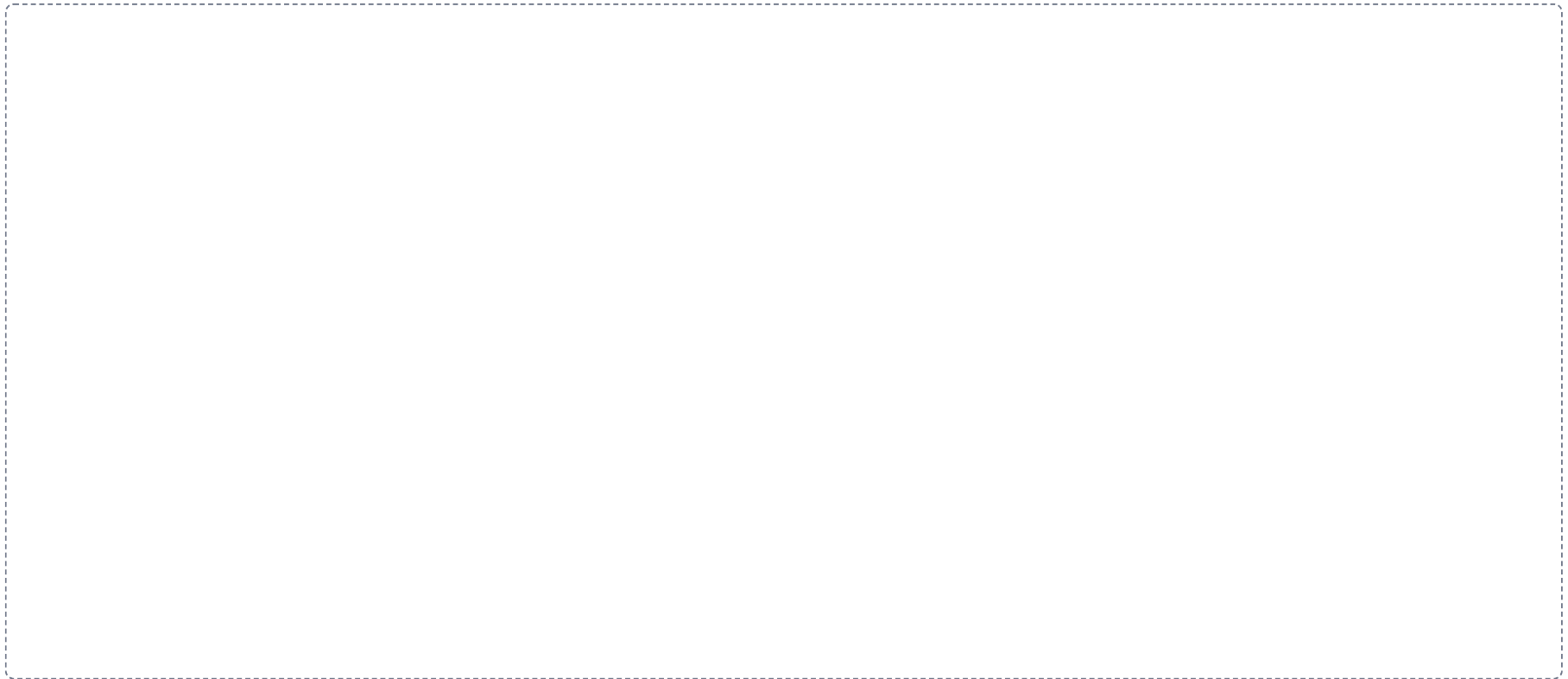
DRAWING 1

The kidney in frontal section, with its vessels

BRAIN DUMP FIRST

Two minutes. Every region, every collecting space, and the arterial chain running through them.

Draw a kidney cut front to back, hilum on the medial side. Lay in the cortex, the medulla with several pyramids, the renal columns between them, and the papilla at the tip of each pyramid. Draw the minor calyces cupping the papillae, the major calyces, the renal pelvis, and the ureter leaving at the hilum. Then draw the arterial chain over the top of it: renal artery at the hilum, segmental and interlobar arteries running in the columns, arcuate arteries arching along the bases of the pyramids, and interlobular arteries branching up into the cortex.



In your notebook, head the page **M4-4 The kidney in frontal section, with its vessels** so you can find it later.
Correct it in a second color.

YOU KNOW IT WHEN

You can put a finger anywhere in your kidney and say whether the fluid there is still inside a tubule or already in a collecting space.

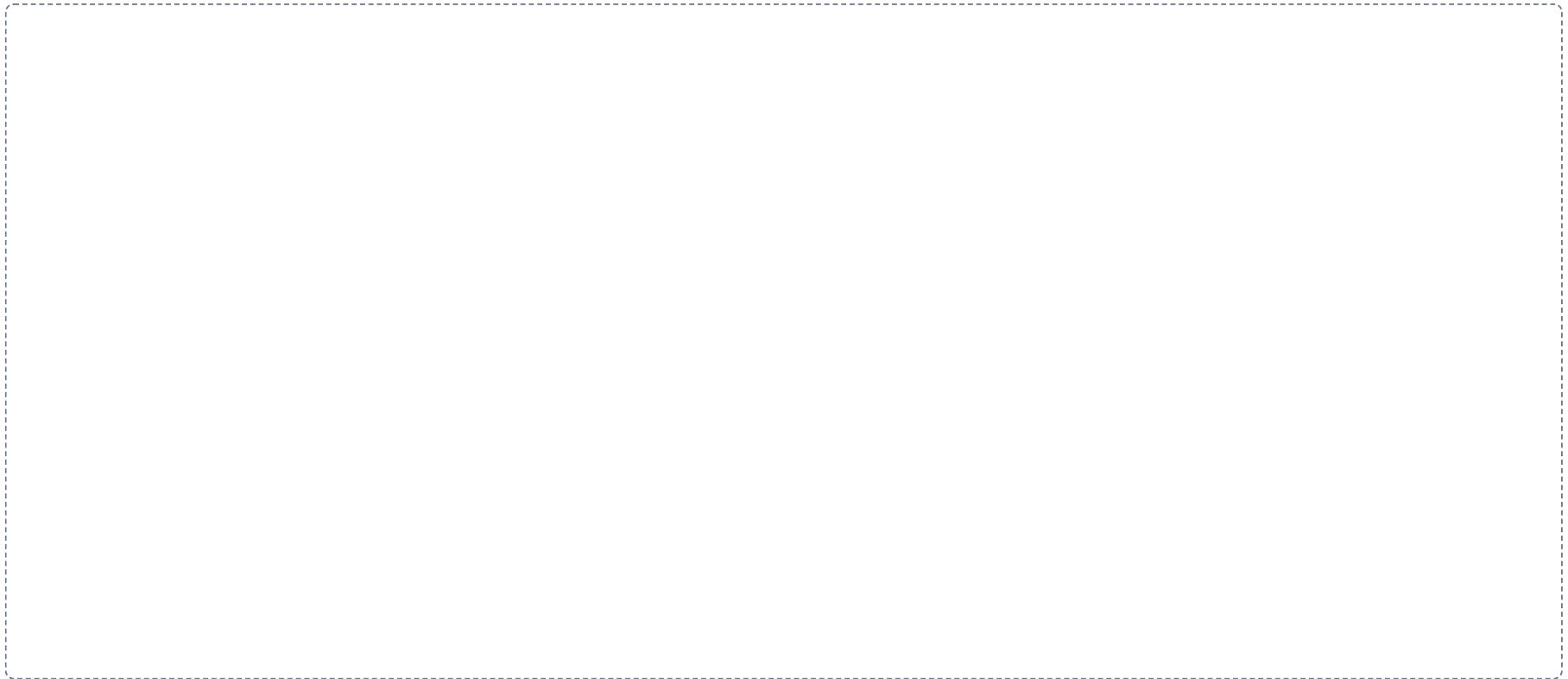
DRAWING 2

One nephron, drawn in place

BRAIN DUMP FIRST

Two minutes. Every segment in order, and which region of the kidney each one sits in.

Draw a wedge of kidney with the cortex above and the medulla below, and the line between them marked. Put a renal corpuscle in the cortex and run the tubule out of it: proximal convoluted tubule coiled in the cortex, nephron loop diving into the medulla and turning back, distal convoluted tubule coiled in the cortex again, then a collecting duct running down the pyramid to the papilla. Wrap peritubular capillaries around the cortical tubules and the vasa recta around the loop. Then draw a second nephron beside the first with a long loop reaching deep, and label which type each one is.



In your notebook, head the page **M4-5 One nephron, drawn in place** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can say for every segment of your nephron whether it lies in the cortex or the medulla, and which of the two nephron types concentrates the urine.

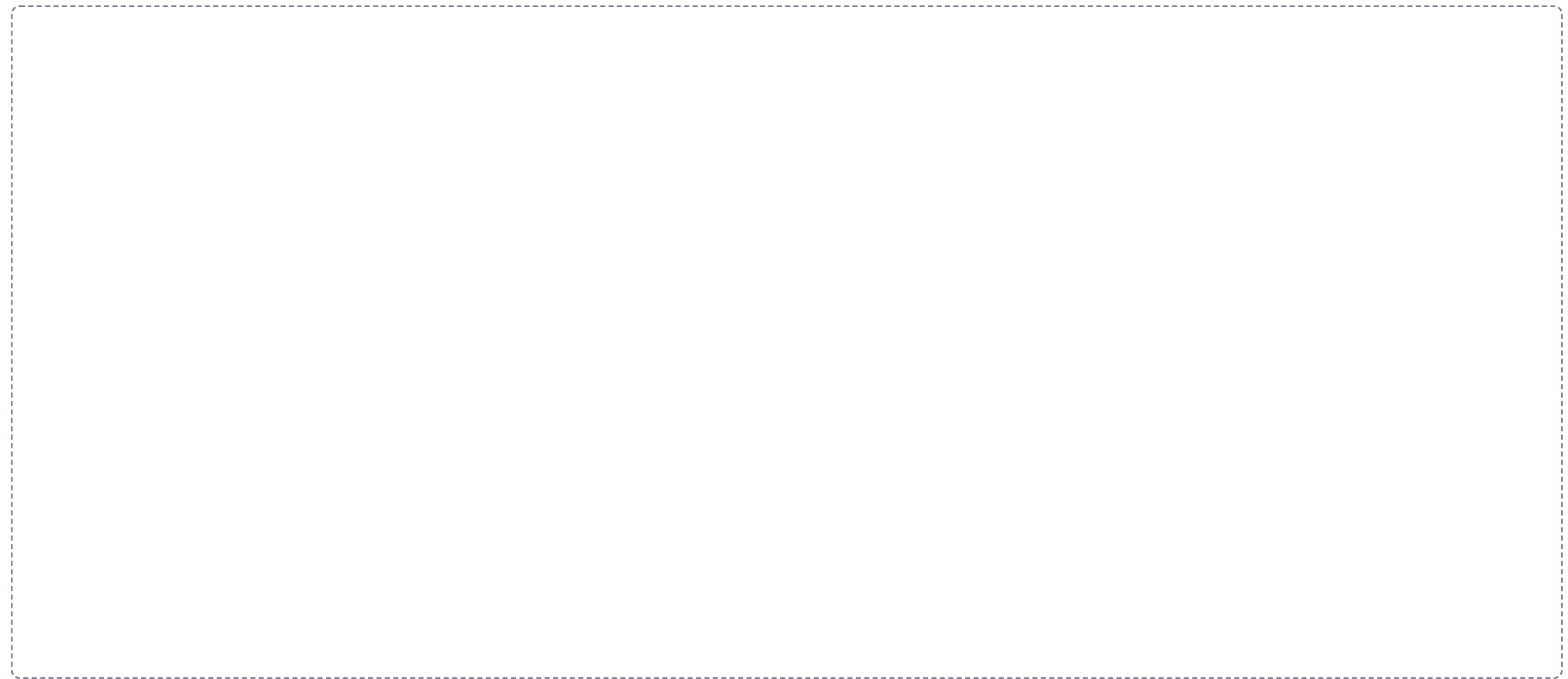
DRAWING 3

The endocrine glands on a body outline

BRAIN DUMP FIRST

Two minutes. Every gland, where it sits, and what the two-part glands are built from.

Draw a body outline from head to pelvis. Mark the hypothalamus and the pituitary hanging beneath it in the sella turcica, the pineal gland at the back of the third ventricle, the thyroid across the trachea with its isthmus and the four parathyroids behind it, the thymus behind the sternum, an adrenal gland capping each kidney, the pancreas behind the stomach, and the ovaries or the testes. Then enlarge two glands in the margin: the pituitary with its anterior and posterior lobes and the infundibulum joining it to the hypothalamus, and the adrenal with its three cortical zones in order and the medulla inside them.



In your notebook, head the page **M4-6 The endocrine glands on a body outline** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can name every gland on the outline with the labels covered, and say for the pituitary and the adrenal which part is nervous tissue and which is glandular.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A urine sample contains albumin, a large plasma protein, but no blood cells. Name the layer of the filtration membrane most likely damaged and explain what rules out the other two.

Answer. The basement membrane.

Reasoning. Read the sample as a sieve test rather than as a disease. Three layers stand between the blood and the capsular space, and each one holds back something of a different size, so what escaped names the layer that failed. The fenestrated endothelium of the glomerular capillary is the layer that stops blood cells, and no cells appear in the sample, so that layer is still doing its job. The slit membranes between the podocyte processes hold back medium-sized proteins, and had they alone failed you would expect smaller proteins through first, not the largest one. Albumin is a large plasma protein, and the layer built to hold back large plasma proteins is the basement membrane lying between the capillary and the podocytes. The size of what got through is the whole of the reasoning.

Set A · Name it

1. The cup-shaped chamber that collects urine draining from a renal papilla is a _____.
2. The cells of the afferent arteriole that release renin when blood pressure is low are the _____.
3. The smooth triangular area at the base of the bladder, marked by three openings, is the _____.
4. The bridge of thyroid tissue crossing the front of the trachea is the _____.
5. Name the three zones of the adrenal cortex in order, from the capsule inward.

Set B · Predict it

1. A patient loses the fat surrounding one kidney. Predict what happens to the position of the organ and to the flow of urine.

2. A pituitary tumor enlarges. Predict the direction it grows and the first symptom, working from the bone around it.

3. A stone forms in the renal pelvis and stays there. Predict when the patient first feels pain and explain why.

4. The stalk between the hypothalamus and the pituitary is cut. Predict which lobe is affected more directly, working from the link each lobe uses.

Set C · Tell it apart

1. Distinguish a cortical from a juxtamedullary nephron by loop length, blood supply, and job.

2. The kidney appears on both the urinary sheet and the endocrine sheet. Defend both placements.

3. Explain why urinary tract infections are more common in women, using one structural difference.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

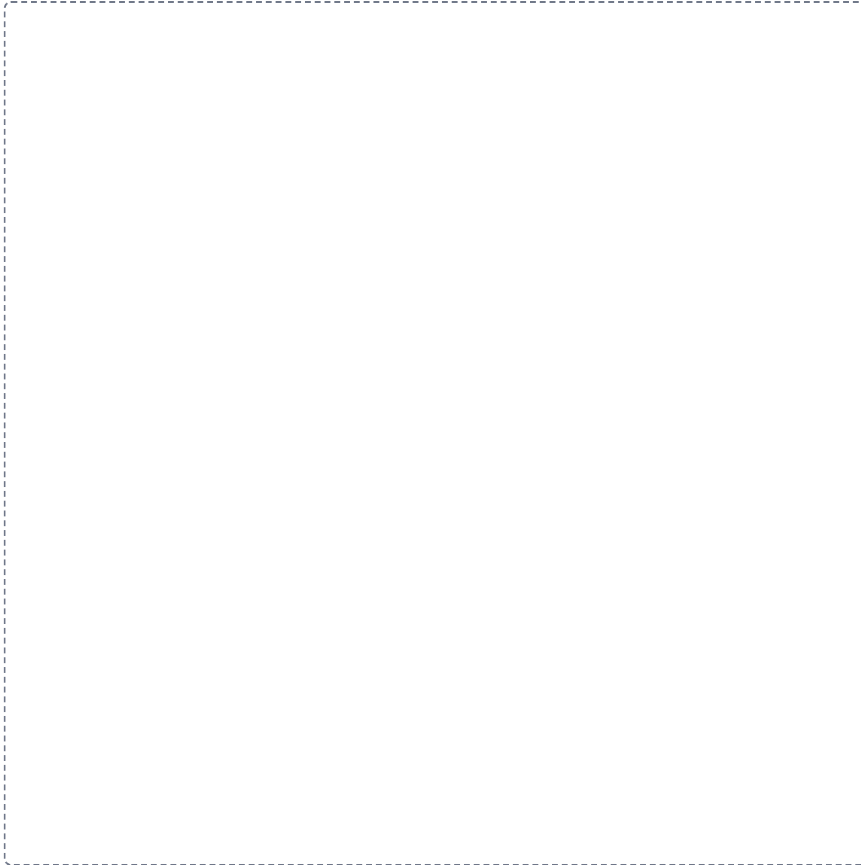
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1

The kidney from the outside in

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

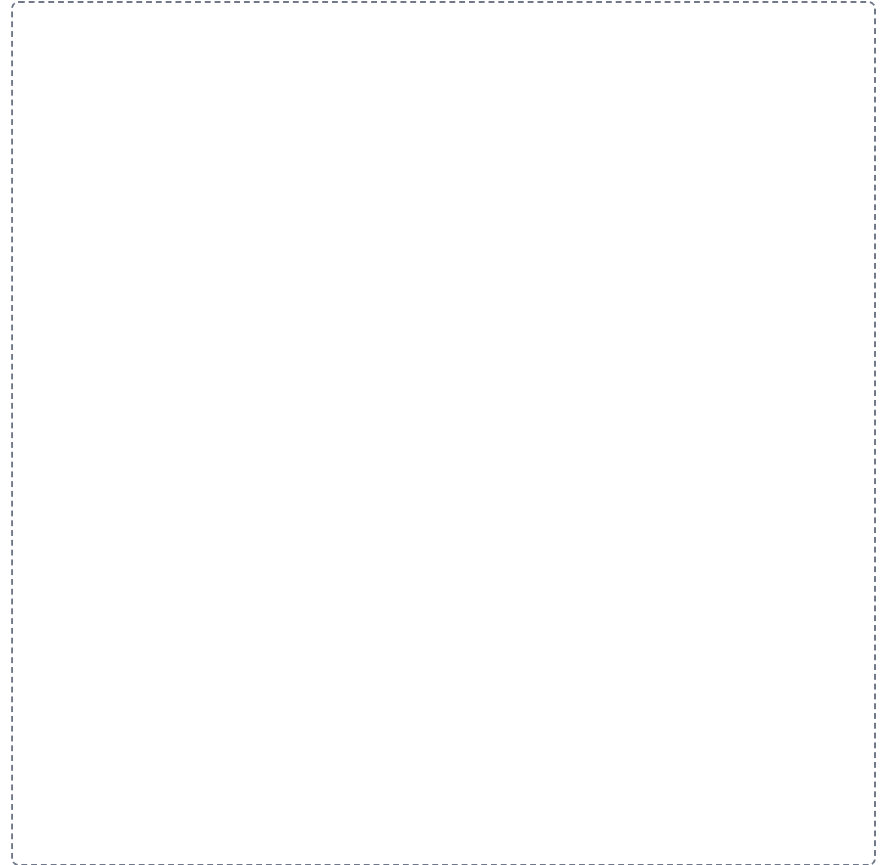


2

The kidney in frontal section

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.



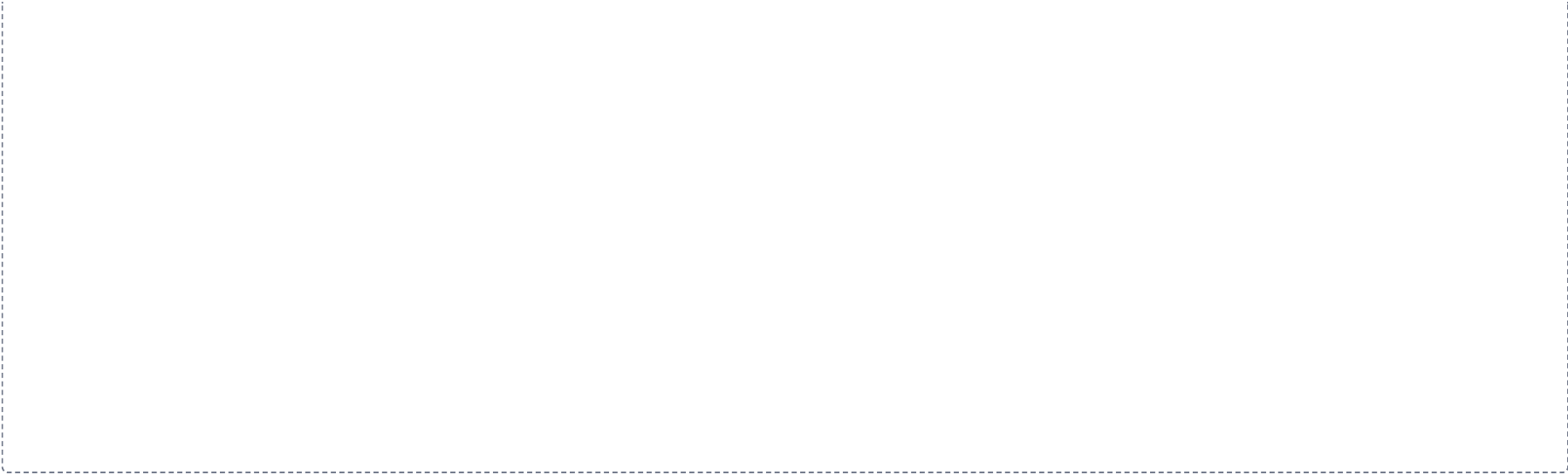
3

The blood supply, aorta to renal vein

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

A large dashed rectangular box intended for the student to draw a flowchart showing the steps from the aorta to the renal vein. The box is empty and occupies most of the page below the instructions.



4

The renal corpuscle and the juxtaglomerular apparatus

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

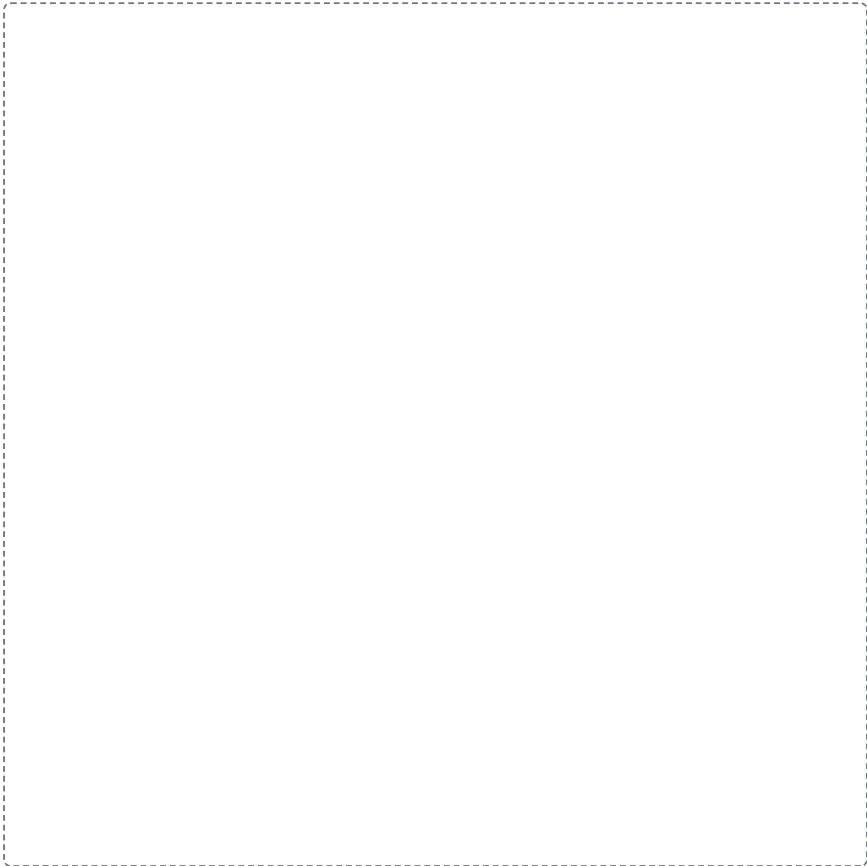


5

The three layers of the filtration membrane

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

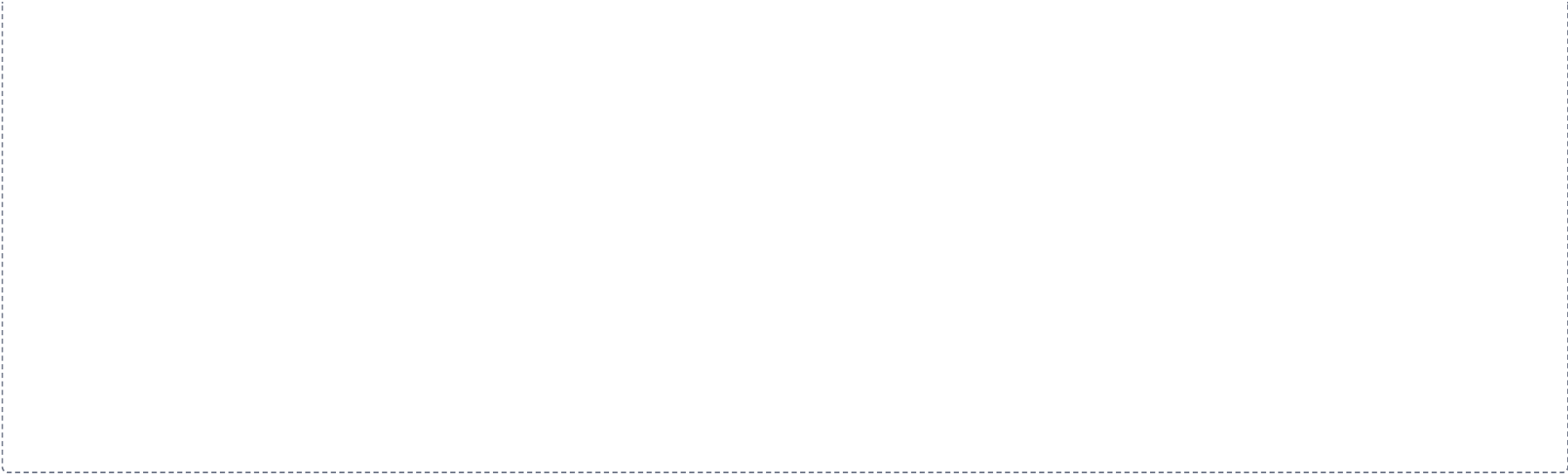


6

From capsular space to the outside

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

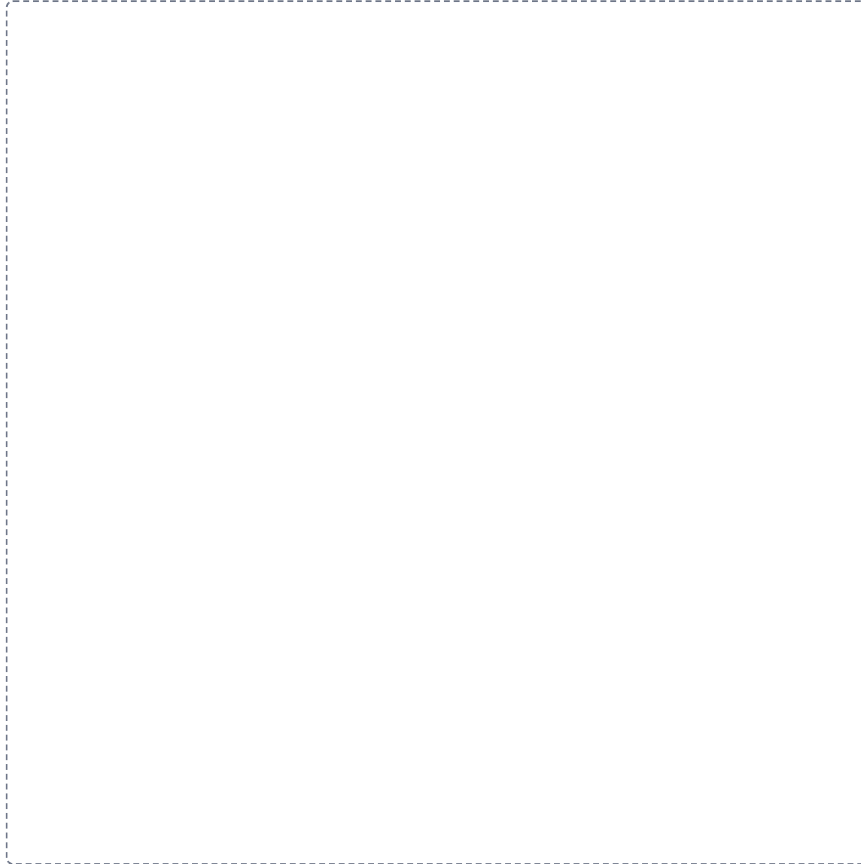


7

Where every endocrine gland sits

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

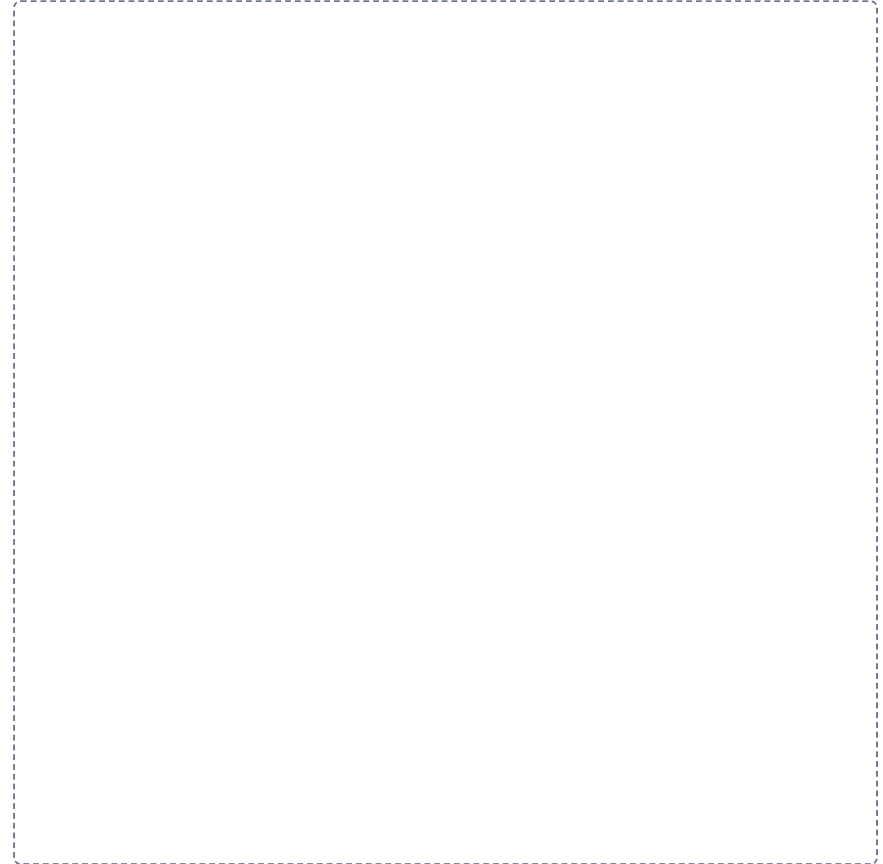


8

The two lobes of the pituitary

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.



Pre-work • Reproductive Systems and Pregnancy

The Male Reproductive System, The Female Reproductive System, and Pregnancy and Birth. Read all three Part A chapters first. The two tracts are built on the same plan, a gonad, a duct system, accessory glands, and an outlet, so build them side by side and the third chapter reads as the sequel to both.

Module 4 • 4 of 5

Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1 Start with the reading.

Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.

2 Organize what you read.

In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.

3 Draw it.

In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.

4 Apply it.

Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.

5 Come back to it later in the week

, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

3 min

PANEL 1 OF 3

Turn the three reproductive chapters into mini visuals. Eight chunks. Two chains carry the two routes, and the second of them runs on past fertilization into the first week of a pregnancy.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 HUB

The testis, from tunic to tubule

One testis at the hub, cut open. Spoke to the tunica vaginalis on the outside, the tunica albuginea and the lobules it divides, the seminiferous tubules coiled inside one lobule, and the epididymis on the posterior surface. Then spoke to the two cell types, sustentacular cells lining the tubule and interstitial cells lying between the tubules, and write the job of each. Add the dartos and the cremaster around the outside, and mark which is smooth muscle and which is skeletal.

2 CHAIN

The path of sperm, tubule to the outside

Five boxes with arrows, seminiferous tubule through to the urethra. Under each box write whether that structure makes, stores, matures, or carries. Only the first box can say makes, and the chain gets easier to hold once you see that.

3 HUB

A sperm cell

One sperm at large scale. Spoke to the head with its nucleus and the acrosome capping it, then to the neck, middle piece, principal piece, and end piece along the tail. Write the one job of each part beside it, and mark the part packed with mitochondria.

4 GRID

The three accessory glands

Gland down the side, three rows. What it adds to the semen, where it empties, and one structural note across the top. Star the one gland the urethra runs straight through.

5 HUB

The ovary in section

One almond-shaped ovary at the hub, cut open. Spoke outward through the germinal epithelium and tunica albuginea to the cortex with its follicles and the medulla with its vessels. Draw a follicle at three sizes around the rim, then the mature follicle, the corpus luteum, and the corpus albicans. Hang the ovarian, suspensory, and broad ligaments off the outside.

6 CHAIN

From ovulation to implantation

Seven boxes with arrows, ovary through to the endometrium. Mark the box where fertilization normally happens and the box where implantation happens, and write above the chain that about a week of travelling and dividing separates the two.

7 LADDER

The three layers of the uterine wall

Three rungs, perimetrium at the top down to endometrium. Beside each write what it is made of and what it does. Split the bottom rung in two, stratum functionalis above stratum basalis, and mark which of the two is shed.

8 GRID

The temporary structures of pregnancy

Structure down the side, six rows. What it is and what it does across the top. Add a third column asking whether it is kept or discarded at birth. Every row answers the same way, and that is the point of the grid.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

3 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

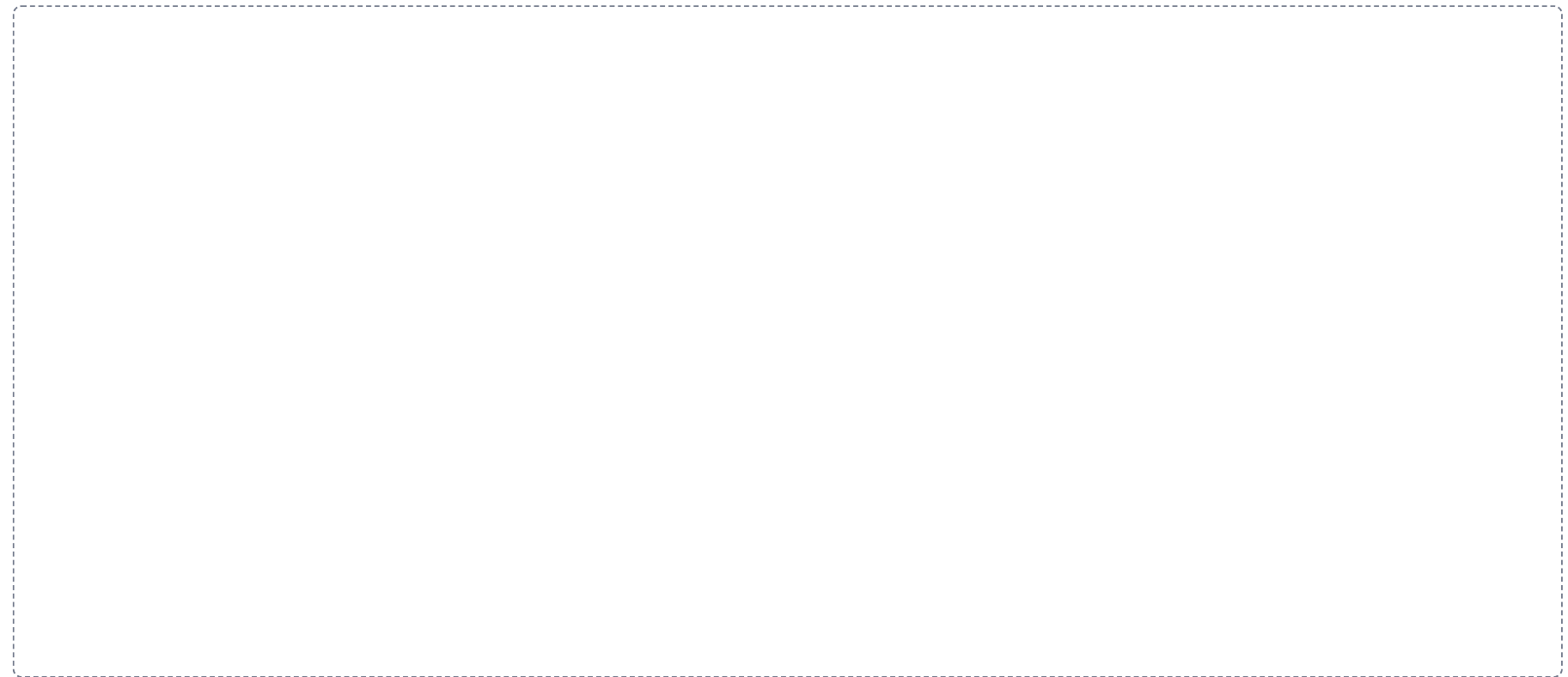
DRAWING 1

The male tract in sagittal section

BRAIN DUMP FIRST

Two minutes. Every duct and every gland in order, from the testis to the outside.

Draw the male pelvis cut down the midline, with the bladder, the pubic symphysis, and the rectum in place. Add the testis in the scrotum, the epididymis on its posterior surface, and the ductus deferens running up in the spermatic cord and down behind the bladder. Draw the seminal vesicle joining it to form the ejaculatory duct, the prostate wrapped around the urethra below the bladder, and the bulbourethral glands beneath that. Follow the urethra through its prostatic, membranous, and spongy parts, then draw the penis in cross section with the two corpora cavernosa and the single corpus spongiosum around the urethra.



In your notebook, head the page **M4-10 The male tract in sagittal section** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can trace the whole route with a finger and say at every junction what has just been added to the sperm.

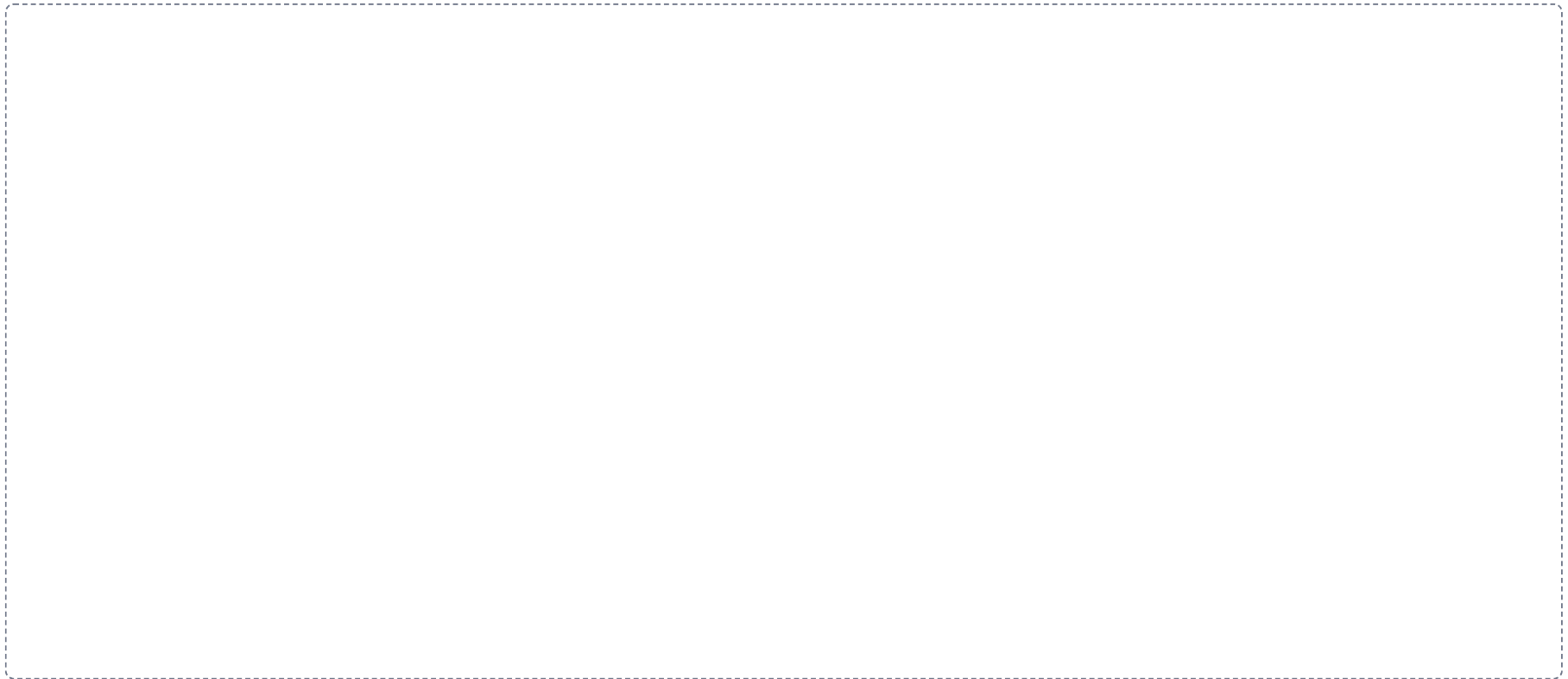
DRAWING 2

The uterus, tubes, and ovaries, in frontal view

BRAIN DUMP FIRST

Two minutes. Both tubes with every region, the uterus with every region, and the wall in layers.

Draw the uterus pear-shaped and cut open on one side. Mark the fundus, body, cervix, uterine cavity, and cervical canal with its internal and external os, and label the three wall layers along the cut edge. Draw both uterine tubes running out from the top and mark the isthmus, ampulla, infundibulum, and fimbriae on each, with the fimbriae reaching over the ovary rather than sealing onto it. Add an ovary at each end and draw the ovarian, suspensory, broad, round, cardinal, and uterosacral ligaments where each one attaches.



In your notebook, head the page **M4-11 The uterus, tubes, and ovaries, in frontal view** so you can find it later.
Correct it in a second color.

YOU KNOW IT WHEN

You can explain from your own drawing how an infection reaches the pelvic cavity from the vagina, and why that open route exists at all.

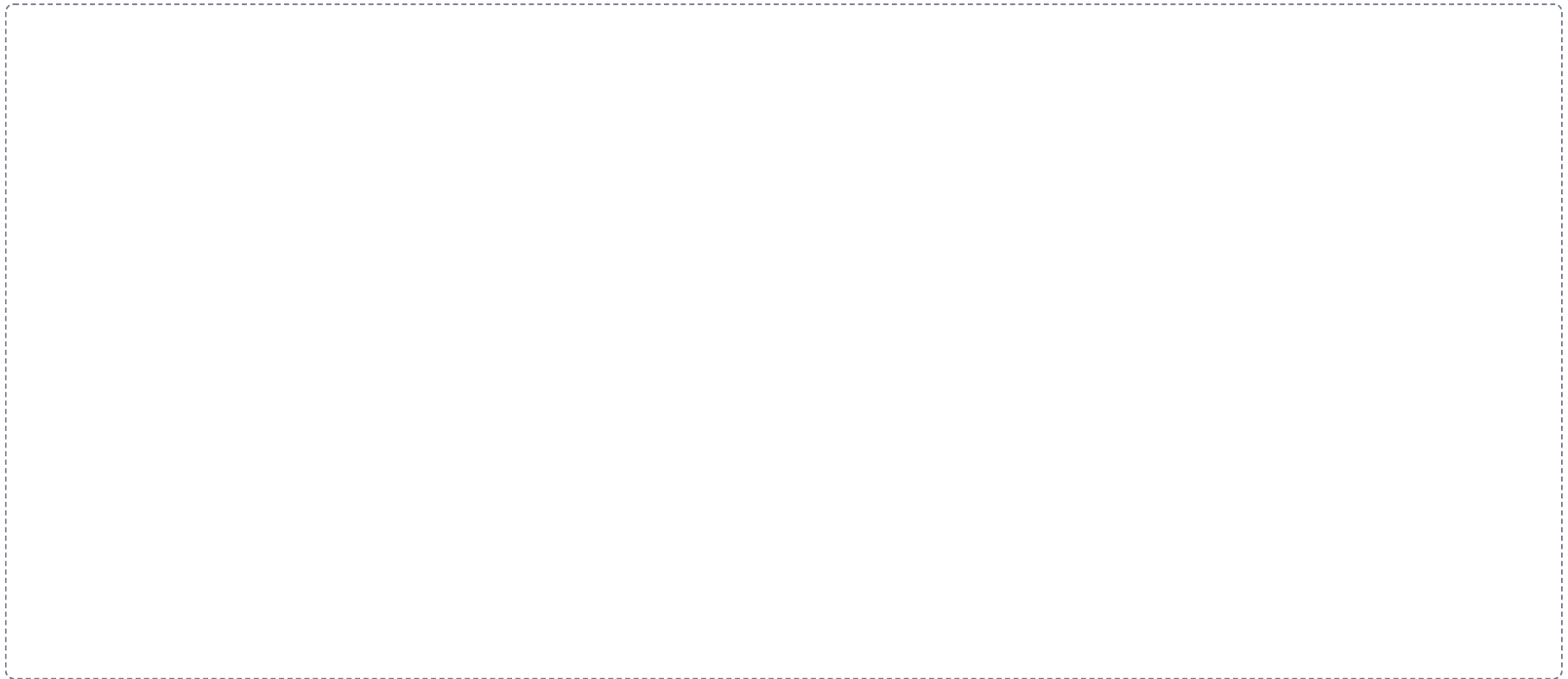
DRAWING 3

The uterus at full term, and the three stages of labor

BRAIN DUMP FIRST

Two minutes. The fetus in position, every temporary structure, and the order of birth.

Draw the gravid uterus in sagittal section, reaching up toward the rib cage, with the fetus in the vertex position, head down at the cervix. Draw the placenta on the uterine wall, the umbilical cord with its two arteries and one vein, the amnion holding the amniotic fluid around the fetus, and the chorion outside that. Then draw three small panels beside it, one for each stage of labor: the cervix thinning and dilating, the fetus being pushed through the cervix and vagina, and the placenta delivered as the afterbirth.



In your notebook, head the page **M4-12 The uterus at full term, and the three stages of labor** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can name every structure that leaves the body at birth and say which of them was built for this pregnancy alone.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A patient with a history of pelvic inflammatory disease has severe pain on one side of the pelvis and a positive pregnancy test, and no embryo is seen in the uterus. Name the most likely site and explain the prediction from structure.

Answer. The ampulla of the uterine tube.

Reasoning. Start with where fertilization normally happens rather than with the pain. A sperm meets the oocyte in the ampulla, the widest and longest part of the tube, and the fertilized ovum then spends about a week travelling toward the uterus while it divides into a morula and then a blastocyst. Anything that slows that journey leaves the blastocyst still inside the tube at the moment it is ready to implant, and scarring left by an earlier infection is exactly that kind of delay. Implantation is not choosy about the site, only about the timing, so the embryo embeds wherever it happens to be, which rules out the idea that it must reach the endometrium first. The ampulla is where it usually is at that moment, and that is why it is the commonest site. A tube a few millimeters wide has no lining built to stretch, which is what makes this a structural emergency rather than a merely misplaced pregnancy.

Set A · Name it

1. The cells lying between the seminiferous tubules that produce testosterone are the _____.
2. The enzyme-filled cap over the head of a sperm cell is the _____.
3. The widest and longest region of the uterine tube, where fertilization usually occurs, is the _____.
4. The outer layer of the blastocyst, which becomes the fetal part of the placenta, is the _____.
5. Name the three stages of labor in order and say what ends each one.

Set B · Predict it

1. The prostate enlarges. Predict the symptom and explain it from the path of the urethra.

2. A testis fails to descend into the scrotum. Predict the effect on sperm production and say why.

3. The ductus deferens is cut and tied on both sides. Predict what is still present in the semen and what is missing.

4. The placenta forms over the opening of the cervix. Predict the problem at delivery and name the condition.

Set C · Tell it apart

1. Distinguish the dartos from the cremaster muscle by tissue type and by what each one moves.

2. Distinguish the stratum functionalis from the stratum basalis of the endometrium by what happens to each one every cycle.

3. Compare an embryo with a fetus by the weeks each name covers and by what is happening to the body in each period.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

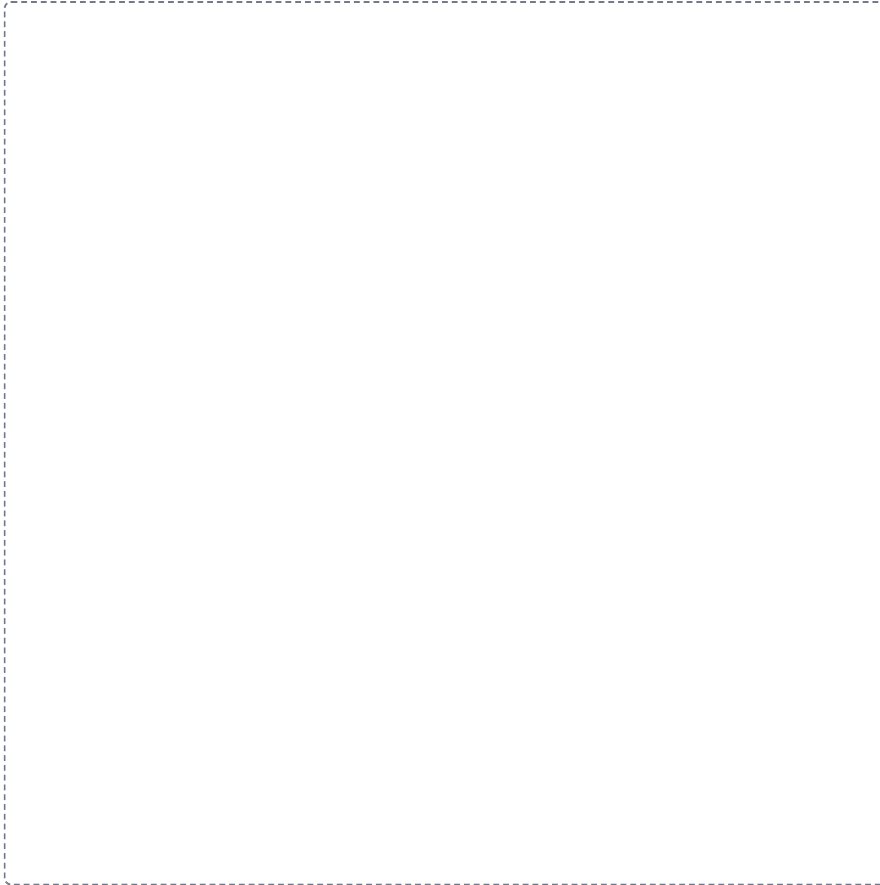
Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

The testis, from tunic to tubule

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

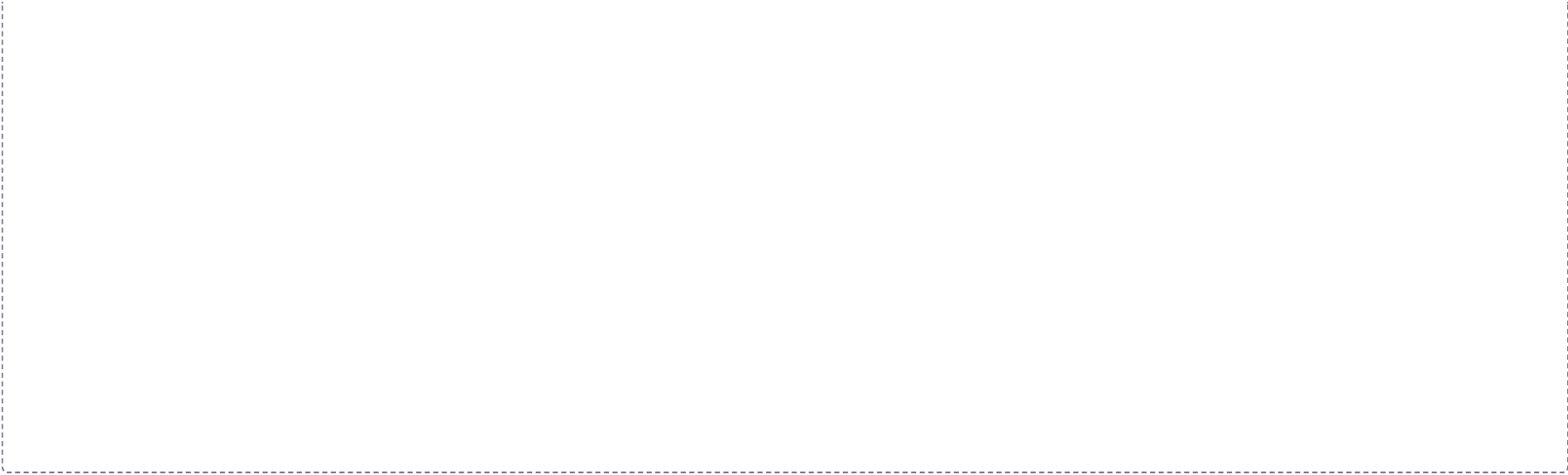


2

The path of sperm, tubule to the outside

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

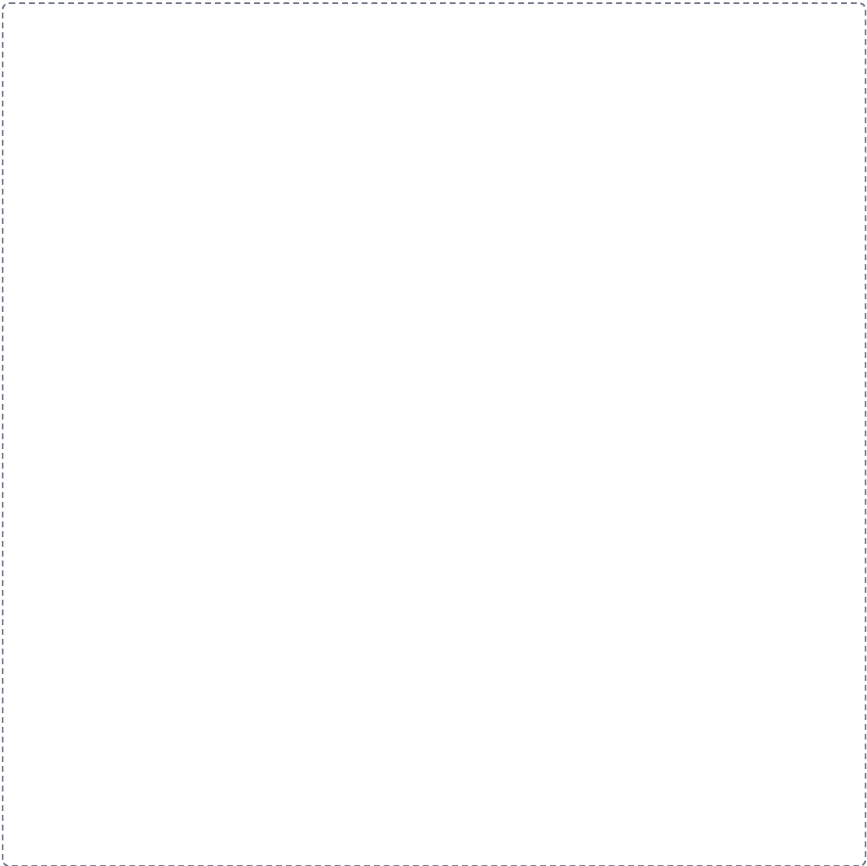


3

A sperm cell

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

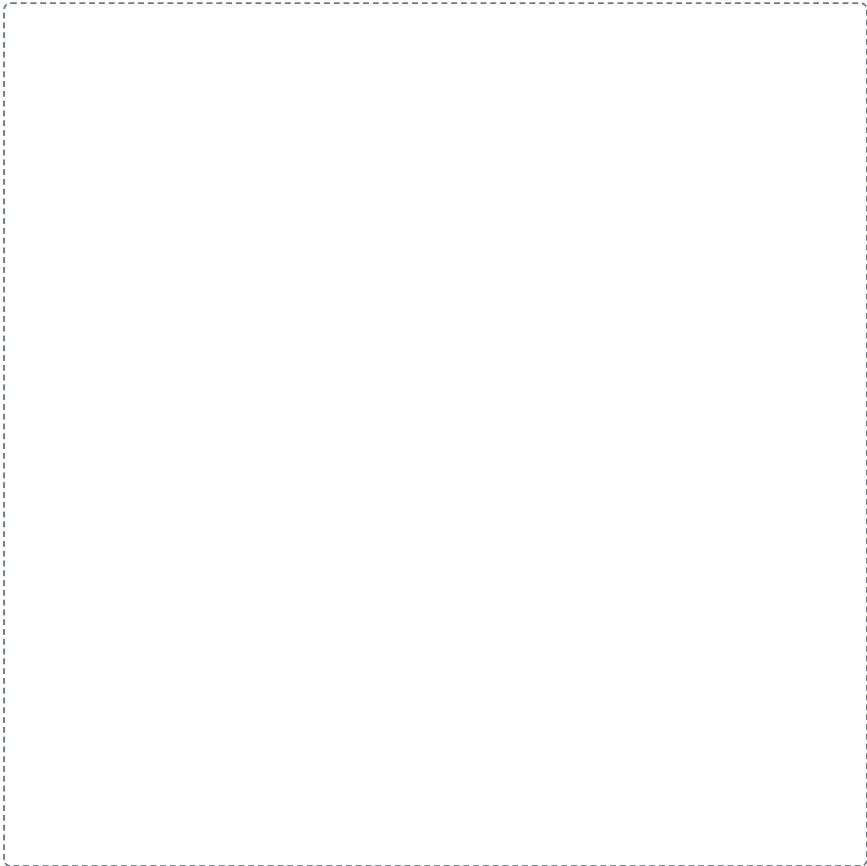


4

The three accessory glands

GRID

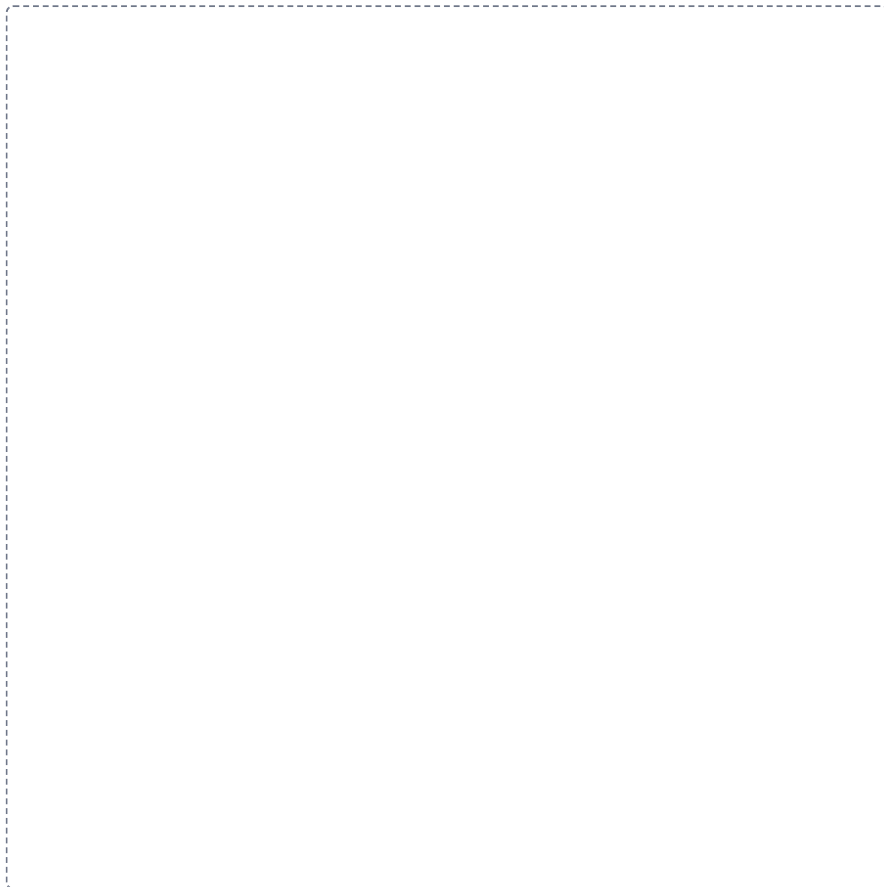
Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.



The ovary in section

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.



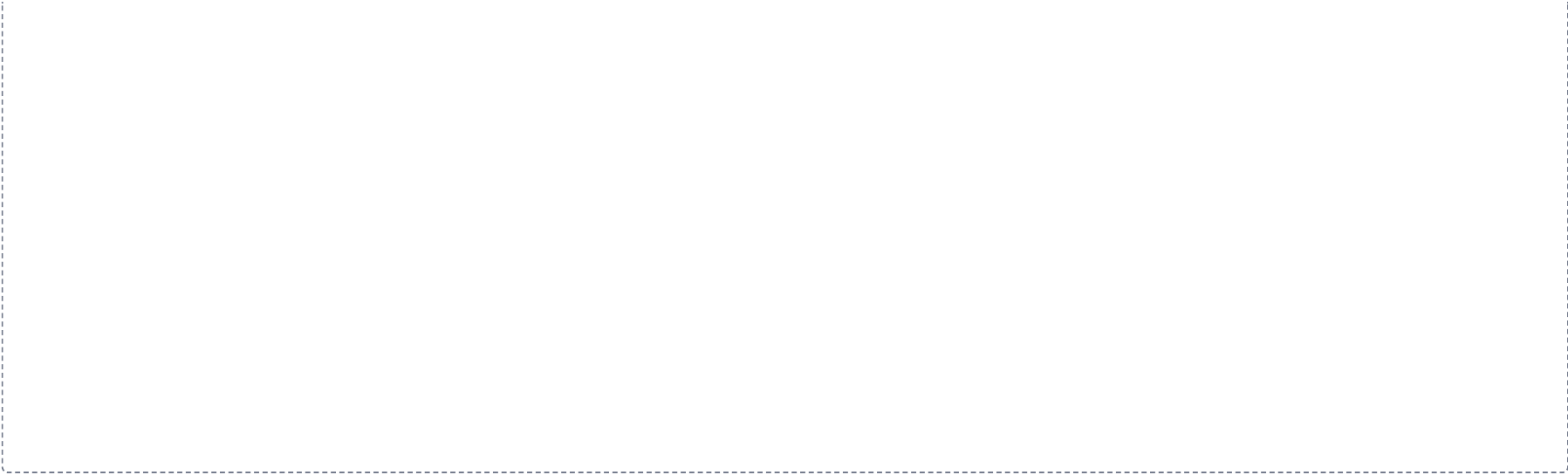
6

From ovulation to implantation

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

A large dashed rectangular box intended for the student to draw a flowchart showing the steps from ovulation to implantation. The box is empty and occupies most of the page below the instructions.

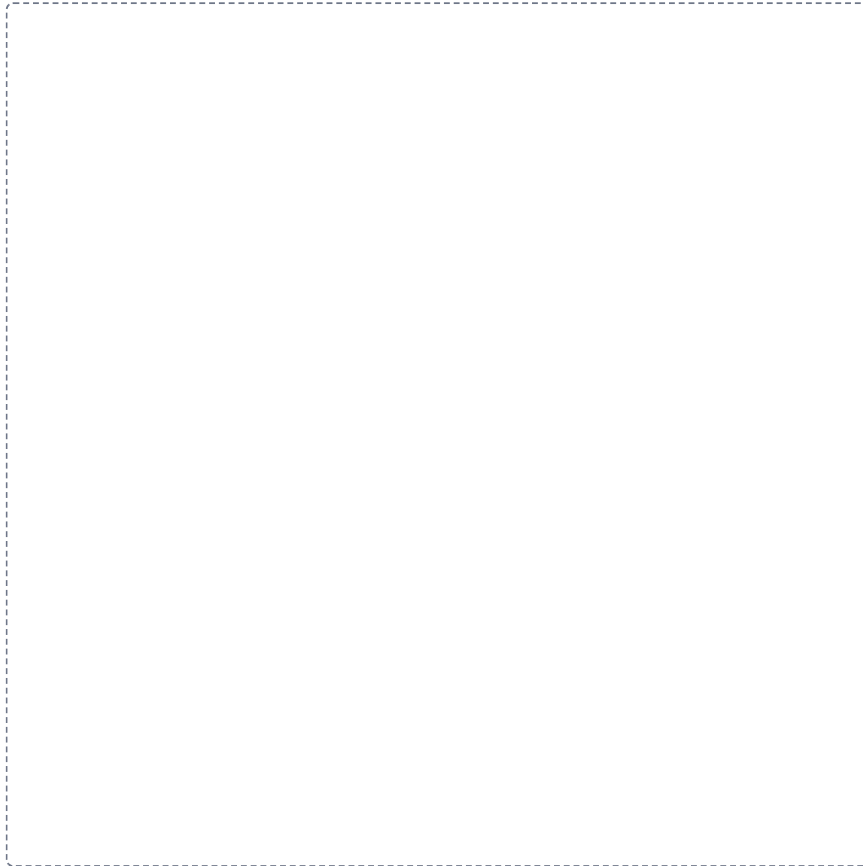


7

The three layers of the uterine wall

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

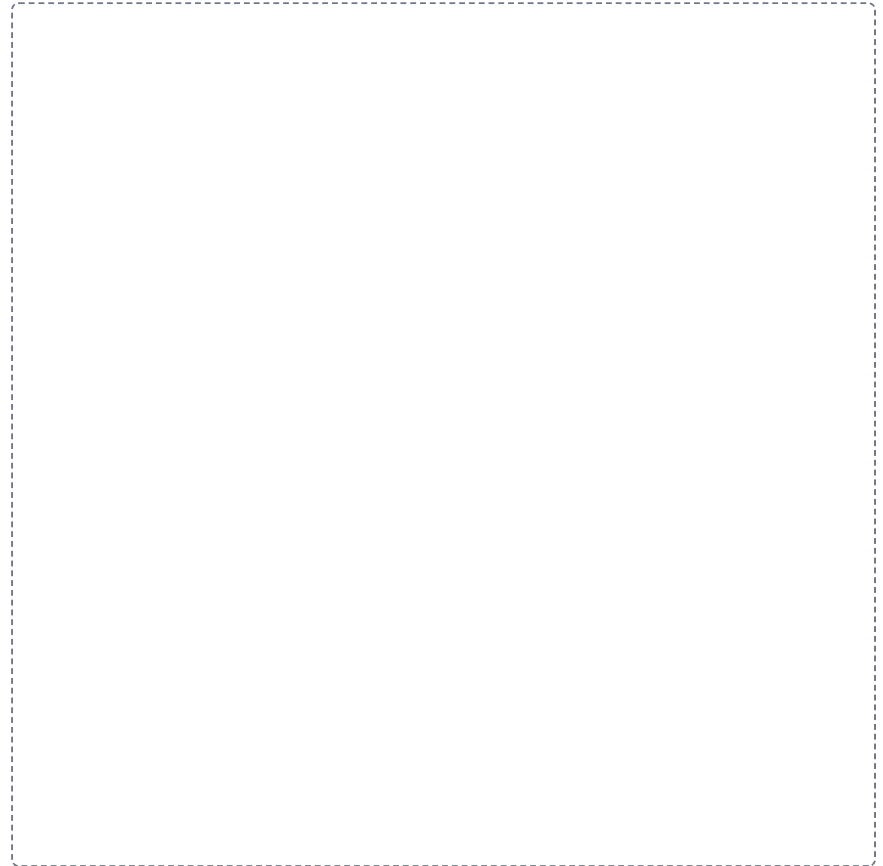


8

The temporary structures of pregnancy

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.



Pre-work • Module 4 Integration

All nine Part A chapters. This is a synthesis sheet, so do not read anything new for it. Work from the four packets you have already built and from your own mini visuals. Every chunk here crosses at least two chapters, and the crossings are where the module is actually tested.

Module 4 • 5 of 5

Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1 Start with the reading.

Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.

2 Organize what you read.

In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.

3 Draw it.

In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.

4 Apply it.

Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.

5 Come back to it later in the week

, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

3 min

PANEL 1 OF 3

This packet cuts across the chapters instead of following them. Eight chunks, each built from at least two systems. If a chunk sends you back to the reading rather than back to your own earlier visuals, that is the material to repair first.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 GRID

Cortex against medulla, four organs

Organ down the side: kidney, adrenal gland, ovary, lymph node. What sits in the cortex and what sits in the medulla across the top. Star the one organ whose two regions are built from entirely different tissues.

2 GRID

The four tubes of the trunk

System down the side: respiratory, digestive, urinary, reproductive. What it carries, where it begins, where it opens to the outside, and the sphincter or valve that guards it across the top. Circle the two rows that share a final passage in the male.

3 CHAIN

One drop of dietary fat, villus to bloodstream

A chain that crosses two chapters. Start in a lacteal inside a villus, then run through the lymphatic collecting vessels, a node, a trunk, the cisterna chyli, the thoracic duct, and into the subclavian vein. Write beside the first box that this is the one absorbed thing that never entered a blood capillary in the intestinal wall.

4 GRID

Organs with two jobs

Organ down the side: pancreas, ovary, testis, liver, kidney, pharynx, male urethra, placenta. The two jobs across the top, and a third column naming the tissue or region that does each one. This grid is the reason the same organ turns up on two different sheets.

5 HUB

The shelf behind the peritoneum

The posterior abdominal wall at the hub. Spoke to every structure Part A places behind the peritoneum: both kidneys, the pancreas, the duodenum, and the parts of the colon. Then add the adrenal gland sitting on top of each kidney and the ureter running down from each one, and write beside every spoke which packet it came from.

6 SPLIT

Exocrine against endocrine

One split. Duct or no duct, where the product is delivered, and how far it acts. Hang three Module 4 examples off each branch, then write the pancreas across the joint between them, because it is both at once.

7 HUB

The same trick in four places, surface area

Write increase the surface at the hub. Spoke to the alveoli of the lung, the circular folds with villi and microvilli of the small intestine, the million nephrons of one kidney, and the follicles of the thyroid gland. Beside each spoke write what has to cross or be stored on that surface.

8 LADDER

Every sphincter and valve, in order

One ordered list down the page, in the order a structure appears along the tubes: upper and lower esophageal, pyloric, sphincter of Oddi, ileocecal, internal and external anal, then internal and external urethral. Beside each write which tube it guards, and circle the two you can control.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

3 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

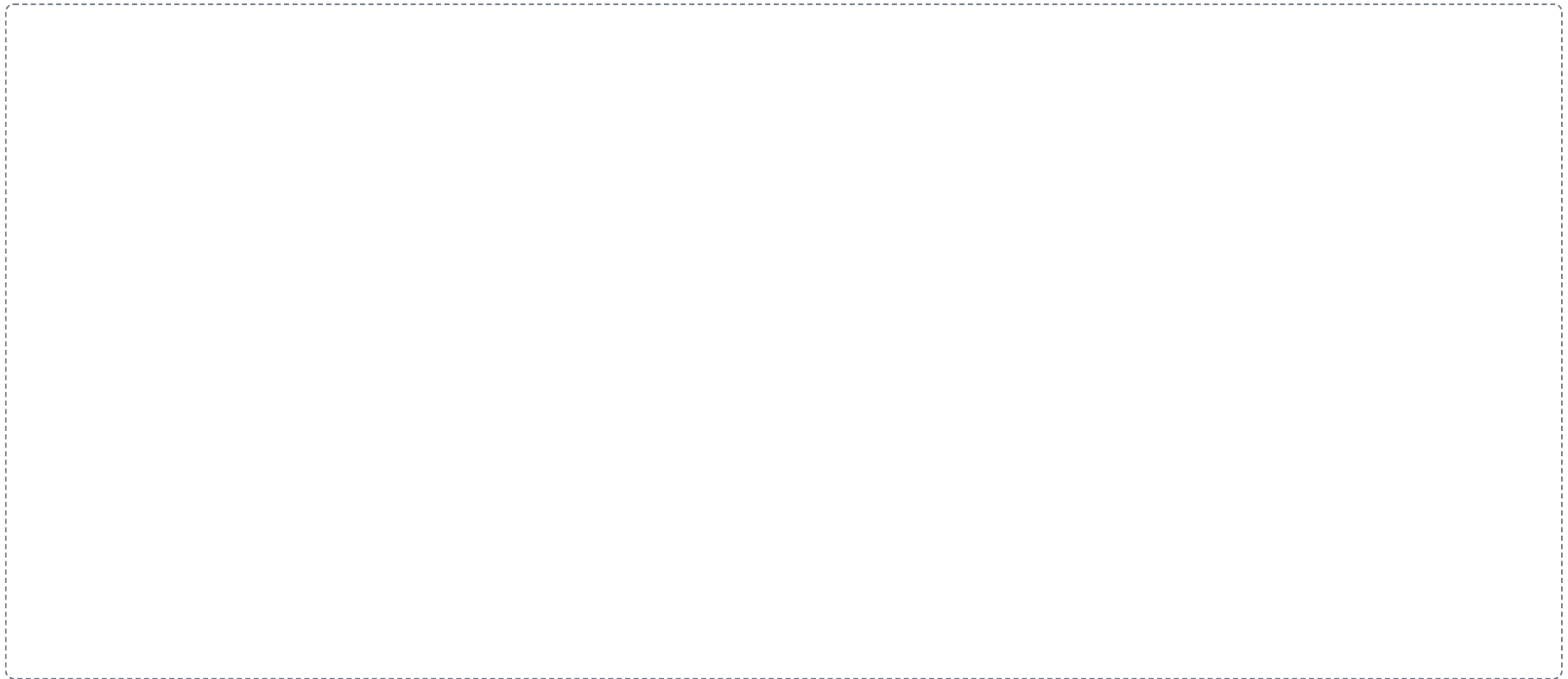
DRAWING 1

The trunk in one sagittal section, four systems at once

BRAIN DUMP FIRST

Three minutes. Every tube of Module 4 in one body, in the right order front to back.

Draw the trunk cut down the midline, from the nasal cavity to the pelvic floor, with the diaphragm across the middle. Lay in the airway in front and the alimentary canal behind it, and mark the one region where the two share a passage. Carry the canal on through the stomach, the intestines, and the rectum. Behind all of that draw one kidney, its ureter, the bladder, and the urethra. Then add the pelvic reproductive organs for one sex and mark the point where their duct joins or crosses the urinary tract.



In your notebook, head the page **M4-13 The trunk in one sagittal section, four systems at once** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can point at any level of your section and name every tube crossing it, front to back, without a label.

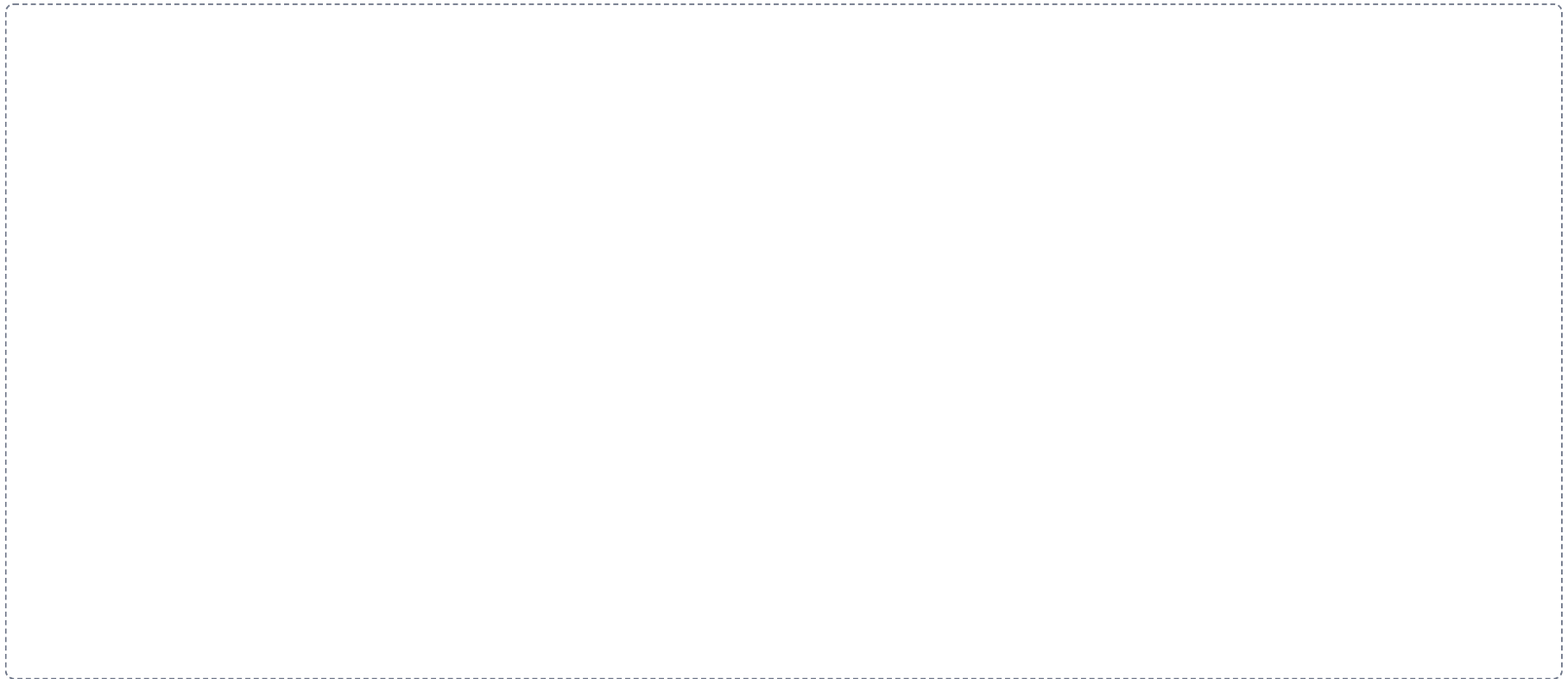
DRAWING 2

Four organs with a cortex and a medulla

BRAIN DUMP FIRST

Two minutes. Four sections side by side, and what fills each region of each one.

Draw four small sections in a row: a kidney, an adrenal gland, an ovary, and a lymph node. Label the capsule, the cortex, and the medulla on each. Then fill each region with what belongs in it: renal corpuscles and convoluted tubules in the renal cortex and pyramids in the renal medulla, three zones in the adrenal cortex and modified nervous tissue in its medulla, follicles in the ovarian cortex and vessels in its medulla, and clustered lymphocytes in the cortex of the node with its hilum on the medullary side.



In your notebook, head the page **M4-14 Four organs with a cortex and a medulla** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can be shown any one of the four in section and name the organ from the contents of its cortex alone.

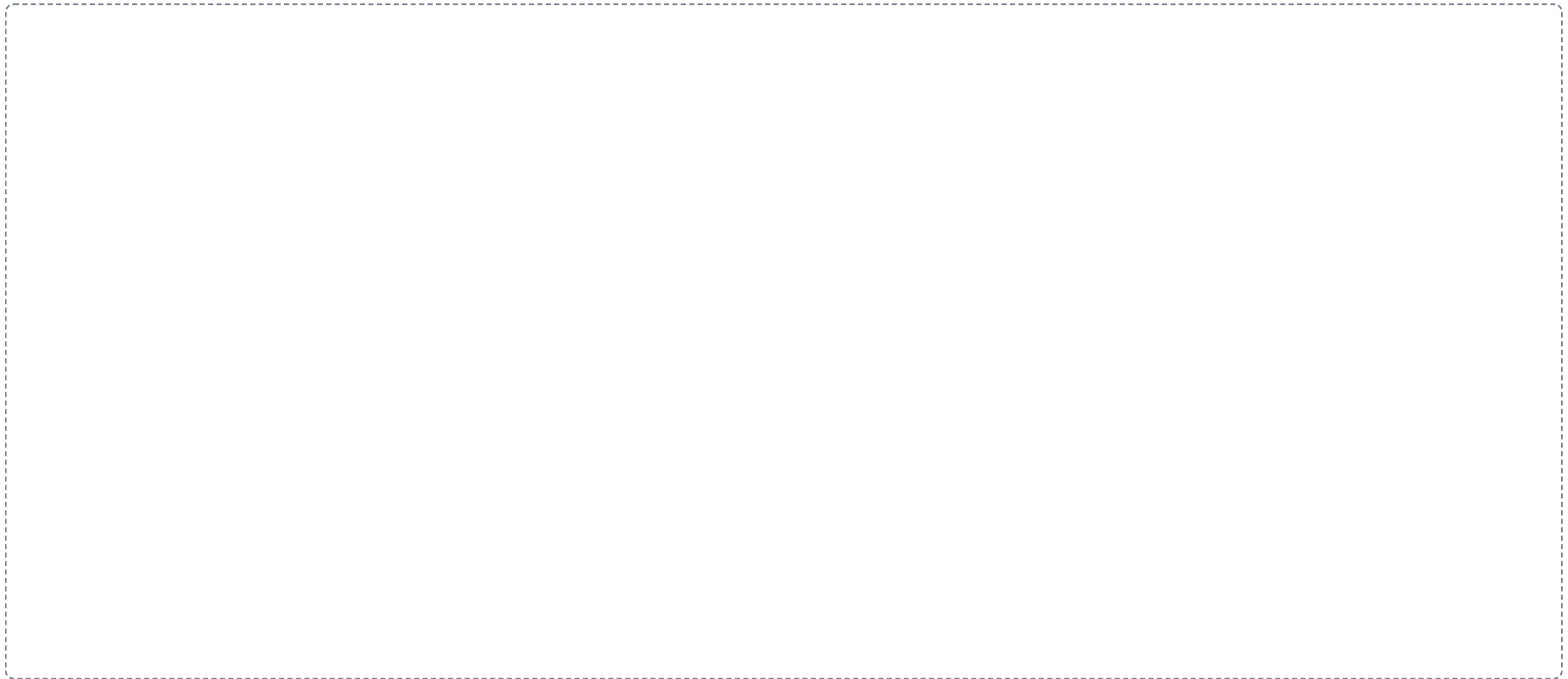
DRAWING 3

Three exchange surfaces, side by side

BRAIN DUMP FIRST

Two minutes. Three barriers drawn at the same scale, with every layer named.

Draw three enlarged details in a row. First an alveolus against a pulmonary capillary, with the alveolar epithelium, the fused basement membranes, and the capillary endothelium of the respiratory membrane. Second one intestinal villus, with its absorptive cells and their microvilli, the capillary bed inside it, and the lacteal down the middle. Third a renal corpuscle, with the fenestrated endothelium, the basement membrane, and the filtration slits between the podocyte processes. Under each one write what crosses and what is held back.



In your notebook, head the page **M4-15 Three exchange surfaces, side by side** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can say for each of the three what stays behind, and why a different barrier is built at each site.

Apply it

3 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A section shows an organ with a capsule, an outer region packed with rounded clusters of cells, and a plainer inner region. Several vessels enter through the capsule and a single vessel leaves at an indentation on one side. Name the organ, and say what rules out the kidney and the ovary.

Answer. A lymph node.

Reasoning. Read the vessels first, because a cortex and a medulla alone fit four organs in this module. Many vessels entering through the capsule and one leaving at a hilum is the traffic pattern of a lymph node, where afferent vessels arrive all around the surface and a single efferent vessel exits at the notch. The kidney also has a cortex, a medulla, and a hilum, but every vessel it owns enters and leaves at that hilum rather than through the capsule, and its medulla is built of cone-shaped pyramids, not a plain inner region. The ovary carries rounded structures in its cortex too, its follicles, but nothing is filtered through it, and its vessels sit in the medulla instead of crossing the capsule. The rounded clusters in the cortex of a node are lymphocytes, and they are exactly why a node reads like an ovary at low power. When two organs share a plan, let the vessels decide.

Set A · Name it

1. The vessels that carry lymph into a node are the _____.
2. Organs such as the kidneys, pancreas, duodenum, and parts of the colon, which lie behind the peritoneum, are described as _____.
3. The one region of the pharynx that carries air only is the _____.
4. The gland that caps the superior pole of each kidney is the _____.
5. Name the four organs in this module that have both a cortex and a medulla.

Set B · Predict it

1. A tube in the trunk is cut and a clear watery fluid drains from it rather than blood. Predict which system it belongs to and explain how you narrowed it down.

2. A single organ is removed and both a digestive secretion and a hormone are lost. Name the organ and predict both losses.

3. The skeletal muscles of the trunk and limbs are inactive for a long period. Predict the effect on lymph return and explain why breathing still helps.

-
-
4. A surgeon operating on the anterior neck must spare four glands the size of grains. Name them and predict what their position puts at risk.

Set C · Tell it apart

1. Three organs in this module are wrapped by a serous membrane. Name them, name each membrane, and say what the space between the two layers is for.
2. Distinguish an exocrine gland from an endocrine gland, then explain why the pancreas is listed as both.
3. The alveolus, the villus, and the renal corpuscle all solve the same problem. State the problem and compare the three solutions.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

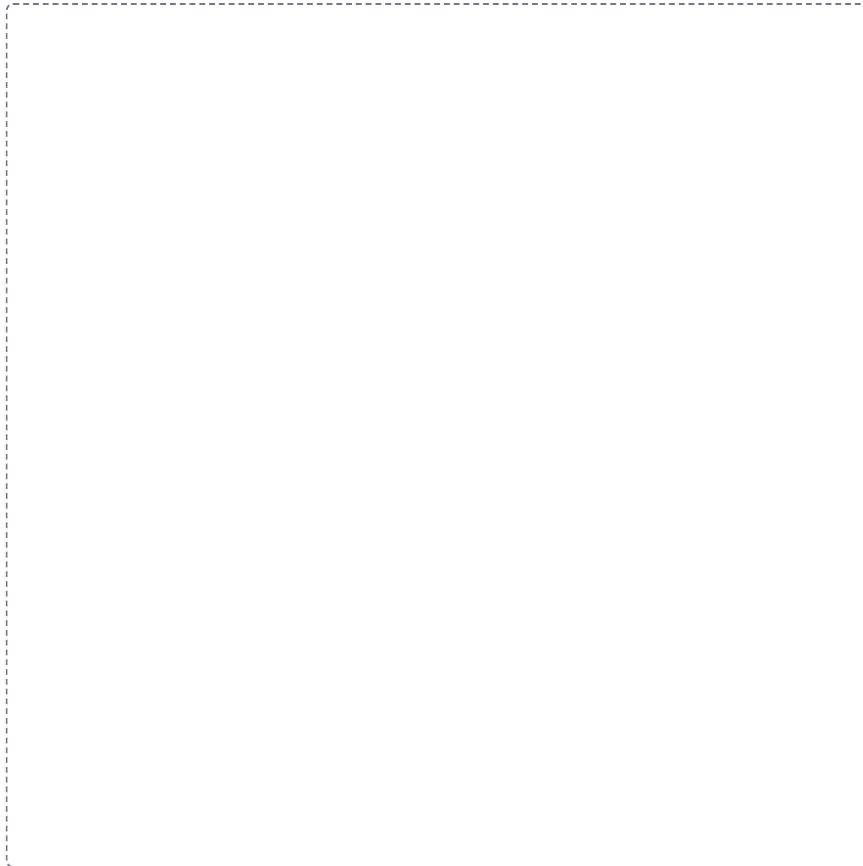
Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

1

Cortex against medulla, four organs

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

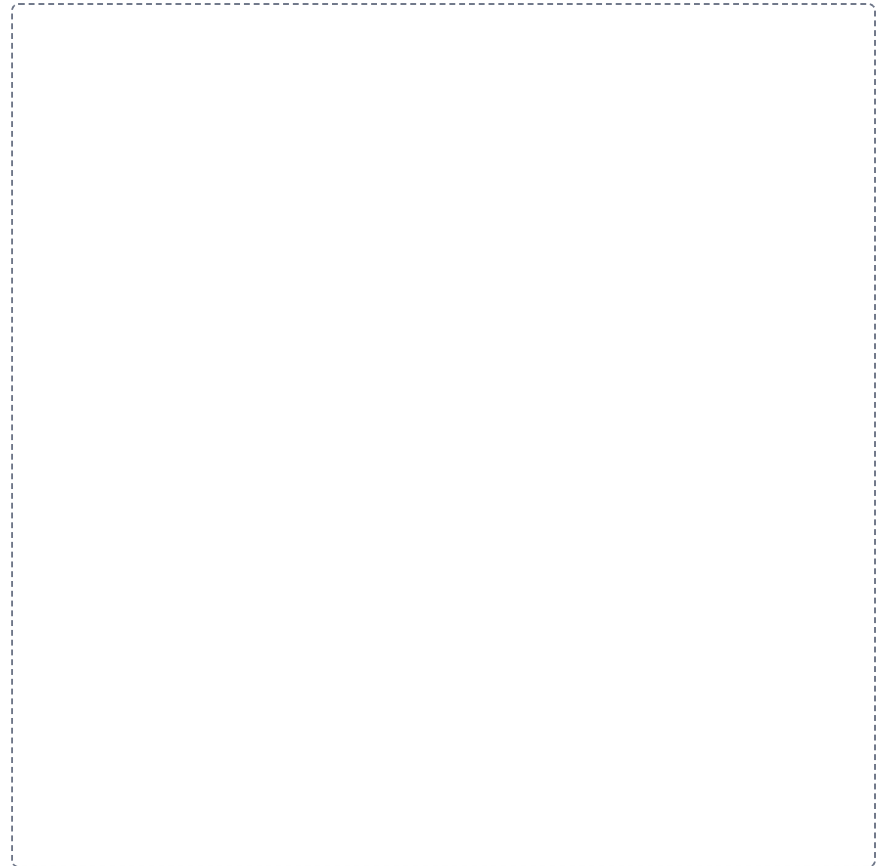


2

The four tubes of the trunk

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.



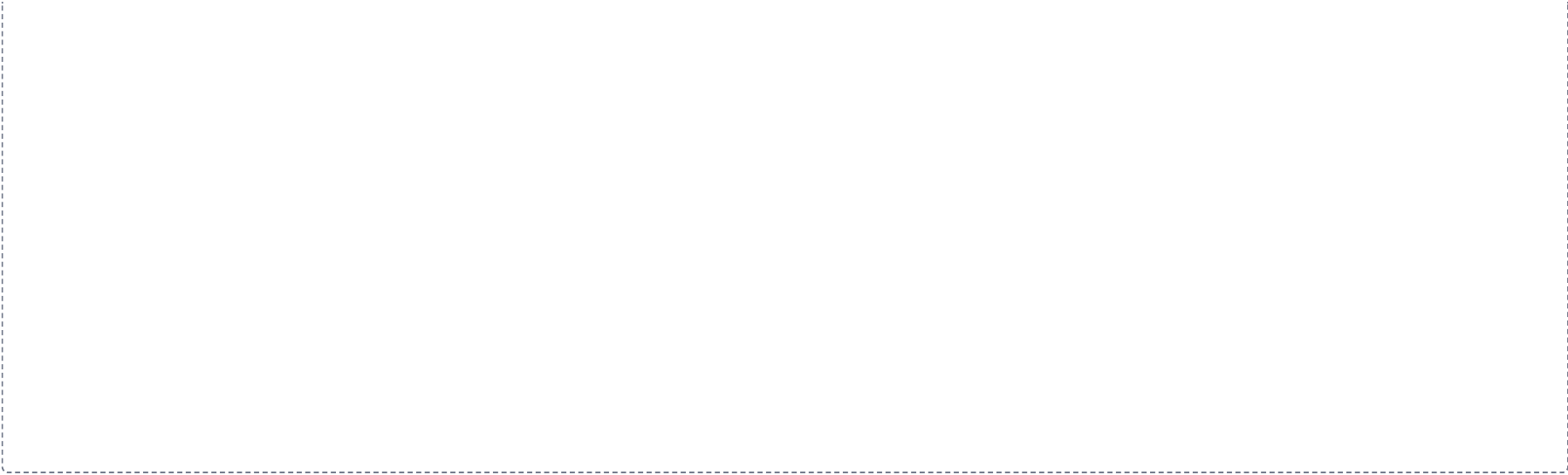
3

One drop of dietary fat, villus to bloodstream

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

A large dashed rectangular box intended for the student to draw a flowchart showing the steps from a villus to the bloodstream. The box is empty and occupies the central portion of the page.

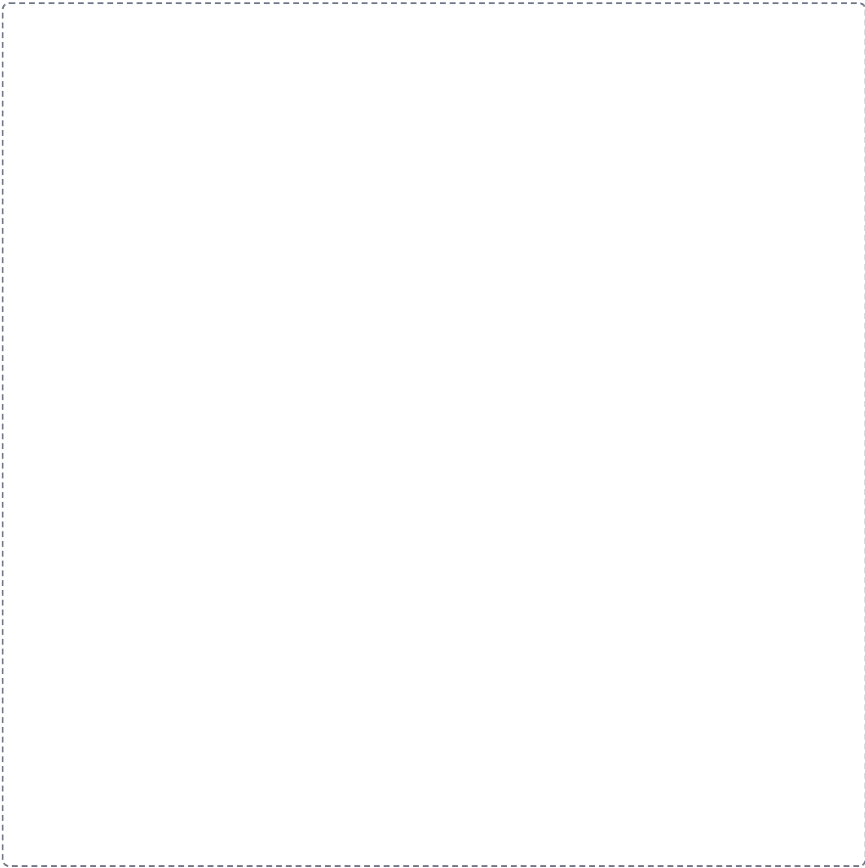


4

Organs with two jobs

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

GRID

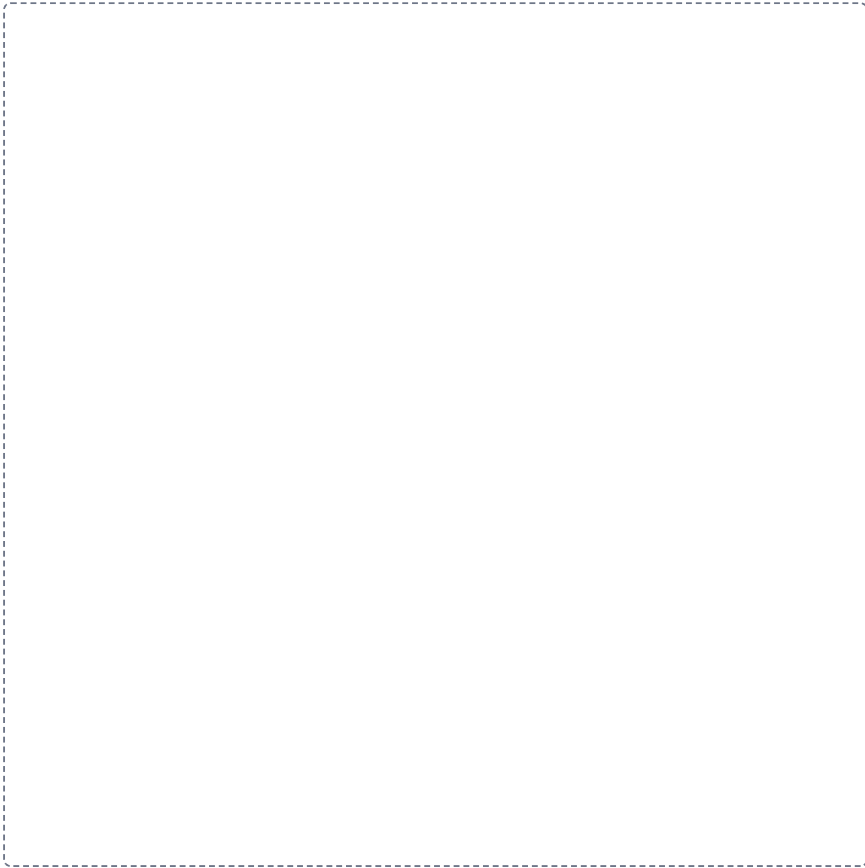


5

The shelf behind the peritoneum

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

HUB

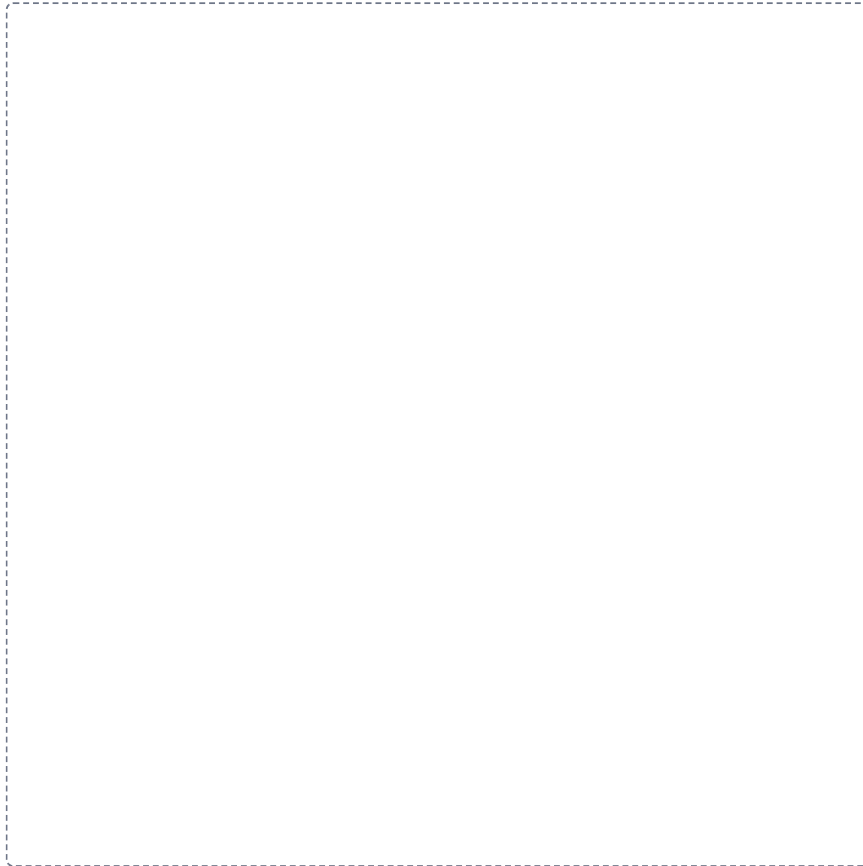


6

Exocrine against endocrine

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

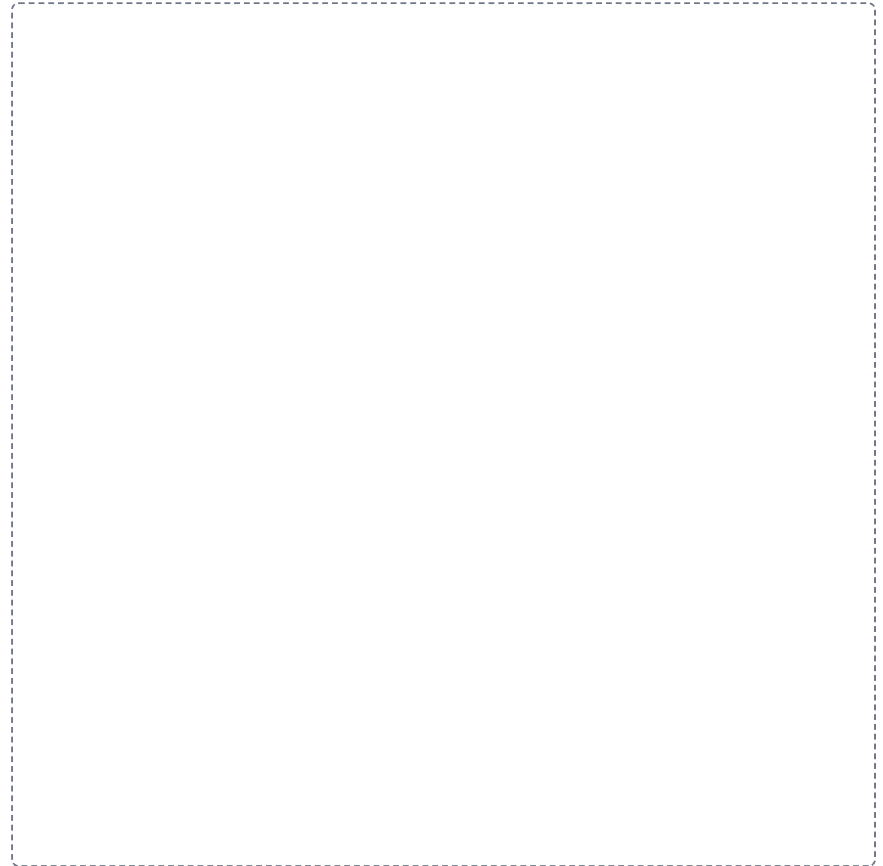


7

The same trick in four places, surface area

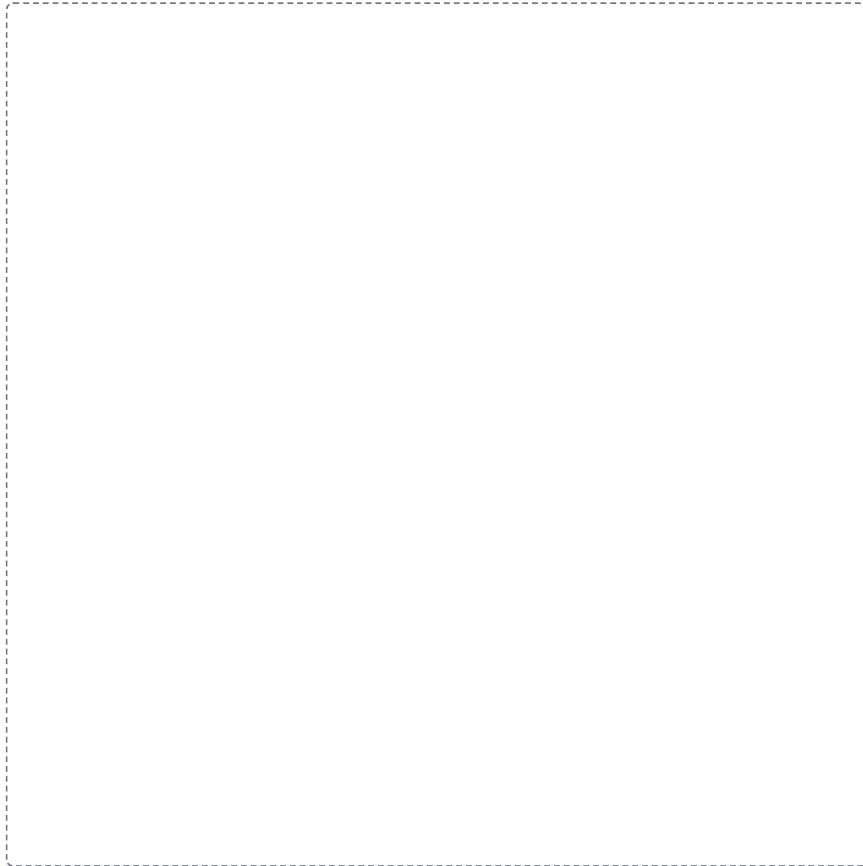
HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.



Every sphincter and valve, in order**LADDER**

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

A large, empty rectangular frame defined by a dashed line, intended for drawing anatomical levels in order from top to bottom.

Lab Structure List.

1 min

PART C

Part C of the Module 4 packet. These are the lab structures and concepts to learn for the lab portion of the Module 4 exam. Treat it as the master list, and cross reference it against every lab worksheet. Be prepared to identify structures on models, images, the cadaver, and slides.

1 The Respiratory System

2 The Lymphatic System

3 The Alimentary Canal

4 The Accessory Digestive Organs

5 The Urinary System

6 The Male Reproductive System

7 The Female Reproductive System

8 The Endocrine System

9 Vessels of the Abdomen and Pelvis

CHAPTER 1

The Respiratory System

3 min

MODELS, CADAVER, AND SLIDES

Follow the air. Every structure in this list sits somewhere along one continuous path, and if you can trace the path you can place any structure on it. The thoracic block at the end is the same cadaver dissection her master list asks for, so work the two together.

☐ External nose and nasal cavity

STRUCTURES TO IDENTIFY

- | | | |
|---|---|--|
| ☐ Root | ☐ Nasal septum | ☐ Inferior nasal concha |
| ☐ Bridge | ☐ Septal cartilage | ☐ Superior meatus |
| ☐ Dorsum nasi | ☐ Perpendicular plate of the ethmoid | ☐ Middle meatus |
| ☐ Apex | ☐ Vomer | ☐ Inferior meatus |
| ☐ Nares (nostrils) | ☐ Superior nasal concha | ☐ Hard palate |
| ☐ Nasal vestibule | ☐ Middle nasal concha | ☐ Soft palate |
| ☐ Nasal cavity | | ☐ Choanae |

☐ Paranasal sinuses

Name the bone each sinus is named for and say where it drains.

STRUCTURES TO IDENTIFY

- | | |
|--|---|
| ☐ Frontal sinus | ☐ Sphenoidal sinus |
| ☐ Ethmoidal air cells | ☐ Maxillary sinus |

☐ Pharynx

Name the epithelium of each region and say whether it carries air, food, or both.

STRUCTURES TO IDENTIFY

- | | | |
|---|--|---|
| ☐ Nasopharynx | ☐ Oropharynx | ☐ Laryngopharynx |
| ☐ Pharyngeal tonsil | ☐ Palatine tonsil | ☐ Uvula |
| ☐ Opening of the auditory tube | ☐ Lingual tonsil | ☐ Soft palate |

☐ Larynx

Identify which single cartilage is elastic rather than hyaline, and which one is the only complete ring.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Thyroid cartilage | ☐ Corniculate cartilages | ☐ Vestibular folds (false vocal cords) |
| ☐ Laryngeal prominence | ☐ Cuneiform cartilages | ☐ Glottis |
| ☐ Cricoid cartilage | ☐ Cricothyroid ligament | ☐ Hyoid bone |
| ☐ Epiglottis | ☐ Vocal folds (true vocal cords) | |
| ☐ Arytenoid cartilages | ☐ Vocal ligaments | |

☐ **Trachea and bronchial tree**

Be prepared to say which main bronchus is wider and more vertical, and why that matters.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---|
| ☐ Trachea | ☐ Right main (primary) bronchus | ☐ Bronchioles |
| ☐ Tracheal cartilages (C rings) | ☐ Left main (primary) bronchus | ☐ Terminal bronchioles |
| ☐ Trachealis muscle | ☐ Lobar (secondary) bronchi | |
| ☐ Carina | ☐ Segmental (tertiary) bronchi | |

☐ **Lungs and pleurae****STRUCTURES TO IDENTIFY**

- | | | |
|--|---|---|
| ☐ Right lung, superior lobe | ☐ Left lung, inferior lobe | ☐ Hilum of the left lung |
| ☐ Right lung, middle lobe | ☐ Cardiac notch | ☐ Root of the lung |
| ☐ Right lung, inferior lobe | ☐ Lingula | ☐ Parietal pleura |
| ☐ Horizontal fissure | ☐ Apex of the lung | ☐ Visceral pleura |
| ☐ Oblique fissure | ☐ Base of the lung | ☐ Right pleural cavity |
| ☐ Left lung, superior lobe | ☐ Hilum of the right lung | ☐ Left pleural cavity |

☐ **The thorax on the cadaver**

Memory trick for the fiber directions: hands in pockets is external, praying hands is internal.

STRUCTURES TO IDENTIFY

- | | | |
|--|------------------------------------|--|
| ☐ Thoracic cavity | ☐ Esophagus | ☐ External intercostals |
| ☐ Mediastinum | ☐ Diaphragm | ☐ Internal intercostals |

Innervation of the diaphragm. _____

Name the three structures that pass through the diaphragm and the cord level of each: T8

_____ T10 _____ T12

☐ **Vessels and nerves of the thorax**

Give the side and say whether each vessel is an artery or a vein before you name it.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---|
| ☐ Brachiocephalic trunk | ☐ Left phrenic nerve | ☐ Left internal jugular vein |
| ☐ Right subclavian artery | ☐ Right brachiocephalic vein | ☐ Right pulmonary artery |
| ☐ Left subclavian artery | ☐ Left brachiocephalic vein | ☐ Left pulmonary artery |
| ☐ Right common carotid artery | ☐ Right subclavian vein | ☐ Right pulmonary veins |
| ☐ Left common carotid artery | ☐ Left subclavian vein | ☐ Left pulmonary veins |
| ☐ Right phrenic nerve | ☐ Right internal jugular vein | |

☐ **The respiratory zone and the alveolus**

FUNCTION _____ **LOCATION** _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | | |
|---|--|---|
| ☐ Respiratory bronchiole | ☐ Alveolus | ☐ Alveolar macrophage |
| ☐ Alveolar duct | ☐ Type I alveolar cell | ☐ Respiratory membrane |
| ☐ Alveolar sac | ☐ Type II alveolar cell | ☐ Pulmonary capillary |

☐ **Respiratory slides**

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | | |
|--|---|---|
| ☐ Pseudostratified ciliated columnar epithelium | ☐ Cilia | ☐ Simple squamous epithelium of the alveolus |
| ☐ Goblet cells | ☐ Hyaline cartilage of the trachea | ☐ Lung tissue, low power |

Trace a breath of air from the nostril to an alveolus, naming every structure it passes.

CHAPTER 2

The Lymphatic System

1 min

MODELS, CADAVER, AND SLIDES

A one-way drainage network. Her master list names the spleen and the three tonsils outright; the vessels, the node, and the organ list come from the lymphatic chapter of this module. Learn the route first and the parts fall onto it.

☐ **Lymphatic vessels, from the tissue spaces to the veins**

Say which duct drains the upper right quadrant and which drains the rest of the body.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Lymphatic capillaries | ☐ Lymphatic trunks | ☐ Subclavian veins |
| ☐ Lacteals | ☐ Right lymphatic duct | ☐ Internal jugular veins |
| ☐ Lymphatic collecting vessels | ☐ Thoracic duct | |
| ☐ Lymphatic valves | ☐ Cisterna chyli | |

☐ **The lymph node**

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | | |
|----------------------------------|---|---|
| ☐ Capsule | ☐ Afferent lymphatic vessels | ☐ Cervical nodes |
| ☐ Cortex | ☐ Efferent lymphatic vessels | ☐ Axillary nodes |
| ☐ Medulla | ☐ Hilum | ☐ Inguinal nodes |

☐ **The lymphatic organs**

Sort each one as a primary or a secondary lymphatic organ.

STRUCTURES TO IDENTIFY

- | | | |
|--|--------------------------------------|----------------------------------|
| ☐ Red bone marrow | ☐ Lymph nodes | ☐ Tonsils |
| ☐ Thymus | ☐ Spleen | ☐ MALT |

☐ **The spleen**

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

☐ Spleen☐ White pulp☐ Red pulp☐ **The tonsils**

Name the region of the pharynx each tonsil guards.

STRUCTURES TO IDENTIFY

☐ Pharyngeal tonsil☐ Palatine tonsil☐ Lingual tonsil

Which two lymphatic organs are primary, and what happens in each?

Trace a drop of interstitial fluid from a tissue space back to the bloodstream.

CHAPTER 3**The Alimentary Canal**

2 min

THE CADAVER, THE TORSO MODEL, AND SLIDES

One continuous tube from the mouth to the anus. Work the peritoneum first, because the folds are what you have to move aside to reach the organs, and then take the canal region by region.

☐ **Body cavities, serous membranes, and the omenta**

STRUCTURES TO IDENTIFY

☐ Abdominal cavity☐ Visceral peritoneum☐ Falciform ligament☐ Pelvic cavity☐ Retroperitoneum☐ Mesentery☐ Peritoneal cavity☐ Greater omentum☐ Mesocolon☐ Parietal peritoneum☐ Lesser omentum

Name the retroperitoneal structures. Memory device: SADPUCKER.

☐ **The wall of the canal, in section**

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

☐ Mucosa☐ Submucosa☐ Myenteric plexus☐ Epithelium☐ Submucosal plexus☐ Serosa☐ Lamina propria☐ Muscularis, inner circular layer☐ Adventitia☐ Muscularis mucosae☐ Muscularis, outer longitudinal layer☐ Lumen

☐ **Mouth, pharynx, and esophagus**

Say which end of the esophagus carries an adventitia rather than a serosa.

STRUCTURES TO IDENTIFY

- | | | |
|--------------------------------------|--|---|
| ☐ Oral cavity | ☐ Palatine tonsil | ☐ Esophagus |
| ☐ Hard palate | ☐ Lingual tonsil | ☐ Esophageal hiatus |
| ☐ Soft palate | ☐ Nasopharynx | ☐ Upper esophageal sphincter |
| ☐ Uvula | ☐ Oropharynx | ☐ Lower (cardiac) esophageal sphincter |
| ☐ Fauces | ☐ Laryngopharynx | |

☐ **The stomach**

The stomach carries one more muscle layer than the rest of the tube. Name it and give its position.

STRUCTURES TO IDENTIFY

- | | | |
|---|--|---|
| ☐ Cardiac region | ☐ Pylorus | ☐ Gastric pits |
| ☐ Fundus | ☐ Pyloric sphincter | ☐ Gastric glands |
| ☐ Body | ☐ Cardiac sphincter | ☐ Muscularis, oblique layer |
| ☐ Pyloric region | ☐ Lesser curvature | ☐ Muscularis, circular layer |
| ☐ Pyloric antrum | ☐ Greater curvature | ☐ Muscularis, longitudinal layer |
| ☐ Pyloric canal | ☐ Gastric rugae | |

☐ **The small intestine**

FUNCTION _____ **LOCATION** _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | | |
|---|---|--|
| ☐ Duodenum | ☐ Ileocecal valve | ☐ Villi |
| ☐ Major duodenal papilla | ☐ Ileocecal sphincter | ☐ Lacteal |
| ☐ Jejunum | ☐ Mesentery | ☐ Microvilli (brush border) |
| ☐ Ileum | ☐ Circular folds (plicae circulares) | |

☐ **The large intestine**

Name the three features that belong to the large intestine and to no other part of the canal.

STRUCTURES TO IDENTIFY

- | | | |
|--|---|--|
| ☐ Cecum | ☐ Splenic flexure (left colic flexure) | ☐ Internal anal sphincter |
| ☐ Vermiform appendix | ☐ Descending colon | ☐ External anal sphincter |
| ☐ Ascending colon | ☐ Sigmoid colon | ☐ Teniae coli |
| ☐ Hepatic flexure (right colic flexure) | ☐ Rectum | ☐ Haustra |
| ☐ Transverse colon | ☐ Anal canal | ☐ Omental appendices |

Trace a bite of food from the mouth to the anus, naming every region it passes through.

CHAPTER 4**The Accessory Digestive Organs**

1 min

THE CADAVER, THE TORSO MODEL, AND SLIDES

The organs that sit alongside the tube and empty into it. Two questions on this sheet are hers outright: the fetal remnant behind the ligamentum teres, and the path of bile.

☐ The tongue and the salivary glands

Give the location of each of the three large salivary glands and say where its duct opens.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|--|
| ☐ Tongue | ☐ Intrinsic tongue muscles | ☐ Submandibular gland |
| ☐ Lingual frenulum | ☐ Extrinsic tongue muscles | ☐ Sublingual gland |
| ☐ Papillae | ☐ Parotid gland | |

☐ The liver

STRUCTURES TO IDENTIFY

- | | | |
|--|---|--|
| ☐ Right lobe | ☐ Caudate lobe | ☐ Ligamentum teres (round ligament) |
| ☐ Left lobe | ☐ Falciform ligament | ☐ Lesser omentum |
| ☐ Quadrate lobe | | |

Ligamentum teres. It is the remnant of which fetal structure? _____

☐ Liver tissue, on the slide and the model

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | | |
|--|--|--|
| ☐ Hepatocytes | ☐ Bile canaliculi | ☐ Branch of the hepatic artery |
| ☐ Hepatic laminae | ☐ Portal triad | ☐ Branch of the hepatic portal vein |
| ☐ Hepatic sinusoids | ☐ Bile duct | ☐ Central vein |

☐ The gallbladder and the biliary tree

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---|
| ☐ Gallbladder | ☐ Right hepatic duct | ☐ Common bile duct |
| ☐ Fundus of the gallbladder | ☐ Left hepatic duct | ☐ Hepatopancreatic ampulla |
| ☐ Body of the gallbladder | ☐ Common hepatic duct | ☐ Major duodenal papilla |
| ☐ Neck of the gallbladder | ☐ Cystic duct | ☐ Sphincter of Oddi |

☐ The pancreas

Say which part of the pancreas is exocrine and which is endocrine, and which duct joins the common bile duct.

STRUCTURES TO IDENTIFY

- | | | |
|---|--|---|
| ☐ Head of the pancreas | ☐ Main pancreatic duct | ☐ Major duodenal papilla |
| ☐ Body of the pancreas | ☐ Accessory pancreatic duct | ☐ Acini |
| ☐ Tail of the pancreas | ☐ Hepatopancreatic ampulla | ☐ Pancreatic islets |

Trace a drop of bile from the hepatocyte that made it to the duodenum.

CHAPTER 5

The Urinary System

1 min

KIDNEY MODEL, CADAVER, AND SLIDES

Her master list runs this system in five steps: the gross kidney, its vasculature, the renal corpuscle, the renal tubule, then the two traces. Follow it in that order and the two traces answer themselves.

☐ Gross anatomy of the urinary system

Say which kidney sits lower and why.

STRUCTURES TO IDENTIFY

- | | | |
|---|--|--|
| ☐ Right kidney | ☐ Renal column | ☐ Renal sinus |
| ☐ Left kidney | ☐ Renal pyramid | ☐ Renal hilum |
| ☐ Suprarenal (adrenal) gland | ☐ Renal papilla | ☐ Ureter |
| ☐ Renal capsule | ☐ Minor calyx | ☐ Urinary bladder |
| ☐ Renal cortex | ☐ Major calyx | ☐ Urethra |
| ☐ Renal medulla | ☐ Renal pelvis | |

☐ Vasculature of the kidney

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---------------------------------------|
| ☐ Renal artery | ☐ Efferent arteriole | ☐ Arcuate vein |
| ☐ Arcuate artery | ☐ Peritubular capillaries | ☐ Renal vein |
| ☐ Cortical radiate artery | ☐ Vasa recta | |
| ☐ Afferent arteriole | ☐ Cortical radiate vein | |

☐ The renal corpuscle

FUNCTION _____ **LOCATION** _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | | |
|--|---|---|
| ☐ Glomerulus | ☐ Parietal layer of the glomerular capsule | ☐ Podocytes |
| ☐ Glomerular (Bowman's) capsule | ☐ Visceral layer of the glomerular capsule | ☐ Capsular space |

☐ The renal tubule and the juxtaglomerular apparatus

Name the segments in the order the filtrate meets them.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Proximal convoluted tubule | ☐ Distal convoluted tubule | ☐ Juxtaglomerular apparatus |
| ☐ Loop of Henle, descending limb | ☐ Collecting duct | ☐ Macula densa cells |
| ☐ Loop of Henle, ascending limb | ☐ Papillary duct | ☐ Granular (juxtaglomerular) cells |

☐ The bladder and urethra

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---|
| ☐ Trigone | ☐ Detrusor muscle | ☐ Male urethra |
| ☐ Ureteral openings | ☐ Internal urethral sphincter | ☐ Female urethra |
| ☐ Internal urethral orifice | ☐ External urethral sphincter | |

Trace the path of urine and filtrate from the proximal convoluted tubule to the outside of the body.

Trace the path of blood through the kidney, from the renal artery to the renal vein.

CHAPTER 6

The Male Reproductive System

1 min

PELVIS MODEL, CADAVER, AND SLIDES

Sort the structures into the gonad, the ducts, the accessory glands, and the supporting structures before you learn them individually. The trace at the end is hers, and it runs straight through the duct group.

☐ The scrotum and the testis

STRUCTURES TO IDENTIFY

- | | | |
|---|--|---|
| ☐ Scrotum | ☐ Tunica albuginea | ☐ Pampiniform plexus |
| ☐ Testis | ☐ Lobules of the testis | ☐ Spermatic cord |
| ☐ Cremaster muscle | ☐ Seminiferous tubules | ☐ Inguinal canal |
| ☐ Dartos muscle | ☐ Testicular artery | |
| ☐ Tunica vaginalis | ☐ Testicular vein | |

☐ Testis tissue, on the slide

FUNCTION _____ LOCATION _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | |
|--|--|
| ☐ Seminiferous tubule | ☐ Interstitial (Leydig) cells |
| ☐ Sustentacular (Sertoli) cells | ☐ Developing sperm cells |

☐ The duct system

Only one structure on this sheet produces sperm. Name it, then say what each of the others does with them.

STRUCTURES TO IDENTIFY

- | | | |
|---|--|--|
| ☐ Epididymis | ☐ Ejaculatory duct | ☐ Membranous urethra |
| ☐ Vas deferens (ductus deferens) | ☐ Prostatic urethra | ☐ Spongy (penile) urethra |

☐ The accessory glands

Say what each gland contributes to semen.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---|
| ☐ Seminal vesicles | ☐ Prostate gland | ☐ Bulbourethral glands |
|---|---|---|

☐ **The penis****STRUCTURES TO IDENTIFY**

- | | | |
|--------------------------------------|--|--|
| ☐ Root | ☐ Prepuce | ☐ Spongy (penile) urethra |
| ☐ Body | ☐ Corpus spongiosum | |
| ☐ Glans penis | ☐ Corpora cavernosa | |

☐ **The pelvic vessels****STRUCTURES TO IDENTIFY**

- | | | |
|--|--|--|
| ☐ Internal iliac artery | ☐ Medial umbilical ligament | ☐ External iliac vein |
| ☐ External iliac artery | ☐ Internal iliac vein | |

Medial umbilical ligament. It is the remnant of which fetal structure?

Trace the path of sperm from the seminiferous tubules to the outside of the body.

CHAPTER 7**The Female Reproductive System**

1 min

PELVIS MODEL, CADAVER, AND SLIDES

The ligaments are the hard part of this sheet, so name what each one connects as you find it. The tract itself is short and runs in one direction, from the ovary to the vaginal orifice.

☐ **The ovary and its attachments**

Say what each ligament attaches the ovary to.

STRUCTURES TO IDENTIFY

- | | | |
|--|---|---------------------------------------|
| ☐ Ovary | ☐ Broad ligament of the uterus | ☐ Ovarian vein |
| ☐ Ovarian ligament | ☐ Mesovarium | |
| ☐ Suspensory ligament | ☐ Ovarian artery | |

☐ **Ovary tissue, on the slide**

FUNCTION _____ **LOCATION** _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | | |
|--|--|--|
| ☐ Germinal epithelium | ☐ Ovarian medulla | ☐ Corpus luteum |
| ☐ Tunica albuginea | ☐ Ovarian follicles | ☐ Corpus albicans |
| ☐ Ovarian cortex | ☐ Mature follicle | |

☐ **The uterine tubes**

Name the region where fertilization usually happens.

STRUCTURES TO IDENTIFY

- | | |
|---------------------------------------|----------------------------------|
| ☐ Fimbriae | ☐ Ampulla |
| ☐ Infundibulum | ☐ Isthmus |

☐ **The uterus**

Name the layer that sheds in menstruation and the layer that rebuilds it.

STRUCTURES TO IDENTIFY

- | | | |
|---|---|--|
| ☐ Fundus | ☐ Cervical canal | ☐ Cardinal ligament |
| ☐ Body | ☐ Round ligament of the uterus | ☐ Perimetrium |
| ☐ Cervix | ☐ Broad ligament of the uterus | ☐ Myometrium |
| ☐ Uterine cavity | ☐ Uterosacral ligament | ☐ Endometrium |

☐ **The vagina and the vulva****STRUCTURES TO IDENTIFY**

- | | | |
|--|---|------------------------------------|
| ☐ Vagina | ☐ Labia majora | ☐ Vestibule |
| ☐ Vaginal orifice | ☐ Labia minora | ☐ Perineum |
| ☐ Fornix | ☐ Clitoris | |
| ☐ Mons pubis | ☐ Urethral orifice | |

☐ **The pelvic vessels****STRUCTURES TO IDENTIFY**

- | | | |
|--|--|--|
| ☐ Internal iliac artery | ☐ Medial umbilical ligament | ☐ External iliac vein |
| ☐ External iliac artery | ☐ Internal iliac vein | |

Medial umbilical ligament. It is the remnant of which fetal structure?

Trace the path of an oocyte from the ovary to the uterine cavity.

CHAPTER 8**The Endocrine System**

1 min

TORSO MODEL, CADAVER, AND SLIDES

The endocrine glands are scattered, so this sheet is a tour of locations first. Her master list names the thyroid gland in the neck and the endocrine pancreas and the suprarenal glands in the abdomen; the rest come from the endocrine chapter of this module.

☐ **Locating the glands**

Give the location of each gland before you name it.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|----------------------------------|
| ☐ Hypothalamus | ☐ Parathyroid glands | ☐ Ovaries |
| ☐ Pituitary gland | ☐ Thymus | ☐ Testes |
| ☐ Pineal gland | ☐ Adrenal (suprarenal) glands | |
| ☐ Thyroid gland | ☐ Pancreas | |

☐ **The pituitary gland**

Say which lobe is nervous tissue and which is glandular epithelium.

STRUCTURES TO IDENTIFY

- | | | |
|--|--|---|
| ☐ Hypophysis | ☐ Infundibulum | ☐ Posterior lobe (neurohypophysis) |
| ☐ Sella turcica | ☐ Anterior lobe (adenohypophysis) | |

☐ **The thyroid and parathyroid glands****STRUCTURES TO IDENTIFY**

- | | | |
|--|---|----------------------------------|
| ☐ Thyroid gland, right lobe | ☐ Isthmus | ☐ Trachea |
| ☐ Thyroid gland, left lobe | ☐ Parathyroid glands | ☐ Larynx |

☐ **Thyroid tissue, on the slide**

FUNCTION _____ **LOCATION** _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | |
|--|---|
| ☐ Thyroid follicles | ☐ Follicular cells |
| ☐ Colloid | ☐ Parafollicular (C) cells |

☐ **The adrenal gland**

FUNCTION _____ **LOCATION** _____

SPECIAL STRUCTURES _____

STRUCTURES TO IDENTIFY

- | | | |
|---|---|---------------------------------|
| ☐ Adrenal (suprarenal) gland | ☐ Zona fasciculata | ☐ Kidney |
| ☐ Adrenal cortex | ☐ Zona reticularis | |
| ☐ Zona glomerulosa | ☐ Adrenal medulla | |

☐ **The endocrine pancreas**

Say which of these two clusters is endocrine and which is exocrine.

STRUCTURES TO IDENTIFY

- | | | |
|-----------------------------------|--|--------------------------------|
| ☐ Pancreas | ☐ Pancreatic islets | ☐ Acini |
|-----------------------------------|--|--------------------------------|

Which gland sits on the superior pole of each kidney, and what are its two parts made of?

CHAPTER 9**Vessels of the Abdomen and Pelvis**

1 min

CADAVER AND TORSO MODEL

Her master list groups these as arteries then veins, with the hepatic portal system kept separate. Learn the three unpaired branches of the abdominal aorta in order from the top, because their order is what the tag is testing.

☐ **Arteries of the abdomen**

Name the three unpaired visceral branches of the abdominal aorta in order from superior to inferior, and say what each one supplies.

STRUCTURES TO IDENTIFY

- | | | |
|--|---|---|
| ☐ Abdominal aorta | ☐ Common hepatic artery | ☐ Inferior mesenteric artery |
| ☐ Celiac trunk | ☐ Superior mesenteric artery | ☐ Right common iliac artery |
| ☐ Left gastric artery | ☐ Renal arteries | ☐ Left common iliac artery |
| ☐ Splenic artery | ☐ Gonadal (testicular or ovarian) arteries | |

☐ **Arteries of the pelvis****STRUCTURES TO IDENTIFY**

- | | | |
|--|--|--|
| ☐ Internal iliac artery | ☐ External iliac artery | ☐ Medial umbilical ligament |
|--|--|--|

☐ **Veins of the abdomen and pelvis****STRUCTURES TO IDENTIFY**

- | | | |
|---|--|---|
| ☐ Inferior vena cava | ☐ Gonadal (testicular or ovarian) veins | ☐ Left common iliac vein |
| ☐ Renal veins | ☐ Right common iliac vein | ☐ Internal iliac vein |
| | | ☐ External iliac vein |

☐ **The hepatic portal system**

Say what makes this a portal system, and name the two capillary beds the blood passes through.

STRUCTURES TO IDENTIFY

- | | | |
|--|---|--|
| ☐ Hepatic portal vein | ☐ Superior mesenteric vein | ☐ Hepatic veins |
| ☐ Splenic vein | ☐ Inferior mesenteric vein | |

Trace a molecule of glucose absorbed by the small intestine from the intestinal capillary to the right atrium.
